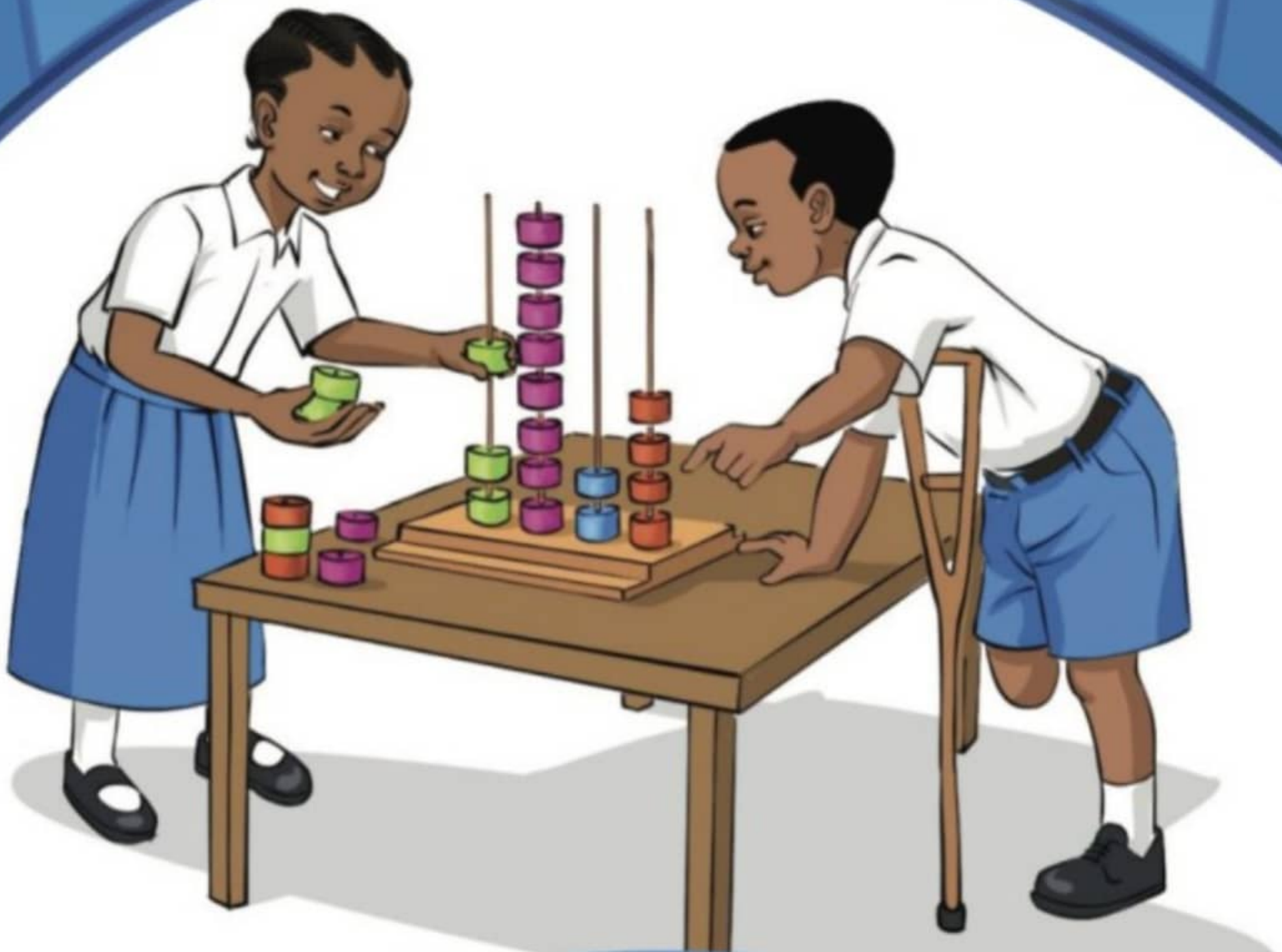
Mathematics

Pupil's Book

Standard Three



Tanzania Institute of Education

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Mathematics

Pupil's Book

Standard Three

THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF EDUCATION,
SCIENCE AND TECHNOLOGY

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Dr Lyabwene M. Mtahabwa
Commissioner for Education

Tanzania Institute of Education

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Dr Aneth A. Komba
Director General
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Introduction

This textbook, Mathematics Pupil's Book for Standard Three, is written in accordance with the 2023 Mathematics Syllabus for Primary Schools issued by the Ministry of Education, Science and Technology. This is the second edition after the improvement of the Standard Three textbook published in 2018, based on the 2016 syllabus.

The book consists of seven chapters, namely, Whole numbers; Addition and subtraction of whole numbers; Measurements of length, mass and volume; Shapes and figures; Fractions; Measurements of time; and Tanzanian currency. The competencies developed will enable you to acquire basic mathematical skills and their applications, logical thinking and solve real life problems.

You are advised to engage yourself and with your friends in all activities and excercises in this book in order to develop the intended competencies.

Additional learning resources are available in the TIE e-Library at https://ol.tie.go.tz or ol.tie.go.tz

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Chapter One
Whole numbers

Introduction

In this chapter, you will learn about whole numbers with more than one digit. You will recognise the place value of a digit in whole numbers, and also read and write whole numbers. You will also compare numbers and recognise sequences of whole numbers. The competencies developed will enable you to use numbers in different contexts.

Place values in whole numbers

The place value of a digit depends on its position in a number. The place value increases from the right to the left of a number. The place value of a digit in a number can be read from right to left as follows:

(a) The first position from right represents **ones**.

(b) The second position from right represents **tens**.

(c) The third position from right represents **hundreds**.

(d) The fourth position from right represents **thousands**.

(e) The fifth position from right represents **ten thousands**.

(f) The sixth position from right represents **hundred thousands**.

(g) The seven position from right represents **millions.**

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Example 1

Write the place value of each digit in 1364.

Answer

4 is in ones
6 is in tens
3 is in hundreds
1 is in thousands

Example 2

Write the place value of each digit in 6948.

Answer

8 is in ones
4 is in tens
9 is in hundreds
6 is in thousands

Example 3

A number has 7 in the thousands place, 3 in the hundreds place, 0 in the tens place, and 4 in the ones place. What is the number?

Answer

The number is 7304

Example 4

Fill in the blanks with the place value of each digit in the given number.

5463 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

Answer

5463 = has 5 thousands, 4 hundreds, 6 tens and 3 ones.

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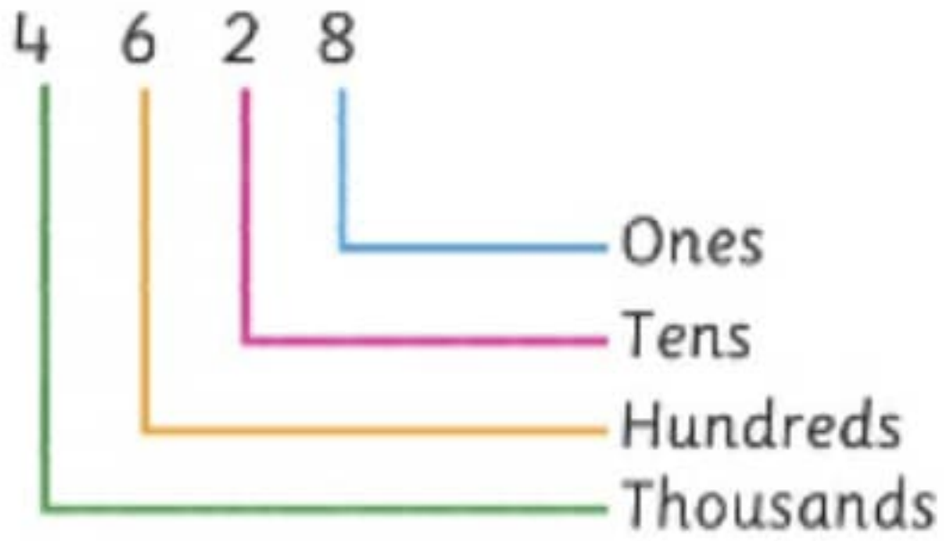
Example 5

Identify the place value of each digit in the following whole numbers:

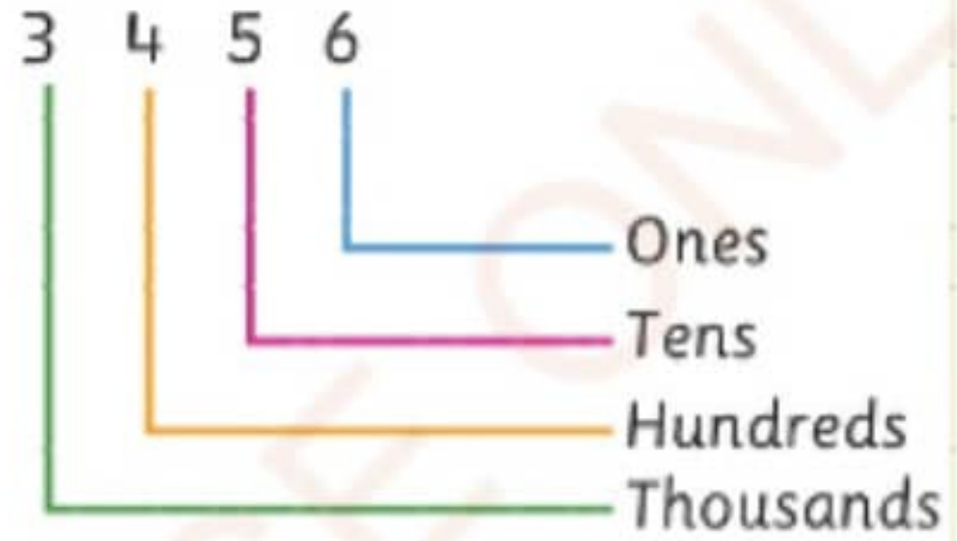
(a) 4628 (b) 3456 (c) 7304 (d) 9000

Answer

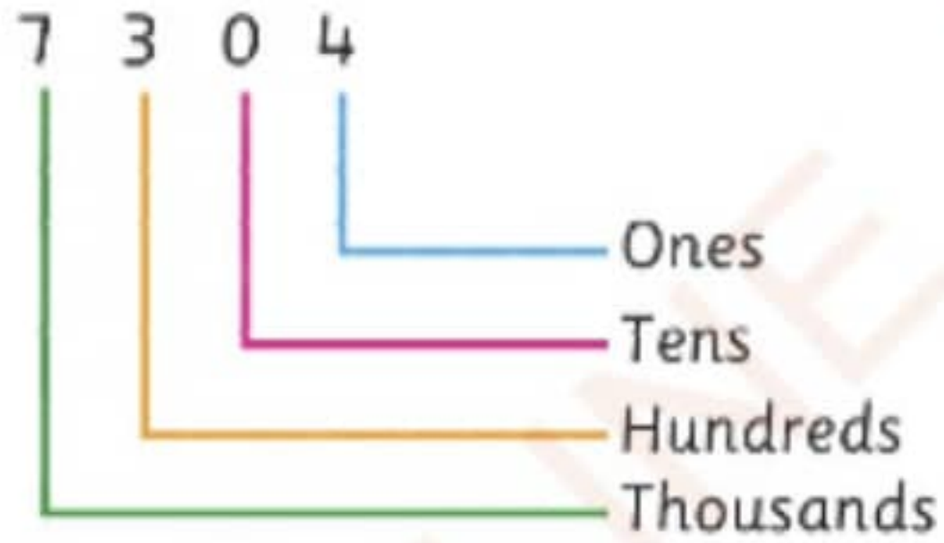
(a) 4628



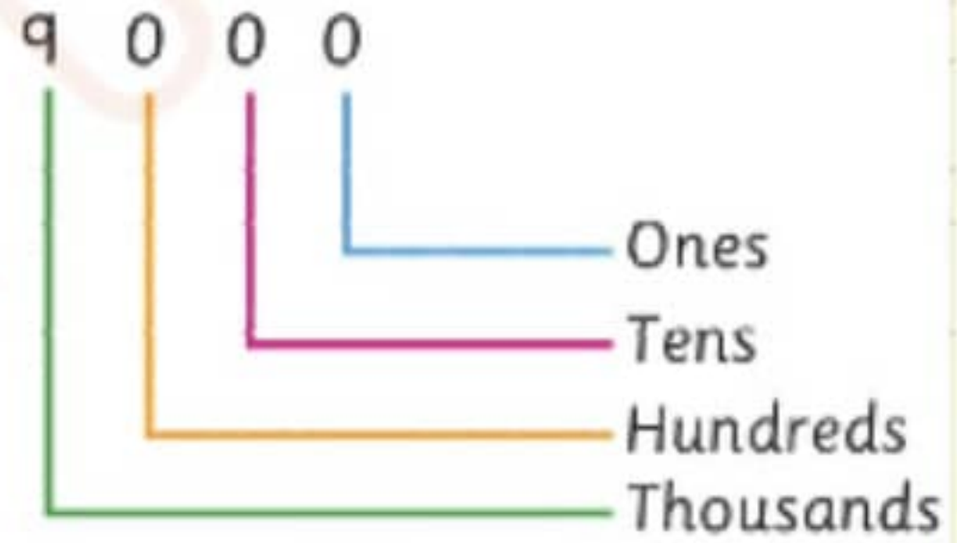
(b) 3456



(c) 7304



(d) 9000



Exercise 1

1. Write the place value of each digit in 9825.

(a) _____ is in the tens place.

(b) _____ is in the ones place.

(c) _____ is in the thousands place.

(d) _____ is in the hundreds place.

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2. Write a number with the following place values:
   (a) 2 hundreds, 9 ones, 3 thousands and 8 tens
   (b) 0 ones, 2 hundreds, 6 tens and 5 thousands
   (c) 6 tens, 6 thousands, 8 hundreds and 2 ones
   (d) 3 thousands, 4 tens, 8 hundreds and 7 ones

3. Fill in the following table the place value of each digit in the given numbers.

Number	Place values			
	Thousands	Hundreds	Tens	Ones
1001				
6666				
4000				
3339				

4. (a) Which digit is in the hundreds place in 7029?
   (b) Which digit is in the thousands place in 8173?
   (c) Identify the digit which is in the tens place in the number 3487.
   (d) A number has 0 in the ones place, 6 in the tens place and 7 in the hundreds place. What is the number?

5. (a) How many hundreds are there in 3287?
   (b) There are 9674 apples in the basket. How many thousands apples are there?

6. Identify the place value of each digit in the following whole numbers:
   (a) 6247 (b) 9999 (c) 7001

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(d) 4444 (e) 3239 (f) 2776

(g) 1111 (h) 8107 (i) 9009

(j) 5108

7. Write the place value of the shaded digit in the following whole numbers:

(a) 10**2**0 (b) **2**010 (c) 168**7**

(d) 7**9**22 (e) 888**8** (f) 1**6**92

(g) **5**401 (h) 91**1**1 (i) 3**3**33

(j) 20**0**0

8. Fill in the blanks with the place values of the given digits.

(a) In 2968:

2 is in _____, 9 is in _____,

6 is in _____ and 8 is in _____.

(b) In 9801:

9 is in _____, 8 is in _____,

0 is in _____ and 1 is in _____.

(c) In 7236:

7 is in _____, 2 is in _____,

3 is in _____ and 6 is in _____.

(d) In 5649:

5 is in _____, 6 is in _____,

4 is in _____ and 9 is in _____.

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9. Write the number represented by the place values in the following table:

Thousands	Hundreds	Tens	Ones	Number
1	0	2	0	
1	1	2	0	
2	4	4	0	
3	3	0	1	
6	8	9	9	
9	2	7	4	
2	9	7	3	
8	1	2	2	
7	1	7	3	
1	2	7	7	
7	2	9	1	
1	0	0	0	
2	3	1	1	

10. Fill in the blanks with the correct place values of the following whole numbers.

(a) 1872 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(b) 2663 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(c) 4793 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

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(d) 5187 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(e) 4002 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(f) 1562 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(g) 7371 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(h) 4572 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(i) 3421 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(j) 2012 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(k) 6682 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(l) 5583 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(m) 5234 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(n) 2709 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(o) 5235 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(p) 9123 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(q) 3410 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

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11. Write the digit in the thousands place value in each of the following whole numbers:

    (a) 7682 (b) 9501

    (c) 8306 (d) 6412

12. Write the place value of 3 in each of the following whole numbers:

    (a) 436 (b) 3900

    (c) 2843 (d) 2301

13. Underline the digit in the hundreds place in each of the following whole numbers:

    (a) 269 (b) 7901

    (c) 3749 (d) 9568

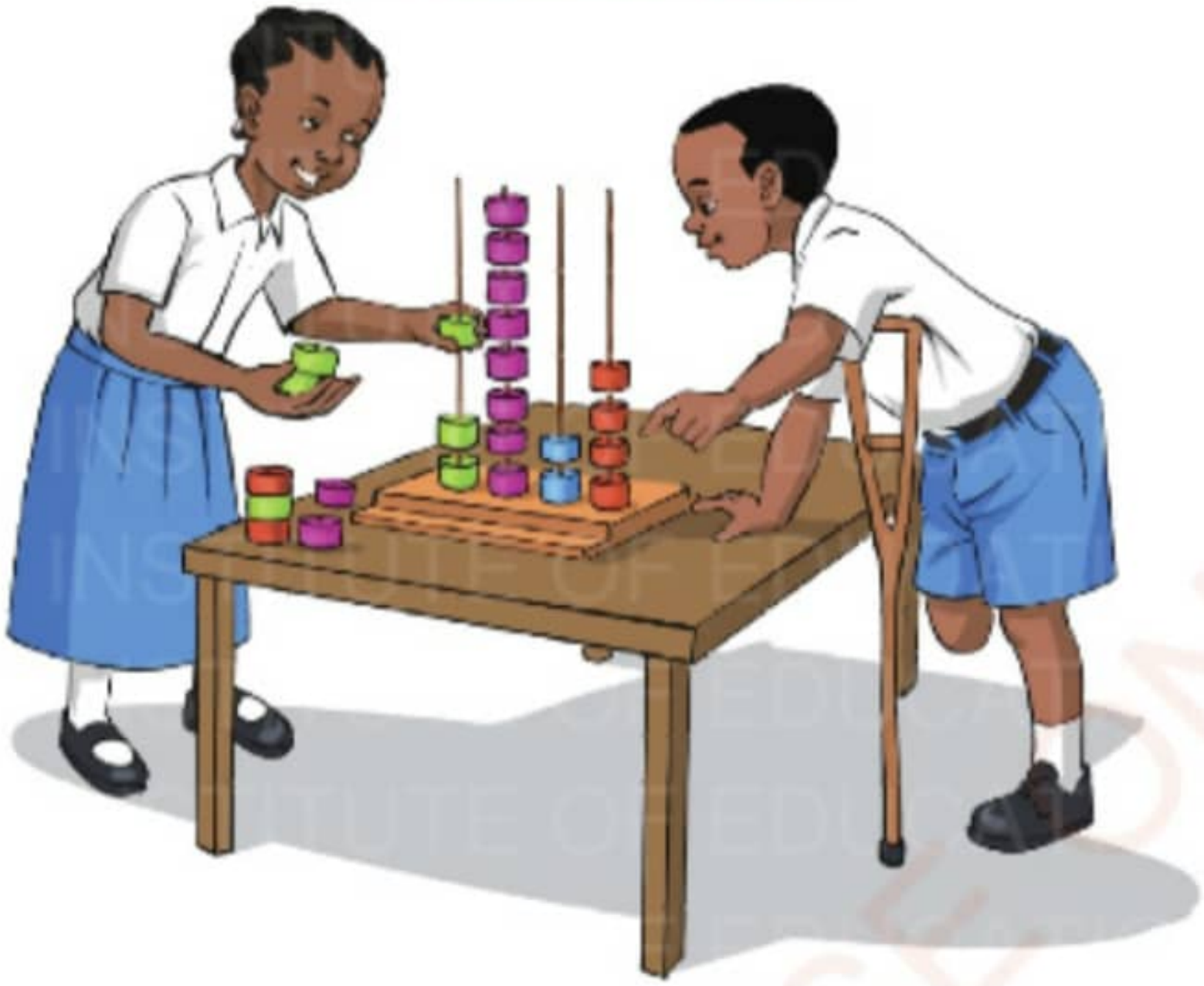
14. (a) I am a two-digit number with 3 in the ones place. The digit in the tens exceeds that in the ones by 4. What number do I represent?

    (b) I am a two-digit number with 9 in the tens place. The digit in the ones is less than the digit in the tens by 3. What number am I?

Writing the place value of a digit in a number using abacus

The place value of digits in a number can be identified using the abacus. The following picture shows pupils placing counters on the abacus to form a number.

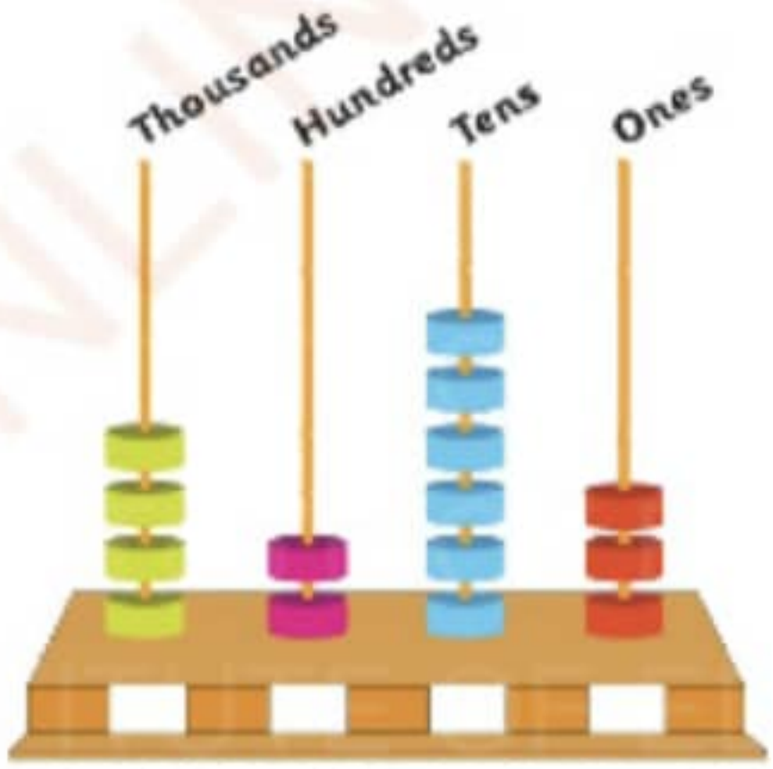
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After forming a number, they can write the place value of each digit in that number.

Example 1

Write the place value of each digit in the following abacus:



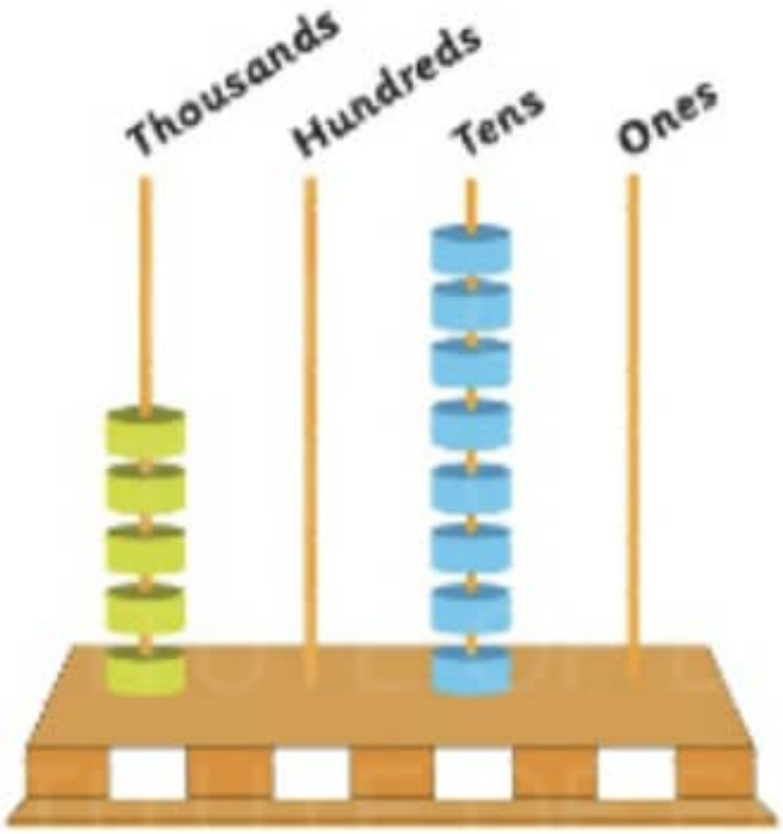
Answer

The abacus represents 4263. There are 4 thousands, 2 hundreds, 6 tens, and 3 ones.

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Example 2

Write the place value of each digit in the following abacus:

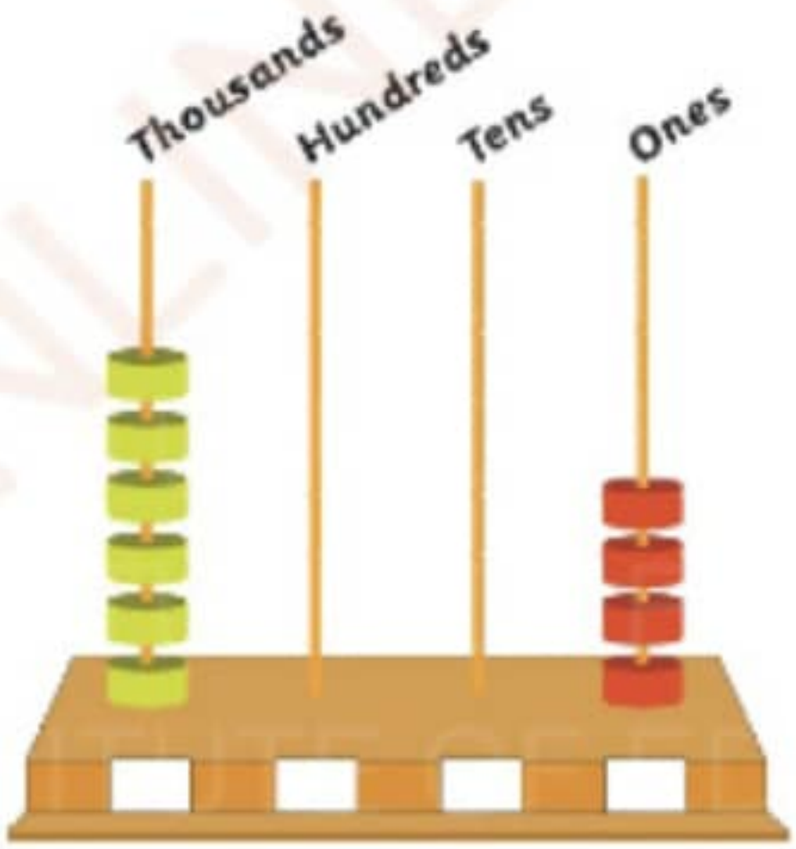


Answer

The abacus represents 5080. There are 5 thousands, 0 hundreds, 8 tens, and 0 ones.

Example 3

Write the place value of each digit in the following abacus:



Answer

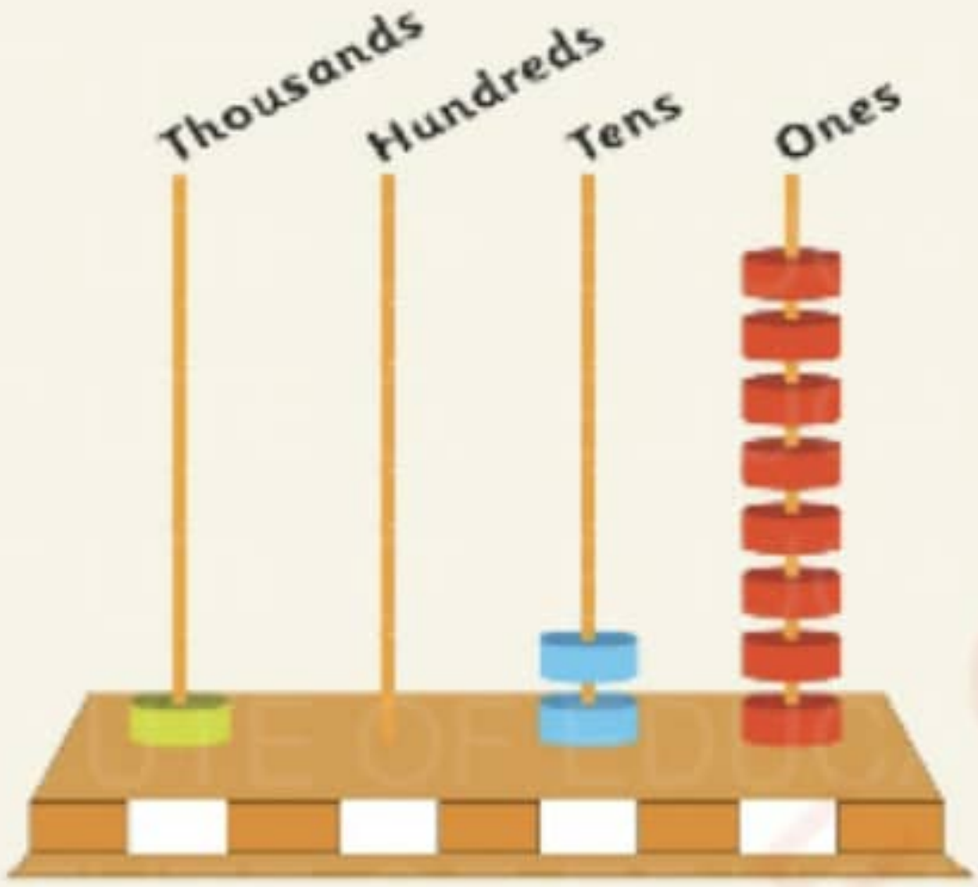
The abacus represents 6004. There are 6 thousands, 0 hundreds, 0 tens, and 4 ones.

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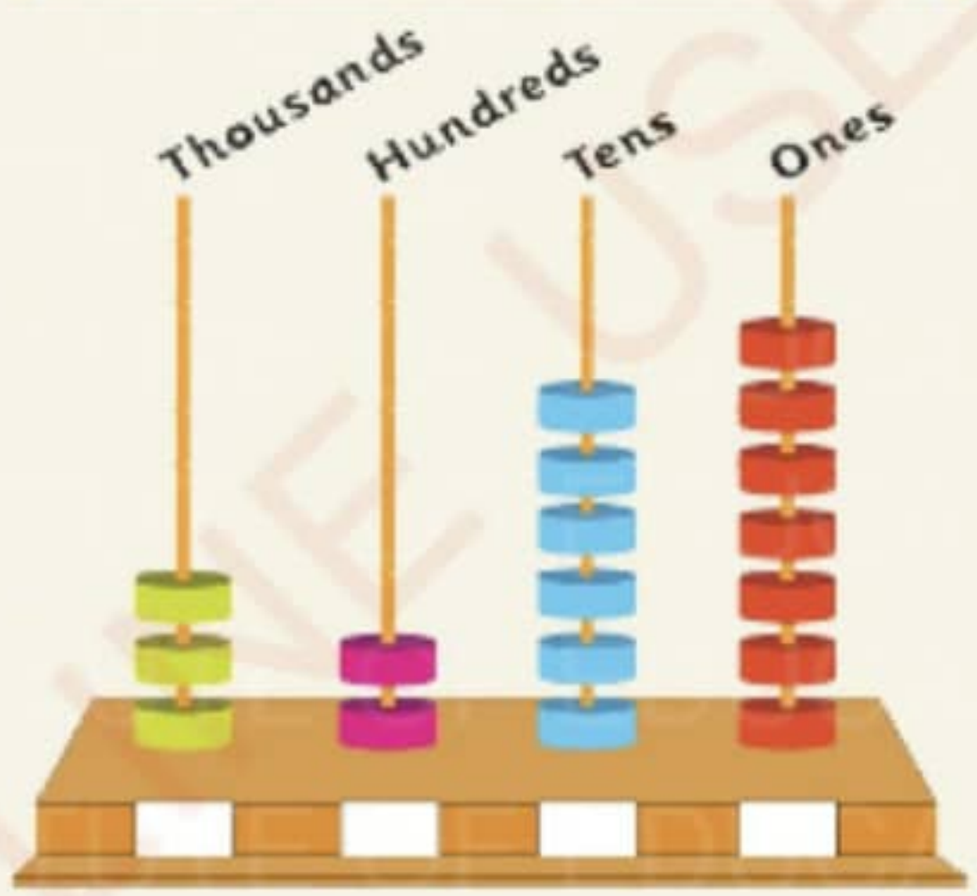
Exercise 2

Carefully study the abaci 1 - 6, then write the number represented by each.

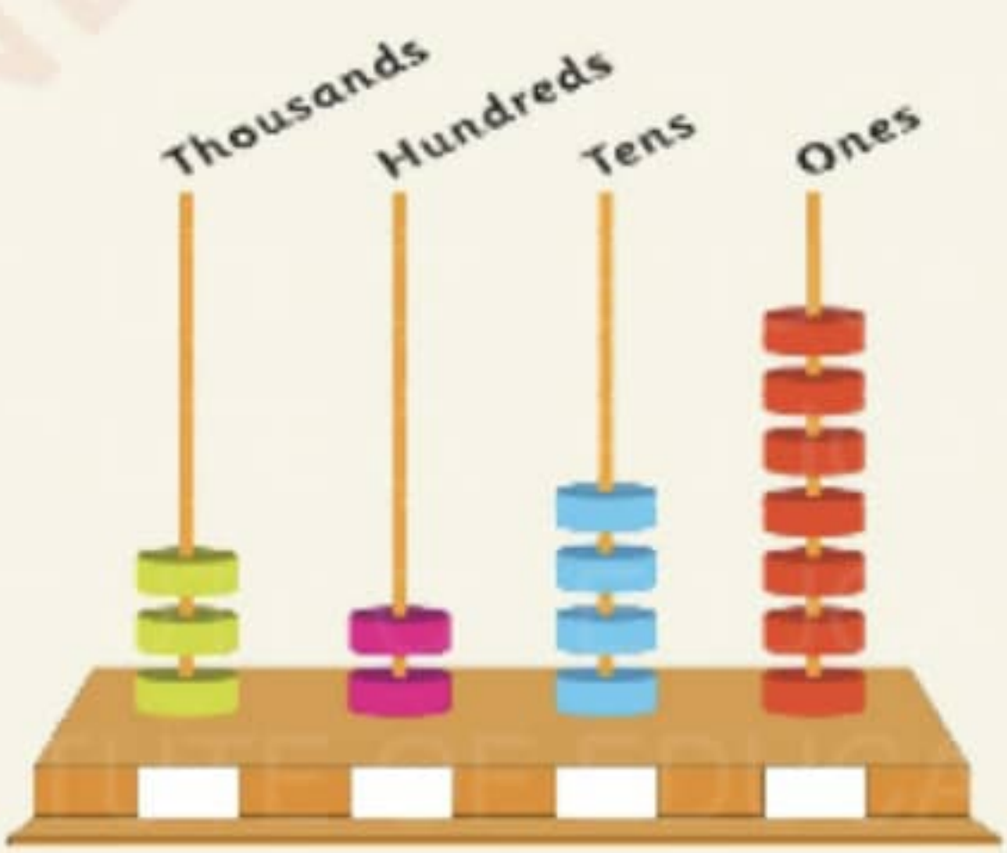
1.



2.

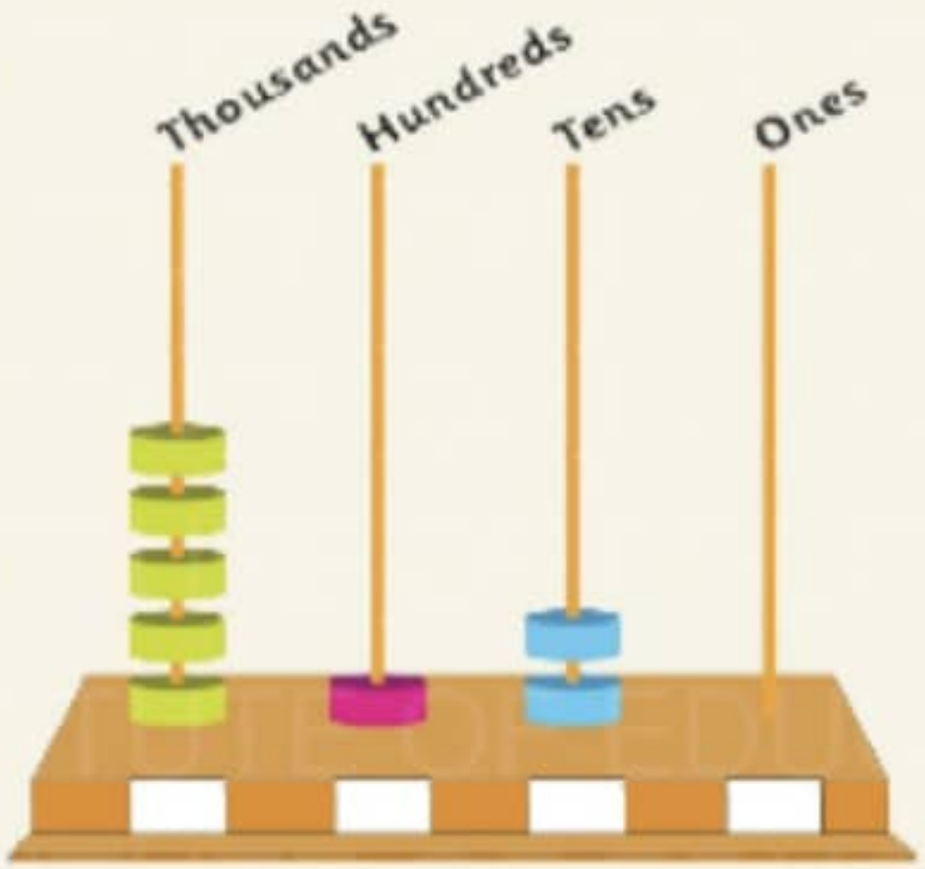


3.

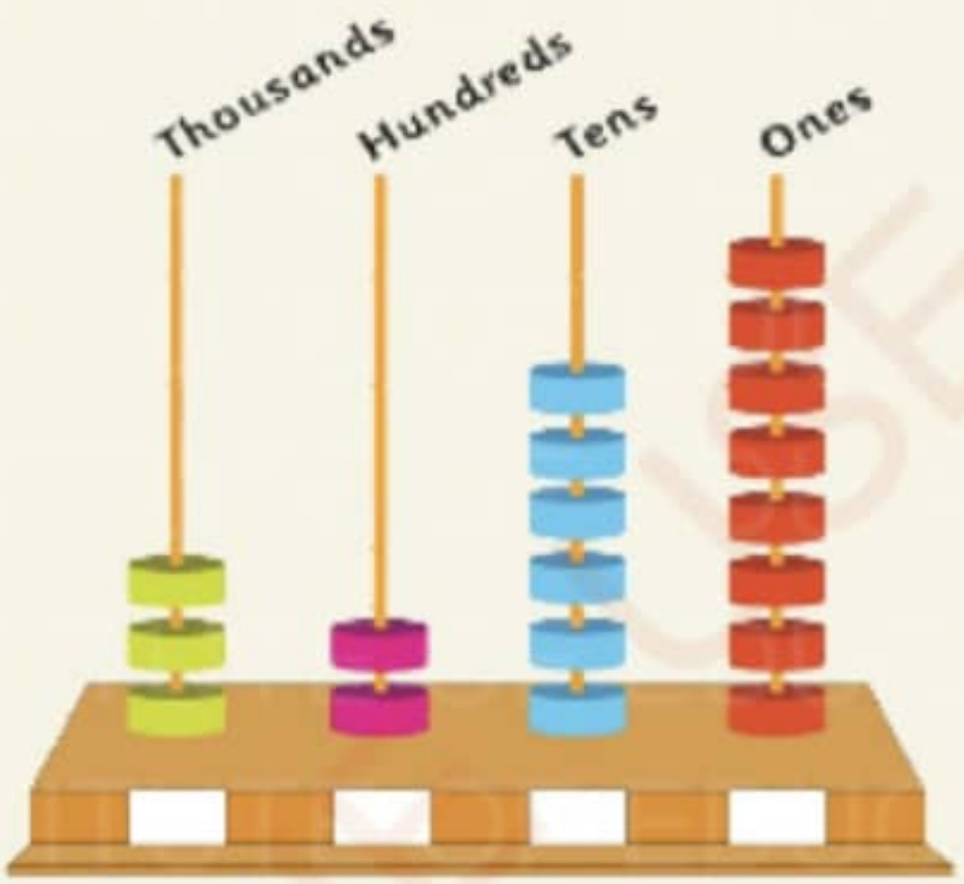


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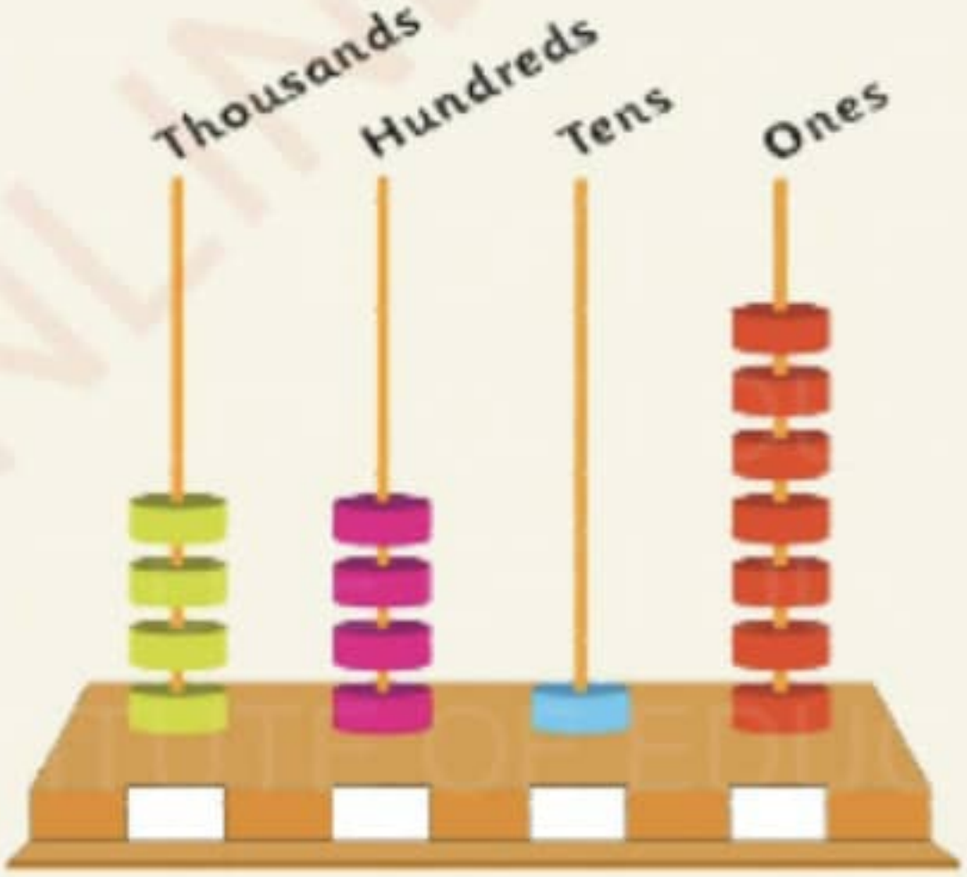
4.



5.



6.



7. Represent the following numbers on the abacus:

(a) 4279 (b) 5802 (c) 6247 (d) 5108

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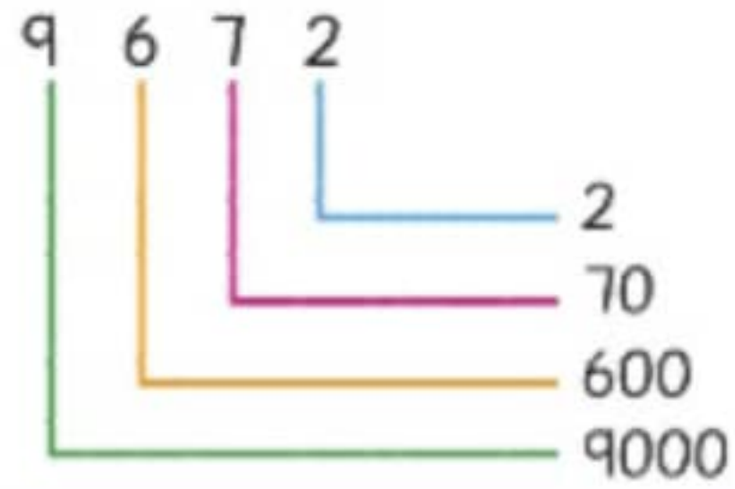
Total value of a digit in a whole number

The total value of a digit in a number increases from right to left. It is obtained by considering the place value of a digit.

Example 1

Write the total value of each digit in 9672.

Answer



Example 2

Write the total value of each underlined digit in 8794.

Answer

The total value of 9 is 90
The total value of 8 is 8000

Example 3

Given the number 7985, write the following:
(a) The place value of each digit
(b) The total value of each digit

Answer

Number	Digit	(a) Place value	(b) Total value
7985	7	Thousands	7000
	9	Hundreds	900
	8	Tens	80
	5	Ones	5

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Example 4

Write the place value of 6 and total value of 4 in 4612.

Answer

The place value of 6 in 4612 is hundreds.
The total value of 4 in 4612 is 4000.

Exercise 3

1. *Write the total value of the underlined digits in the following numbers:*
   (a) 4$\underline{7}$65 (b) $\underline{3}$926 (c) 186$\underline{7}$ (d) 45$\underline{1}$9
2. *Write the total value of each digit in the following numbers:*
   (a) 2345 (b) 1246 (c) 8769 (d) 9876
3. *Write the total value of 5 in the following numbers:*
   (a) 5672 (b) 3564 (c) 9851 (d) 1695
4. *Given the number 9864, write the following:*
   (a) *The place value of each digit*
   (b) *The total value of each digit.*

Expressing numbers in expanded form

A number is in expanded form when it is written as the sum of the total values of its digits. This form identifies the total value of each digit in a number.

Example 1

Write each of the following numbers in expanded form:

(a) 9732 (b) 5080
(c) 4000 (d) 3334

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Answer

(a) 9732 = 9000 + 700 + 30 + 2

(b) 5080 = 5000 + 0 + 80 + 0

(c) 4000 = 4000 + 0 + 0 + 0

(d) 3334 = 3000 + 300 + 30 + 4

Example 2

Write each of the following numbers in short form:

(a) 4000 + 500 + 30 + 1 =

(b) 4000 + 800 + 40 + 2 =

(c) 9000 + 700 + 0 + 2 =

(d) 5000 + 0 + 0 + 1 =

Answer

(a) 4000 + 500 + 30 + 1 = 4531

(b) 4000 + 800 + 40 + 2 = 4842

(c) 9000 + 700 + 0 + 2 = 9702

(d) 5000 + 0 + 0 + 1 = 5001

Exercise 4

1. *Write each of the following numbers in short form:*

(a) 9000 + 800 + 70 + 2 =

(b) 8000 + 800 + 80 + 0 =

(c) 1000 + 0 + 0 + 6 =

(d) 5000 + 600 + 50 + 8 =

(e) 9000 + 700 + 0 + 2 =

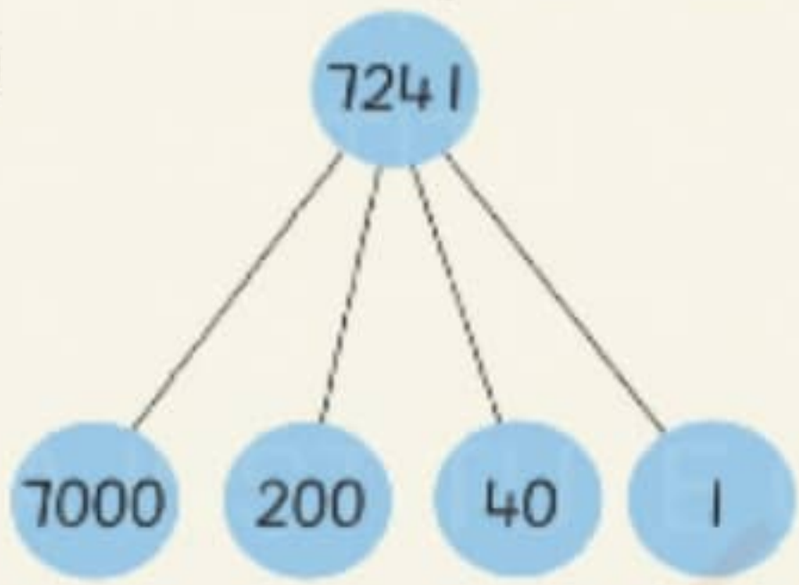
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2. Complete the following table. Take the first row as an example.

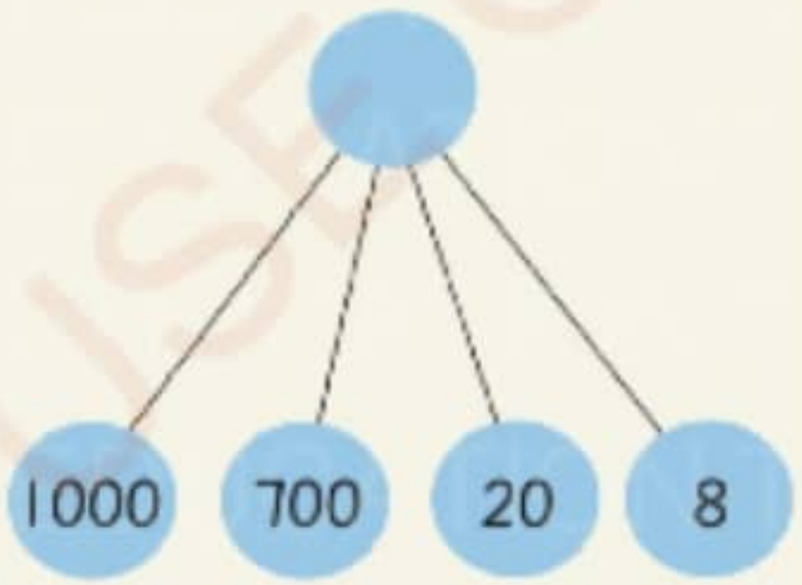
8543	=	8000	+	500	+	40	+	3
3891	=		+		+		+	
1654	=		+		+		+	
4200	=		+		+		+	
2901	=		+		+		+	

3. In the diagrams (a) to (f), one circle has four branches of circles. The sum of numbers in the branches gives the number in short form. Taking 3(a) as an example, write down the missing numbers.

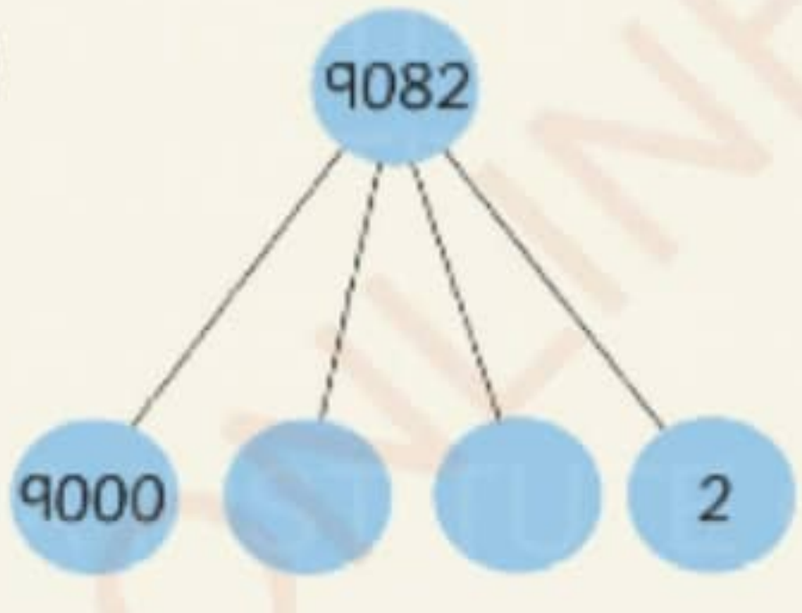
(a)



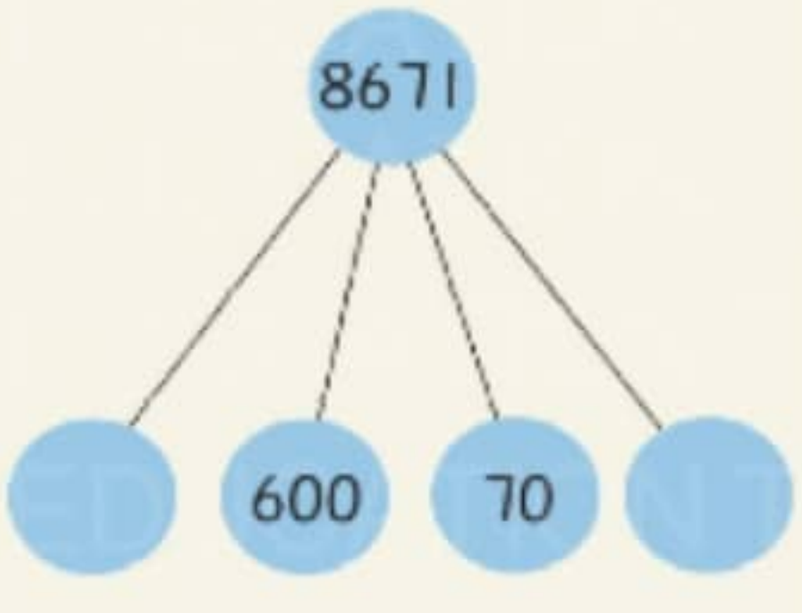
(b)



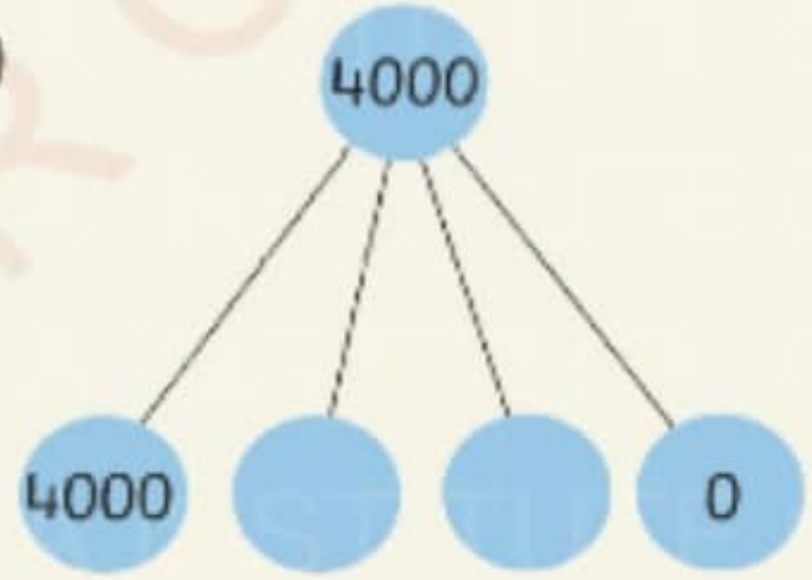
(c)



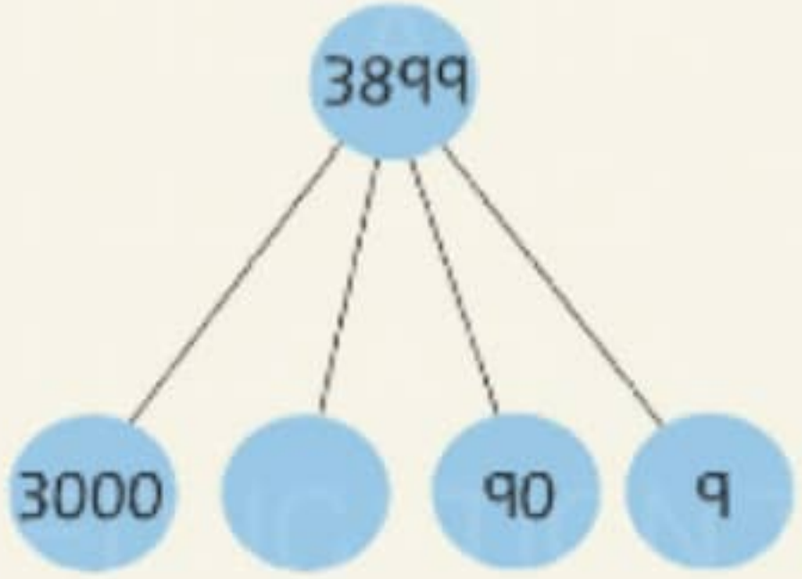
(d)



(e)



(f)



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4. Write each of the following numbers in expanded form:

(a) 7889 (b) 2043 (c) 8075

(d) 9690 (e) 3245 (f) 9770

(g) 3891 (h) 1654

Reading and writing whole numbers

Reading and writing whole numbers involves interpreting numerical symbols and expressing quantities in written form. In this section, you will learn the techniques for reading and writing whole numbers.

Reading whole numbers

Reading whole numbers depends on the number of digits and the total value of the digits in the corresponding number. Start identifying the ones, tens, hundreds, thousands, and so on. Thereafter, read the number from left to right by considering the total value of the corresponding digit. That is, read a number beginning with a digit that has the highest place value. Numbers are read based on the value of the digits from left to right.

Example 1

Read the following numbers:

(a) 1001 (b) 1086 (c) 1100

Answer

(a) 1001 is read as one thousand and one.

(b) 1086 is read as one thousand and eighty-six.

(c) 1100 is read as one thousand and one hundred.

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Exercise 5

Read the following numbers:

1001	1002	1003	1004	1005	1006	1007	1008	1009	1010
1011	1012	1013	1014	1015	1016	1017	1018	1019	1020
1021	1022	1023	1024	1025	1026	1027	1028	1029	1030
1031	1032	1033	1034	1035	1036	1037	1038	1039	1040
1041	1042	1043	1044	1045	1046	1047	1048	1049	1050
1051	1052	1053	1054	1055	1056	1057	1058	1059	1060
1061	1062	1063	1064	1065	1066	1067	1068	1069	1070
1071	1072	1073	1074	1075	1076	1077	1078	1079	1080
1081	1082	1083	1084	1085	1086	1087	1088	1089	1090
1091	1092	1093	1094	1095	1096	1097	1098	1099	1100

Example 2

Read each of the following numbers:

(a) 1350 (b) 1572 (c) 9700

(d) 9540 (e) 9998

Answer

(a) 1350 is read as one thousand three hundred and fifty.

(b) 1572 is read as sixteen thousand five hundred and seventy-two.

(c) 9700 is read as nine thousand and seven hundred.

(d) 9540 is read as nine thousand five hundred and forty.

(e) 9998 is read as nine thousand nine hundred and ninety.

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Exercise 6

Read the numbers in the following table:

1100	1200	1300	1400	1500	1600	1700	1800	1900	2000
2100	2200	2300	2400	2500	2600	2700	2800	2900	3000
3100	3200	3300	3400	3500	3600	3700	3800	3900	4000
4100	4200	4300	4400	4500	4600	4700	4800	4900	5000
5100	5200	5300	5400	5500	5600	5700	5800	5900	6000
6100	6200	6300	6400	6500	6600	6700	6800	6900	7000
7100	7200	7300	7400	7500	7600	7700	7800	7900	8000
8100	8200	8300	8400	8500	8600	8700	8800	8900	9000
9100	9200	9300	9400	9500	9600	9700	9800	9900	10000

Exercise 7

1. Read the following numbers horizontally and vertically:

1110	1120	1130	1140	1150	1160	1170	1180	1190	1200
1210	1220	1230	1240	1250	1260	1270	1280	1290	1300
1310	1320	1330	1340	1350	1360	1370	1380	1390	1400
1410	1420	1430	1440	1450	1460	1470	1480	1490	1500
1510	1520	1530	1540	1550	1560	1570	1580	1590	1600
1610	1620	1630	1640	1650	1660	1670	1680	1690	1700
1710	1720	1730	1740	1750	1760	1770	1780	1790	1800
1810	1820	1830	1840	1850	1860	1870	1880	1890	1900
1910	1920	1930	1940	1950	1960	1970	1980	1990	2000

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2. Read numbers in the following chart:

1111	1112	1113	1114	1115	1116	1117	1118	1119	1120
1121	1122	1123	1124	1125	1126	1127	1128	1129	1130
1131	1132	1133	1134	1135	1136	1137	1138	1139	1140
1141	1142	1143	1144	1145	1146	1147	1148	1149	1150
1151	1152	1153	1154	1155	1156	1157	1158	1159	1160
1161	1162	1163	1164	1165	1166	1167	1168	1169	1170
1171	1172	1173	1174	1175	1176	1177	1178	1179	1180
1181	1182	1183	1184	1185	1186	1187	1188	1189	1190
1191	1192	1193	1194	1195	1196	1197	1198	1199	1200

3. Read the following numbers:

(a) 6546 (b) 7865 (c) 2755
(d) 9118 (e) 4006 (f) 8009
(g) 7016 (h) 9468

Writing whole numbers

A whole number is written from left to right by considering the place values of the digits. It can be written in numerals or in words. A number in numerals shows all the digits in it and their corresponding place values.

Example 1

Write the following numbers in words:

(a) 1000 (b) 1001 (c) 1002 (d) 1003
(e) 1004 (f) 1005 (g) 1006 (h) 1007
(i) 1008 (j) 1009 (k) 1010

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Answer

	Number in numerals	Number in words
(a)	1000	One thousand
(b)	1001	One thousand and one
(c)	1002	One thousand and two
(d)	1003	One thousand and three
(e)	1004	One thousand and four
(f)	1005	One thousand and five
(g)	1006	One thousand and six
(h)	1007	One thousand and seven
(i)	1008	One thousand and eight
(j)	1009	One thousand and nine
(k)	1010	One thousand and ten

Example 2

Write the following numbers in words:

(a) 1018 (b) 4216 (c) 5691

Answer

(a) One thousand and eighteen

(b) Four thousand two hundred and sixteen

(c) Five thousand six hundred and ninety-one

Example 3

Write each of the following numbers in numerals:

(a) Thirteen thousand one hundred and fifty-five

(b) Three hundred six hundred and eight

(c) Nine thousand four hundred and twelve

Answer

(a) 13155 (b) 3608 (c) 9412

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Exercise 8

1. Carefully study the following table and fill in the blanks. The first row is provided as an example.

Thousands	Hundreds	Tens	Ones	Number in numerals	Number in words
1	0	0	1	1001	One thousand and one
				1019	
				7070	
					One thousand three hundred and fifty
				1572	
				9540	
9	9	9	8		
				1086	
1	1	0	0		
				5102	Five thousand one hundred and two

2. Write each of the following numbers in words:

(a) 1250 (b) 1393 (c) 7288
(d) 3425 (e) 4537 (f) 7998
(g) 9663 (h) 8888 (i) 9999
(j) 6660 (k) 3618 (l) 2222
(m) 9768 (n) 7808 (o) 3030

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3. Write each of the following numbers in numerals:
   (a) One thousand and ten
   (b) Two thousand one hundred and sixty
   (c) Three thousand six hundred and eighteen
   (d) Four thousand seven hundred and sixty-three
   (e) Five thousand six hundred and eighty-three
   (f) Seven thousand
   (g) Nine thousand two hundred and thirty-five
   (h) Six thousand one hundred and twenty-two
   (i) Nine thousand nine hundred and seventy-three
   (j) Eight thousand
   (k) Eight thousand nine hundred and ninety-two
   (l) Three thousand five hundred and thirty-seven

4. Fill in the blanks in the following table.

Number in numerals	**Number in words**
	One thousand and fifty-two
1201	
	Two thousand
3000	
	Four thousand and three hundred
4506	
	Five thousand five hundred and fifty
6000	
	Six thousand and six hundred
7001	
	Eight thousand and eighty
9010	
	Nine thousand seven hundred

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5. (a) Put a tick (✓) if a statement is correct, and a cross (✗) if it is incorrect. The first row is provided as an example.

Number in numerals	Number in words	Symbol
6000	Six thousand	✓
3618	Three thousand six hundred and eighteen	
7288	Seven thousand eight hundred and twenty-eight	
3435	Three thousand four hundred and fifty-three	
5633	Five thousand six hundred and thirty-three	
9009	Nine hundred and nine thousand	
6660	Six thousand sixty and six hundred	
2086	Two thousand and eighty-six	
3239	Three thousand twenty-three hundred and nine	
2776	Two thousand seven hundred and seventy-six	
7304	Seven thousand and four	

(b) Write correctly all the incorrect numbers in words from question 5(a).

6. Study figure 1 and 2, read their captions, and then answer the questions that follow:

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Figure 1: *The Old Fort in Zanzibar was built in the year one thousand six hundred and ninety-nine*



Figure 2: *Kilimanjaro mountain is five thousand eight hundred and ninenty-five metres high*

(a) Write in numerals the numbers you have read from each caption.

(b) Write the place values of each digit of the numbers you have read.

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Reading and writing whole numbers exceeding 10000

In order to easily read or write whole numbers exceeding 10000, identify the place value of each digit of a given whole number.

Example 1

Write the place value of each digit in 61270, then write the number in words.

Answer

The place values of the digits in 61270 are as follows:

0 is in ones

7 is in tens

2 is in hundreds

1 is in thousands

6 is in ten thousands

Therefore, 61270 is read as sixty-one thousand two hundred and seventy.

Example 2

Write the place value of each digit in 876543, then write the number in words.

Answer

3 is in ones

4 is in tens

5 is in hundreds

6 is in thousands

7 is in ten thousands

8 is in hundred thousands

Therefore, 876543 is read as eight hundred seventy-six thousand five hundred and forty-three.

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Example 3

Write the place value of each digit in 9261514, then write the number in words.

Solution

4 is in ones
1 is in tens
5 is in hundreds
1 is in thousands
6 is in ten thousands
2 is in hundred thousands
9 is in millions

Therefore, 9261514 is read as nine million two hundred sixty-one thousand five hundred and fourteen.

Example 4

Write the following numbers in expanded form:
(a) 24765 (b) 273453 (c) 2261514

Answer

(a) $24765 = 20000 + 4000 + 700 + 60 + 5$
(b) $273453 = 200000 + 70000 + 3000 + 400 + 50 + 3$
(c) $2261514 = 2000000 + 200000 + 60000 + 1000 + 500 + 10 + 4$

Example 5

Write the following whole numbers in numerals:
(a) Ten thousand three hundred and fifty-five
(b) One million nine hundred eighty thousand three hundred and ninety-five

Answer

(a) 10355 (b) 1980395

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Example 6

Write the total value of each digit in 9876542.

Answer

Number	Digits	Total value of the digit
9876542	9	9000000
	8	800000
	7	70000
	6	6000
	5	500
	4	40
	2	2

Exercise 9

1. Write each of the following numbers in words:

   (a) 28749 (b) 38951 (c) 76187

   (d) 97000 (e) 311008 (f) 7896246

   (g) 89365 (h) 50050 (i) 821834

   (j) 31425 (k) 200512 (l) 3437000

2. Write each of the following numbers in numerals:

   (a) Thirty-five thousand six hundred and eleven

   (b) Seventy thousand

   (c) Ninety thousand nine hundred and eight

   (d) Eighty thousand two hundred and seven

   (e) One hundred twenty-seven thousand and nine hundred

   (f) Eight hundred thousand and eight

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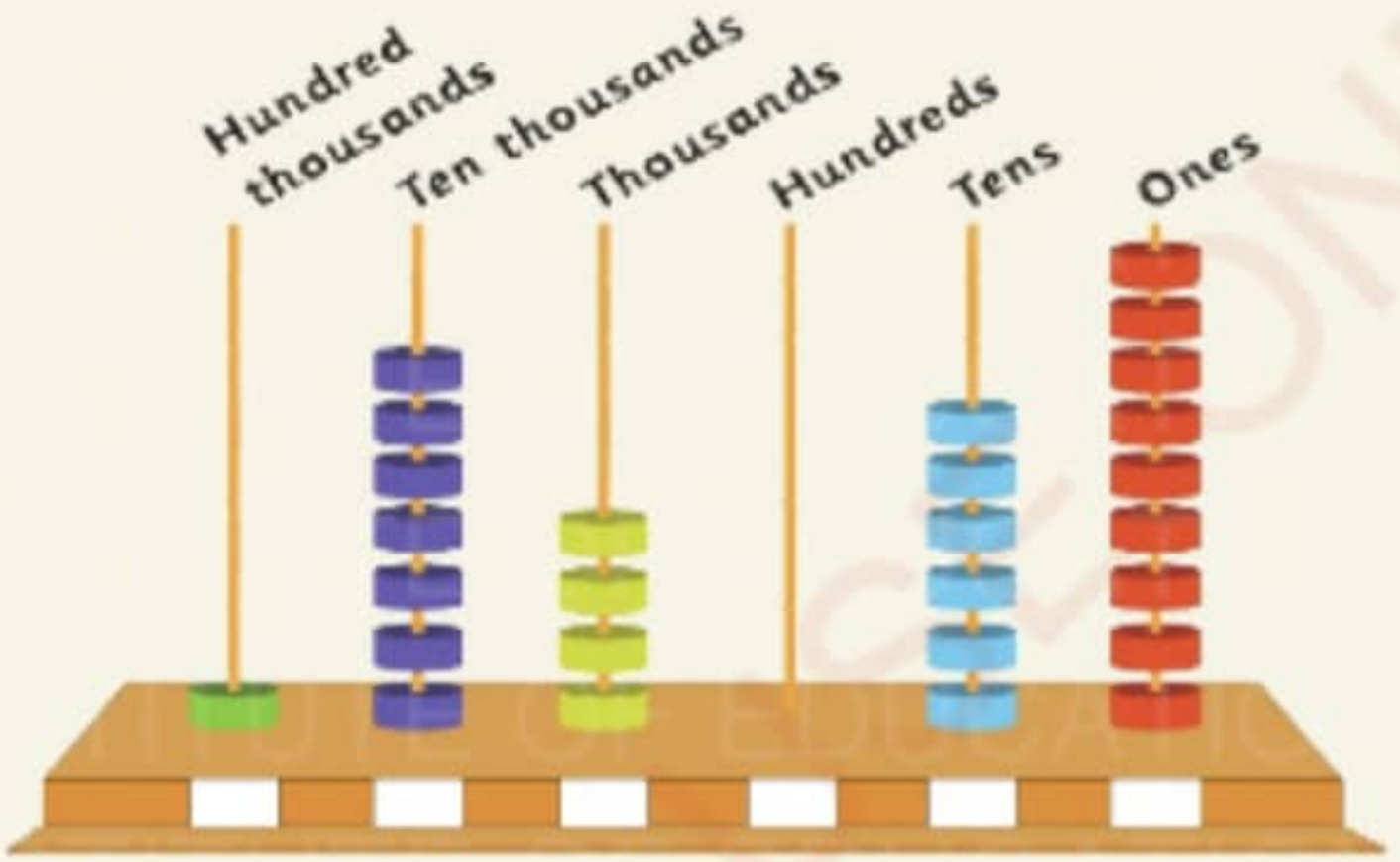
(g) Eight million seven hundred ninety thousand seven hundred and seventy-nine

(h) One million three hundred and forty

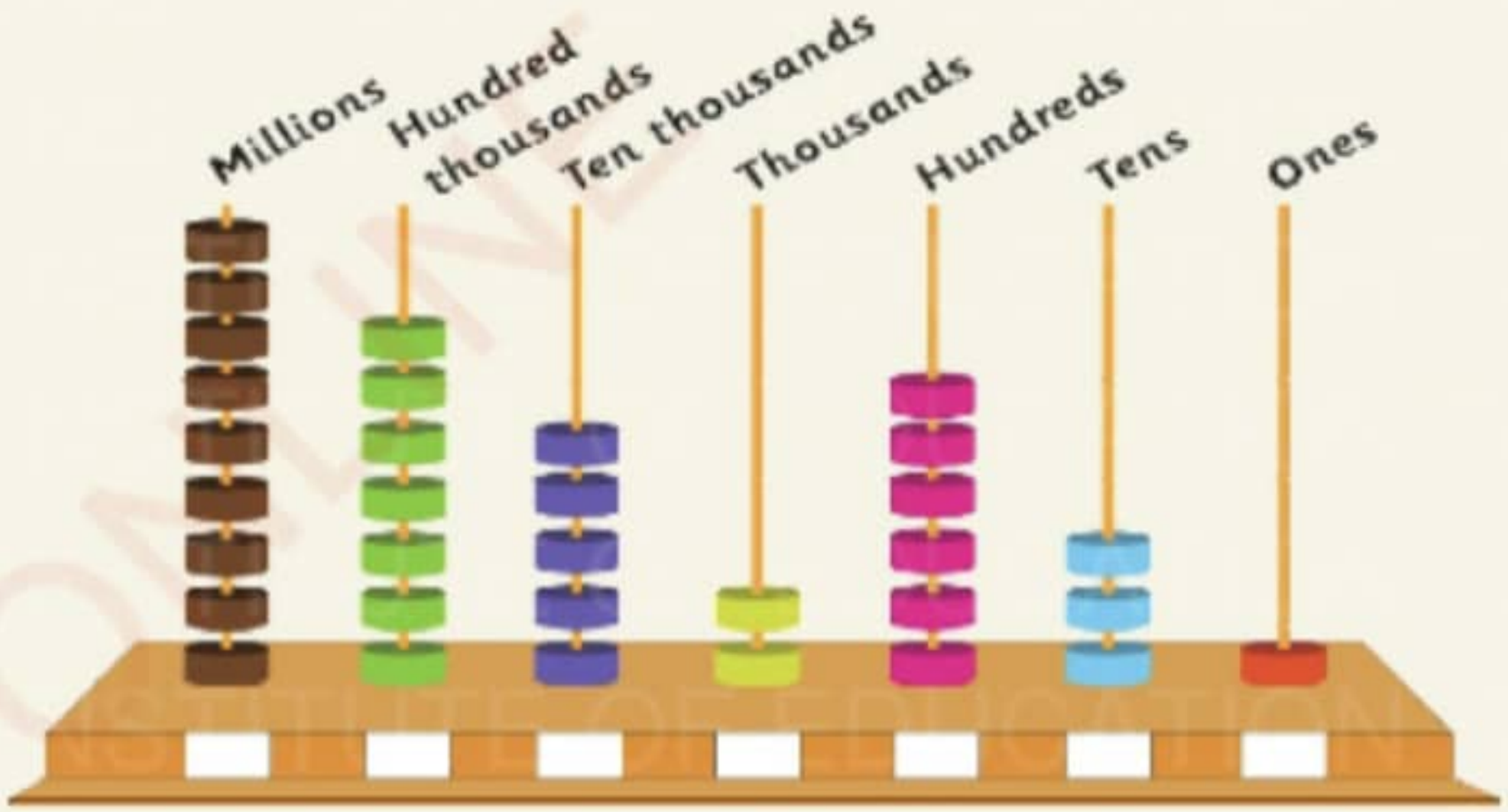
(i) Five thousand nine hundred ninety-nine thousand four hundred and fifty-six

3. Write the number represented by each of the following abaci:

(a)



(b)



4. Write the place value of 6 in each of the following whole numbers:

(a) 645218 (b) 446128 (c) 21610

(d) 699222 (e) 364974 (f) 239364

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5. Write the total value of the underlined digits in each of the following numbers:

(a) 4<u>9</u>002 (b) 6<u>2</u>49356 (c) 54<u>1</u>341

(d) <u>4</u>755964 (e) <u>7</u>215549 (f) <u>3</u>6314

6. Write each of the following numbers in expanded form:

(a) 79684 (b) 645218 (c) 8321483

7. Write each of the following numbers in short form:

(a) 60000 + 5000 + 400 + 30 + 2

(b) 800000 + 70000 + 6000 + 0 + 20 + 9

(c) 7000000 + 600000 + 40000 + 5000 + 400 + 30 + 7

8. Write the place value of the underlined digit in each of the following numbers:

(a) 6<u>4</u>371 (b) 8<u>9</u>2161 (c) <u>2</u>464313 (d) <u>6</u>72314

Comparing whole numbers

Whole numbers can be compared when they are equal, greater than, or less than another. When we have two whole numbers or quantities to compare, we use three basic symbols, which are "equality symbol" (=), "greater than" (>), and "less than" (<).

Example

Use the symbols <, = and > to compare the following numbers:

(a) 8900 _____ 8090 (b) 3220 _____ 3220

(c) 3240 _____ 3420 (d) 715 _____ 517

Answer

(a) 8900 > 8090 (b) 3220 = 3220

(c) 3240 < 3420 (d) 715 > 517

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Exercise 10

1. Use the symbols <, = and > to compare the numbers in the following table. (a) is provided as an example.

(a)	12	<	21	(b)	15		20	(c)	14		7
(d)	20		10	(e)	18		13	(f)	19		91

2. Use the symbols <, = and > to compare the given numbers:

(a) 2968 _____ 8962 (b) 972 _____ 972

(c) 3932 _____ 3239 (d) 999 _____ 1000

(e) 8671 _____ 8761 (f) 5678 _____ 5678

(g) 9002 _____ 902 (h) 1000 _____ 100

3. In the Mathematics test, Tony scored 91, John scored 78, and Emma scored 85. Use the less than sign (<) to compare Tony's and John's scores, then John's and Emma's scores.

4. The following chart shows different fruits and their quantities.

Type of fruits	Oranges	Pineapples	Bananas
Number of fruits	6851	1685	8516

Use the symbols <, > or = to compare the number of fruits as follows:

(a) Oranges (_____) bananas

(b) Oranges (_____) pineapples

(c) Bananas (_____) oranges (_____) pineapples

5. Baraka has a card written nine thousand five hundred and forty. Furaha has a card written 9450. Between the two, who has the card with a smaller number?

6. Halima has a card numbered 2010 and Yusufu has a card numbered 1577. Between the two, who has the card with a larger number?

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7. Carefully study the following pictures and compare the number of the objects:



Box A



Box B

(a) How many pencils are in box A?
(b) How many pencils are in box B?
(c) Which box has more pencils than the other?
(d) Use the symbols =, < or > to compare the number of pencils.

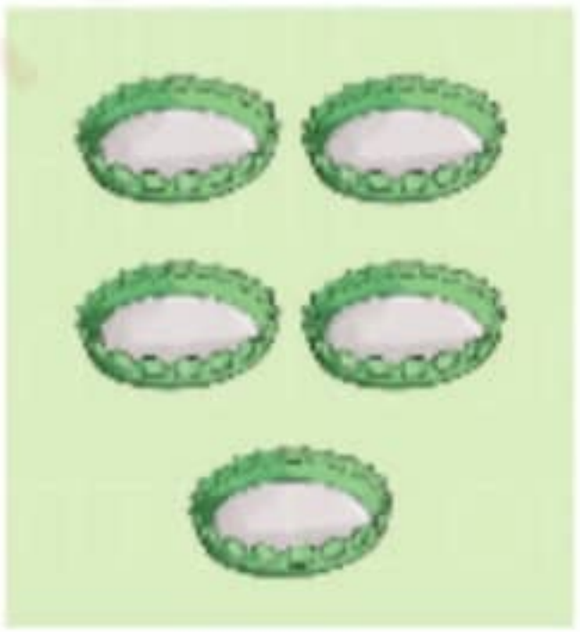
Sequences of whole numbers

Numbers can be arranged in a defined pattern. When whole numbers are arranged in a defined pattern, they form a sequence. It is possible to determine the next numbers in a given sequence.

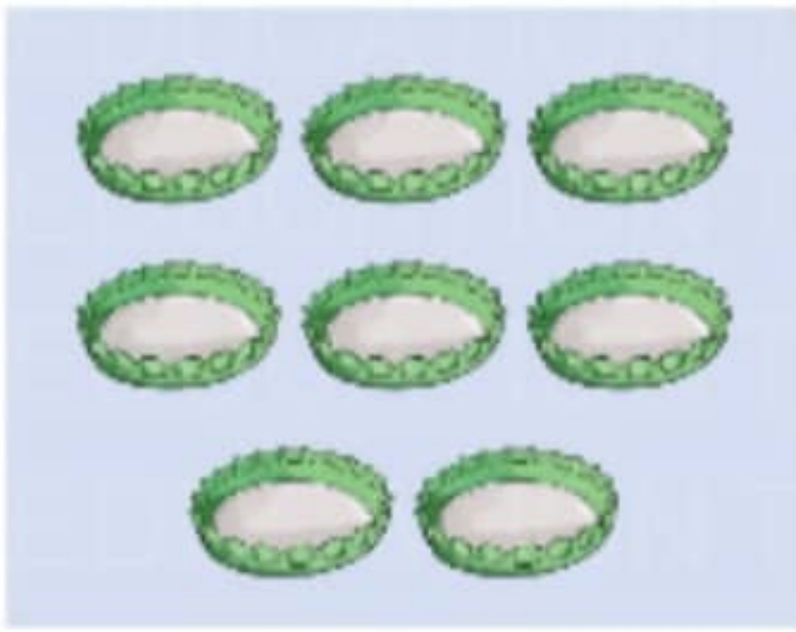
Carefully study the number of bottle tops in group A, B and C, then answer the questions that follow.



Group A



Group B



Group C

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Group A has 2 bottle tops. Group B has 5 bottle tops, Group C has 8 bottle tops. The number of bottle tops in the groups can be listed as 2, 5, 8. There is an increase of 3 bottle tops from Group A to Group B. Similarly, there is an increase of 3 bottle tops from Group B to Group C. This means that 2, 5, 8, is a sequence of numbers. If you form another group of bottle tops the group will have 11 bottle tops. That is $8 + 3 = 11$. In this case, the sequence will be 2, 5, 8, 11.

Example 1

Write the missing numbers in the following sequence:
1, 4, 7, 10, 13, 16, 19, __, __

Solution

In this sequence, the numbers are increasing by 3 in each step:

Thus,

$1 + 3 = 4$
$4 + 3 = 7$
$7 + 3 = 10$
$10 + 3 = 13$

$13 + 3 = 16$
$16 + 3 = 19$
$19 + 3 = 22$
$22 + 3 = 25$

Therefore, the answer is 1, 4, 7, 10, 13, 16, 19, 22, 25.

Example 2

Fill in the missing numbers in the following sequence:
1, 3, 5, 7, 9, __, ____.

Solution

This sequence shows that the numbers are increasing by 2 in each step.

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Thus,

1 + 2 = 3
3 + 2 = 5
5 + 2 = 7

7 + 2 = 9
9 + 2 = 11
11 + 2 = 13

Therefore, the answer is 1, 3, 5, 7, 9, 11, 13.

Example 3

Fill in the missing numbers in the following sequence:
25, 23, 21, ___, ___

Solution

This sequence shows that the numbers decrease by 2 in each step.

Thus,

25 − 2 = 23
23 − 2 = 21
21 − 2 = 19
19 − 2 = 17

Therefore, the answer is 25, 23, 21, 19, 17.

Example 4

Fill in the missing number in the following sequence:
750, 600, 450, 300, 150, ____

Solution

In this sequence, the numbers are decreasing by 150 in each step.

Thus,

750 − 150 = 600
600 − 150 = 450
450 − 150 = 300

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300 – 150 = 150
150 – 150 = 0

Therefore, the answer is 750, 600, 450, 300, 150, 0.

Example 5

Fill in the missing number in the following sequence:
45, 41, 37, 33, _____

Solution

In this sequence, the numbers are decreasing by 4 in each step.
Thus,

45 – 4 = 41
41 – 4 = 37
37 – 4 = 33
33 – 4 = 29

Therefore, the answer is 45, 41, 37, 33, 29.

Exercise 11

1. *Write the numbers given in the following table in the ascending order:*

22	7	15	14	10
11	28	18	20	24

7, _____, _____, _____, _____, _____, _____, _____, _____, _____, _____

2. *Write the numbers given in the following table in the descending order:*

29	23	37	41	35
16	20	26	44	32

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44, ____, ____, ____, ____, ____, ____, ____, ____, ____, ____

3. Write the missing numbers in each of the following sequences:

(a) 0, 2, 4, 6, ____, ____, ____

(b) 11, 22, 33, 44, ____, ____, ____

(c) 2, 6, 10, 14, 18, ____, ____, ____

(d) 3, 8, 13, 18, 23, 28, 33, ____, ____, ____

(e) 465, 466, 467, ____, ____, ____

(f) 210, 220, 230, 240, 250, 260, ____, ____, ____

4. Write the next numbers in each of the following sequences:

(a) 14, 12, 10, 8, ____, ____, ____

(b) 23, 20, 17, 14, 11, ____, ____

(c) 56, 54, 52, ____, ____, ____

(d) 69, 60, 51, 42, 33, ____, ____, __

(e) 95, 85, 75, 65, 55, 45, ____, ____, ____

(f) 260, 250, 240, 230, 220, 210, ____, ____, ____

5. Write the missing numbers in each of the following sequences:

(a) 23, 30, ____, ____, 51, ____

(b) 433, 430, ____, 424, ____, 418, 415, ____

(c) 1027, 1023, ____, 1015, ____, 1007, ____

(d) 4, 10, ____, 22, ____, 34, 40, 46, ____, 58

(e) 15, 18, ____, 24, ____, 30, 33, ____

(f) 1011, 1014, ____, 1020, 1023, ____, 1029

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6. Carefully study the following sequences, then for each sequence state the operation used. Take (a) as an example.

	Sequence of numbers	**Operation used**
(a)	1, 3, 5, 7, 9, 11, 13, 15, 17, 19	The numbers are increased by 2 in each step
(b)	30, 25, 20, 15, 10, 5	
(c)	10, 20, 30, 40, 50, 60, 70	
(d)	500, 600, 700, 800	

7. The heights of 4 mountains were recorded as follows: Shira is 3,626 metres, Oldeani 3,188 metres, Meru 4,566 metres and Klute 3,952 metres high. Arrange the heights of the mountains from the lowest to the highest mountain.

8. Majura sold 15 cups of milk on the first day. He sold 8 more cups on each subsequent day. How many cups of milk did he sell on the fourth day?

9. The factory printed 50 books on the first day. It printed 20 more books on each subsequent day. How many books were printed on the third day?

10. Pamela bought 3 mangoes on the first day. She bought 2 more mangoes on each subsequent day. How many mangoes did she buy on the fourth day?

11. Mkeya has 5 children. The difference in ages between two successive children is 4 years. If the first born is 39 years old, how old is the lastborn?

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Activity

Use online programmes to learn more on whole numbers.

Descriptions

Explore how to do the following:

(a) Read and write whole numbers in numerals and words
(b) Find the place values and total values of whole numbers
(c) Write whole numbers in expanded form
(d) Formulate sequence of whole numbers
(e) Compare the values of whole numbers

Summary

1. Numbers can be written in two different ways, that is, in numerals and in words.
2. When reading numbers, the place value and total value of a digit must be taken into consideration.
3. The equality symbol (=), greater than (>), and smaller than (<) are used to compare the relationship of the values of different objects or numbers.
4. A sequence of numbers always follows a certain pattern.
5. It is important to find out the operation involved in a given sequence in order to get a correct pattern.

Vocabulary

Inequality symbols	These are the symbols "smaller than" (<) "greater than" (>)
Sequence	Numbers arranged in a defined pattern

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Numeral	This is a number written by using digits.
Digits	These are the first ten numbers, that is, 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.

Revision exercise

1. Write the number obtained from the following place values:

   (a) Seven thousands, six hundreds, two tens, and two ones

   (b) Seven ten thousands, zero thousands, six hundreds, eight tens and three ones

   (c) Two ten thousands, three thousands, four hundreds, six tens and six ones

   (d) Four hundred thousands, two ten thousands, seven thousands, three hundreds, one tens and four ones

   (e) One millions, seven hundred thousands, four ten thousands, one thousands, one hundreds, two tens, and zero ones

   (f) Three millions, seven hundred thousands, six ten thousands, five thousands, three hundreds, one tens and three ones

2. Write each of the following numbers in short form.

   (a) 4000 + 200 + 50 + 1 =

   (b) 9000 + 800 + 30 + 3 =

   (c) 30000 + 3000 + 300 + 30 + 3 =

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(d) 600000 + 40000 + 4000 + 300 + 0 + 2 =

(e) 2000000 + 600000 + 40000 + 0 + 700 + 90 + 9 =

3. Use the following number to fill in the blanks with the place value of each digit.

(a) 2063 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(b) 8544 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(c) 3090 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(d) 5713 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

(e) 7226 has _____ thousands, _____ hundreds, _____ tens and _____ ones.

4. Write each of the following numbers in numerals:

(a) One thousand and seven

(b) Five thousand five hundred and three

(c) Six thousand six hundred and ninety-nine

(d) One hundred eighty thousand five hundred and forty-one

(e) Two million six hundred ninety-eight thousand six hundred and fifty-two

5. Write each of the following numbers in words:

(a) 2022 (b) 4969 (c) 8008

(d) 291649 (e) 1889773

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6. Write each of the following numbers in expanded form:

   (a) 7007 (b) 5712 (c) 9669

   (d) 64645 (e) 303067

7. Write each of the following numbers in the ascending order:

   (a) 5, 2, 6, 12, 19, 3, 1, 20,17

   (b) 36, 21, 15, 7, 10, 28, 45, 11

   (c) 789, 63, 32, 981, 252, 172, 426

8. Write each of the following numbers in the descending order:

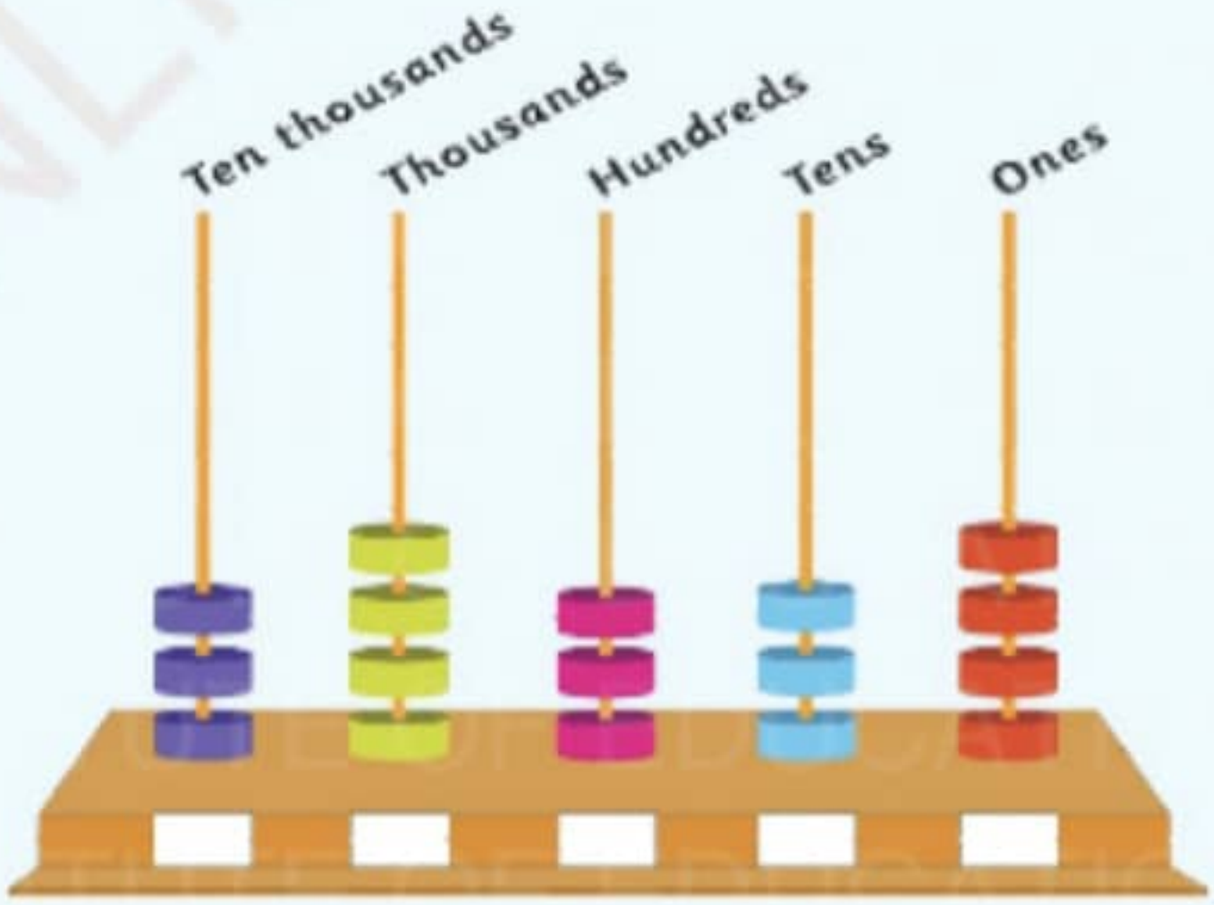
   (a) 125, 78, 111, 89, 305, 32, 120, 360, 115, 92, 412, 76, 61, 27, 56

   (b) 200, 903, 730, 341, 81, 199, 189, 800, 463, 72, 326, 247

   (c) 2500, 1200, 5000, 3300 4500, 1300, 6000, 4600

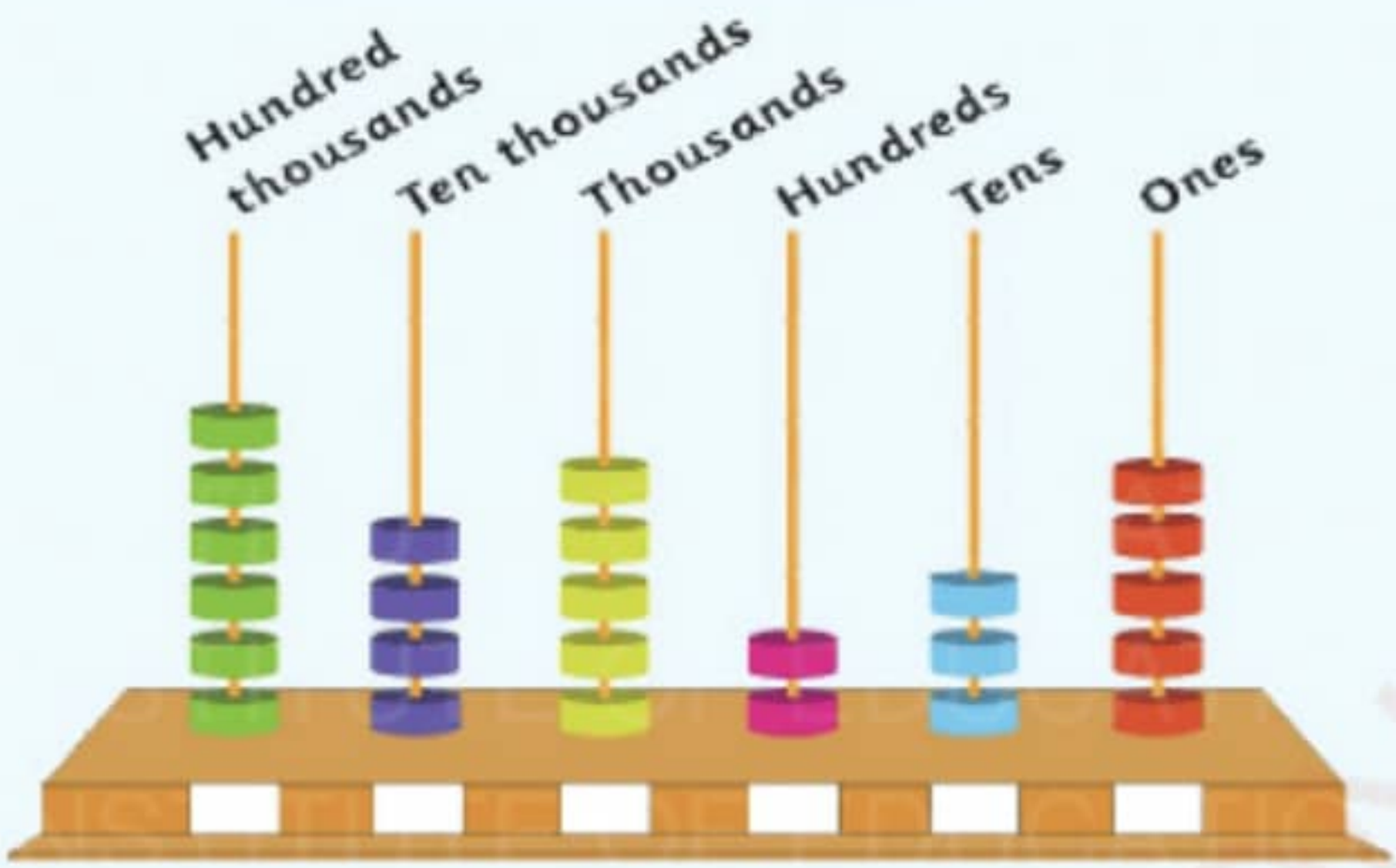
9. Write the numbers represented by each of the following abaci:

   (a)

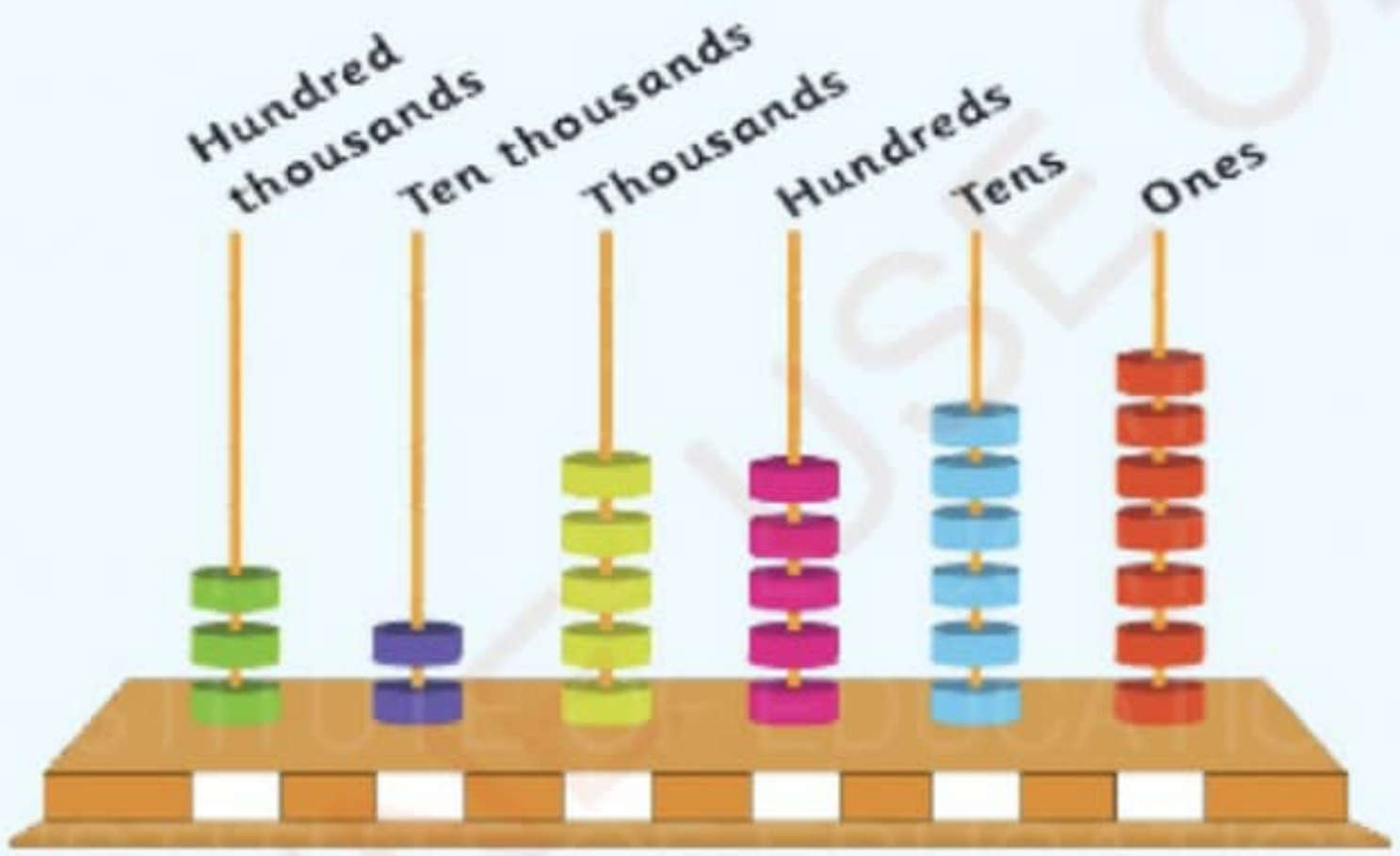


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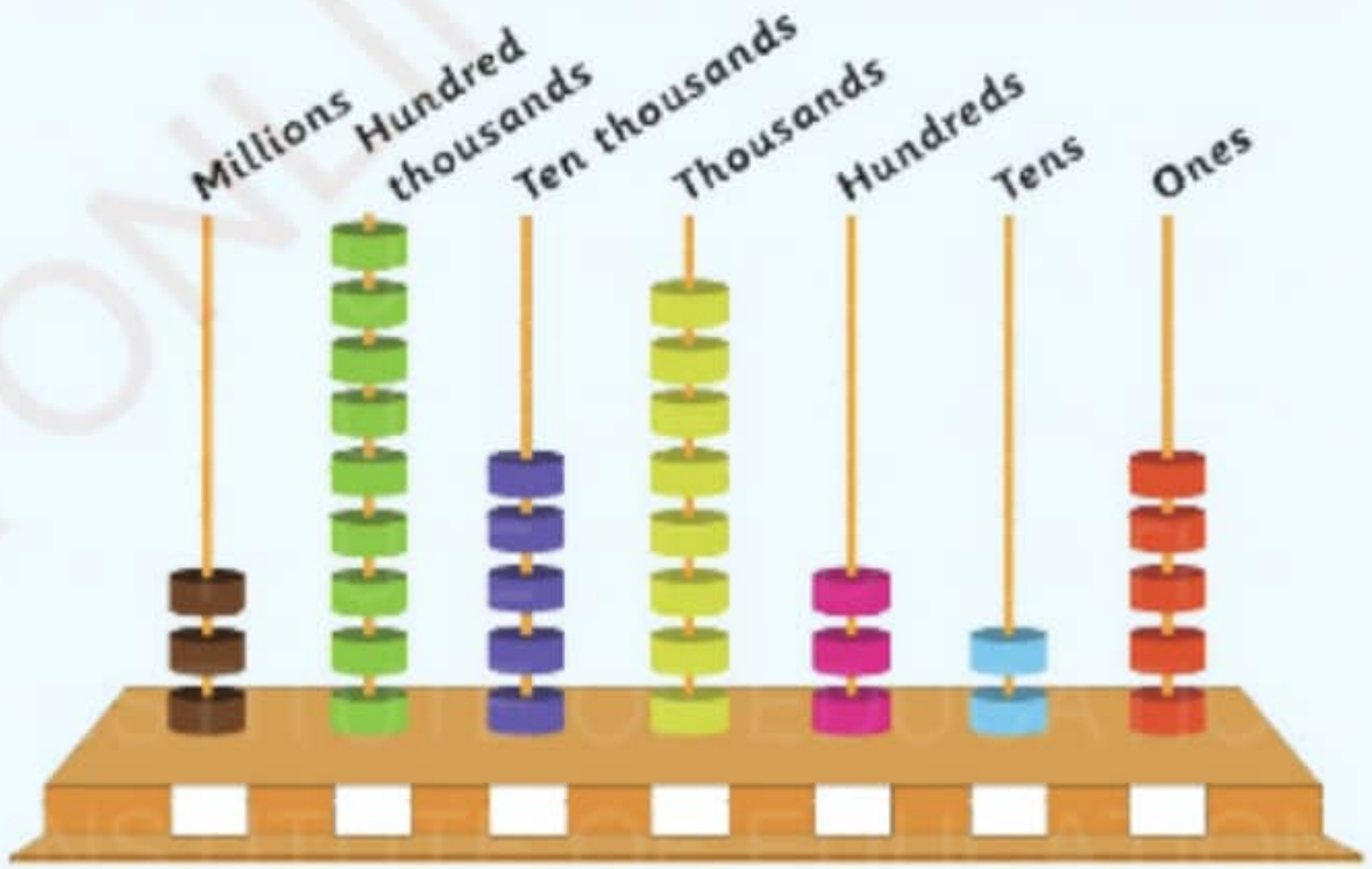
(b)



(c)



(d)



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10. Use the symbols <, = and > to compare the following numbers:

    (a) 8765 and 8576
    (b) 2341 and 2341
    (c) 2222 and 222
    (d) 180541 and 180514
    (e) 1234 and 2134
    (f) 23466 and 24366

11. Fill in the missing numbers in each of the following sequences:

    (a) 71, 66, 61, 56, 51, 46, ____, ____, 31.
    (b) 100, 75, ____, 25, ____.
    (c) 100000, ____, 300000, 400000, ____.
    (d) 729600, 729590, ____, ____, ____, 729550.
    (e) 70, 61, 52, 43, 34, ____, ____, ____.

12. Write the operation used in each of the following sequences. Question (a) is provided as an example.

    (a) 1, 2, 3, 4, 5, 6, 7, 8 <u>one is added in each step</u>
    (b) 0, 4, 8, 12, 16, 20, 24, 28 _____
    (c) 68, 60, 52, 44, 36, 28, 20 _____
    (d) 100, 90, 80, 70, 60, 50, 40, 30, 20, 10, 0 _____
    (e) 1444, 2444, 3444, 4444, 5444 _____

13. Find in the missing numbers in each of the following sequences:

    (a) 2398, 2390, 2382, ____, ____, 2358, 2350, ____.
    (b) 52552, 51552, ____, ____, 48552, 47552, ____, 45552.
    (c) ____, ____, ____, 840000, ____, 860000, 870000.

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(d) 2342, 2350, 2358, _____, _____, 2382, 2390, 2398, _____.

(e) 45562, 46562, _____, _____, 49562, 50562, _____, 52562.

14. A teacher arranged numbers in a sequence as follows: 45, 41, 37, 33, _____, _____. Find the next two numbers.

15. The first number in a sequence is 50. If 3 is subtracted in each step, what is the value of the fifth number?

16. Pupils wrote a sequence starting with 3. If they added 5 in each step, what was the value of the sixth number?

17. Write a sequence of numbers from 0 to 50 with a step of 5.

18. Write a sequence with an interval of 6 in each step from 72 to 0.

19. The attendance of standard three pupils increases by three students each week. If there were 123 pupils in the first week, how many pupils will attend in the fourth week?

20. Write 895674 by using:

(a) The place value of each digit

(b) The total value of each digit

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Chapter Two

Addition and subtraction of whole numbers

Introduction

In this chapter, you will learn about addition and subtraction of whole numbers. You will add and subtract whole numbers aligned horizontally and vertically. You will also learn how to solve word problems involving addition and subtraction of whole numbers.The competencies developed will enable you to find sums or differences of various objects.

Addition of whole numbers

Whole numbers can be added without regrouping or by regrouping, depending on the nature of the numbers being added.

Addition of whole numbers without regrouping

When adding numbers, the place values of digits in the number shoud be considered. Start by adding ones, tens and so on.

Adding whole numbers horizontally

Example 1

3246 + 2312 =

Solution

Align the numbers according to the place value of each digit. This is done as follows:

Thousands	Hundreds	Tens	Ones
3	2	4	6

+

Thousands	Hundreds	Tens	Ones
2	3	1	2

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Add the digits from right to left.

Steps

1. Add the ones and write the answer in the ones place.
2. Add the tens and write the answer in the tens place.
3. Add the hundreds and write the answer in the hundreds place.
4. Add the thousands and write the answer in the thousands place.



Therefore, 3246 + 2312 = 5558.

Example 2

68942 + 30051 =

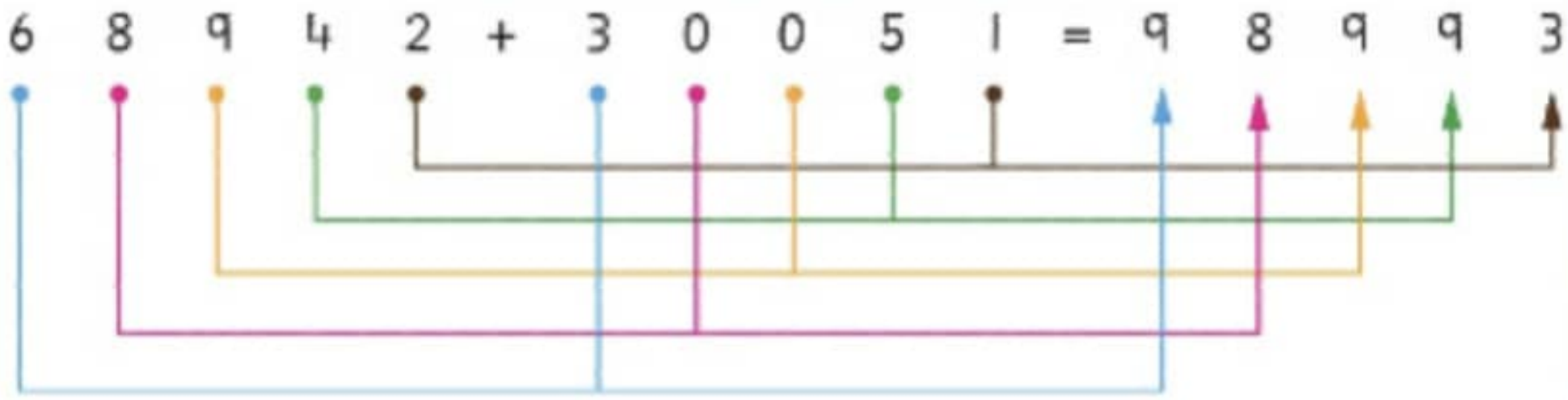
Solution

Add the digits from right to left.

Steps

1. Add the ones: 2 + 1 = 3. Write 3 in the ones place.
2. Add the tens: 4 + 5 = 9. Write 9 in the tens place.
3. Add the hundreds: 9 + 0 = 9. Write 9 in the hundreds place.
4. Add the thousands: 8 + 0 = 8. Write 8 in the thousands place.
5. Add the ten thousands: 6 + 3 = 9. Write 9 in the ten thousands place.

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Therefore, 68942 + 30051 = 98993.

Example 3

352411 + 136587 =

Steps

1. Add the ones and write the answer in the ones place.
2. Add the tens and write the answer in the tens place.
3. Add the hundreds and write the answer in the hundreds place.
4. Add the thousands and write the answer in the thousands place.
5. Add the ten thousands and write the answer in the ten thousands place.
6. Add hundred thousands and write the answer in the hundred thousands place.



Therefore, 352411 + 136587 = 488998.

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Exercise 1

1. 4561 + 17 =
2. 9111 + 888 =
3. 705 + 1261 =
4. 2016 + 2161 =
5. 54622 + 44245 =
6. 53999 + 2000 =
7. 2141 + 42 =
8. 2411 + 136 =
9. 1418 + 2361=
10. 6138 + 1=
11. 631594 + 246305 =
12. 230000 + 144348 =
13. 80021 + 17734 =
14. 54622 + 44245 =
15. 54351 + 12045 =

Adding whole numbers vertically

Example 1

$$\begin{array}{r} 6118 \\ +\ 1211 \\ \hline \\ \hline \end{array}$$

Solution

Align the numbers vertically according to the place values. This can be done as follows:

Steps

1. Add the ones: 8 + 1 = 9. Write 9 in the ones place.
2. Add the tens: 1 + 1 = 2. Write 2 in the tens place.
3. Add the hundreds: 1 + 2 = 3. Write 3 in the hundreds place.
4. Add the thousands: 6 + 1 = 7. Write 7 in the thousands place.

Therefore, the answer is 7329.

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Example 2

$$\begin{array}{r} 45248 \\ +\ 20231 \\ \hline \\ \hline \end{array}$$

Steps

1. Add the ones: 8 + 1 = 9. Write 9 in the ones place.
2. Add the tens: 4 + 3 = 7. Write 7 in the tens place.
3. Add the hundreds: 2 + 2 = 4. Write 4 in the hundreds place.
4. Add the thousands: 5 + 0 = 5. Write 5 in the thousands place.
5. Add the ten thousands: 4 + 2 = 6. Write 6 in the thousands place.

Therefore, the answer is 65479.

Example 3

$$\begin{array}{r} 162792 \\ +\ 531203 \\ \hline \\ \hline \end{array}$$

Solution

Align the numbers vertically according to the place values as follows:

	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones
	1	6	2	7	9	2
+	5	3	1	2	0	3
	6	9	3	9	9	5

Therefore, the answer is 693995.

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Example 4

$$\begin{array}{r} 1222221 \\ +\ 777777 \\ \hline \\ \hline \end{array}$$

Solution

$$\begin{array}{r} 1222221 \\ +\ 777777 \\ \hline 1999998 \\ \hline \end{array}$$

Exercise 2

1. $\begin{array}{r} 4612 \\ +\ 1321 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 3592 \\ +\ 401 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 2227 \\ +\ 12 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 2363 \\ +\ 2111 \\ \hline \\ \hline \end{array}$

5. $\begin{array}{r} 4350 \\ +\ 5646 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 82350 \\ +\ 5213 \\ \hline \\ \hline \end{array}$

7. $\begin{array}{r} 59261 \\ +\ 40712 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 84743 \\ +\ 12251 \\ \hline \\ \hline \end{array}$

9. $\begin{array}{r} 54721 \\ +\ 33222 \\ \hline \\ \hline \end{array}$

10. $\begin{array}{r} 818819 \\ +\ 71110 \\ \hline \\ \hline \end{array}$

11. $\begin{array}{r} 834532 \\ +\ 154063 \\ \hline \\ \hline \end{array}$

12. $\begin{array}{r} 140546 \\ +\ 221121 \\ \hline \\ \hline \end{array}$

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13. $679448 + 110301$

14. $489660 + 10118$

15. $751210 + 12166$

16. $323456 + 1676543$

17. $1456317 + 333572$

18. $999879 + 110$

19. $751201 + 128780$

20. $2013424 + 1976365$

Addition of whole numbers by regrouping

Adding whole numbers horizontally

Example 1

$6327 + 3496 =$

Solution

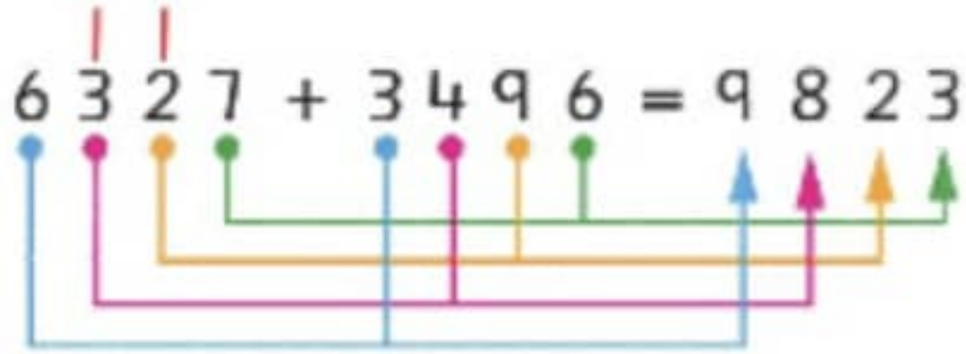
Add the digits from right to left.

Steps

1. Add the ones: $7 + 6 = 13$. Write 3 in the ones place. Regroup 10 ones into 1 tens and shift it to the tens place.
2. Add the tens: $1 + 2 + 9 = 12$. Write 2 in the tens place. Regroup 10 tens into 1 hundreds and shift it to the hundreds place.
3. Add the hundreds: $1 + 3 + 4 = 8$. Write 8 in the hundreds place.

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4. Add the thousands: 6 + 3 = 9. Write 9 in the thousands place.



Therefore, 6327 + 3496 = 9823.

Example 2

58271 + 32989 =

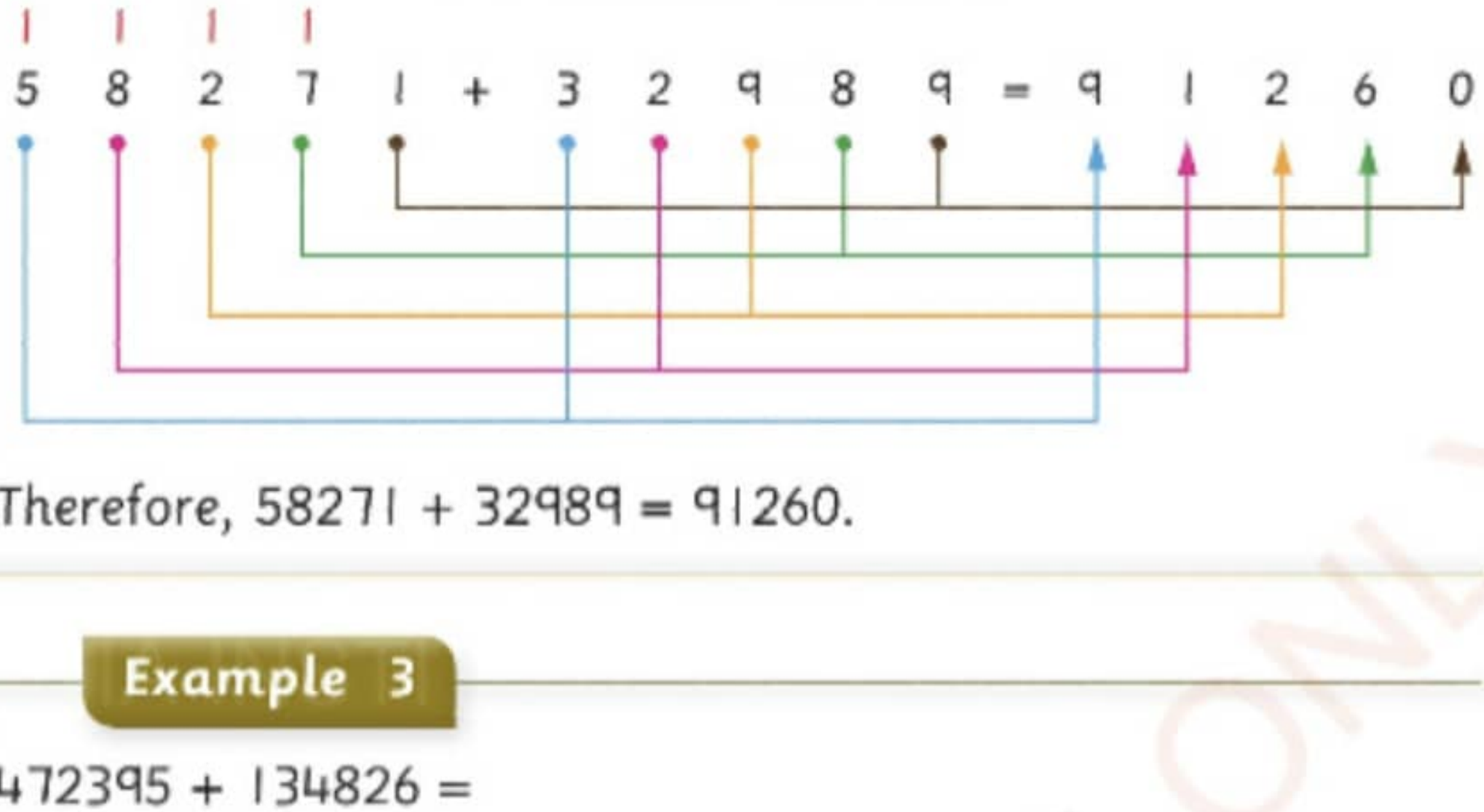
Solution

Add the digits from right to left.

Steps

1. Add the ones: 1 + 9 = 10. Write 0 in the ones place. Regroup 10 ones into 1 tens and shift it to the tens place.
2. Add the tens: 1 + 7 + 8 = 16. Write 6 in the tens place. Regroup 10 tens into 1 hundreds and shift it to the hundreds place.
3. Add the hundreds: 1 + 2 + 9 =12. Write 2 in the hundreds place. Regroup 10 hundreds into 1 thousands and shift it to the thousands place.
4. Add the thousands: 1 + 8 + 2 = 11. Write 1 in the thousands place. Regroup 10 thousands into 1 ten thousands and shift it to the ten thousands place.
5. Add the ten thousands: 1 + 5 + 3 = 9. Write 9 in the ten thousands place.

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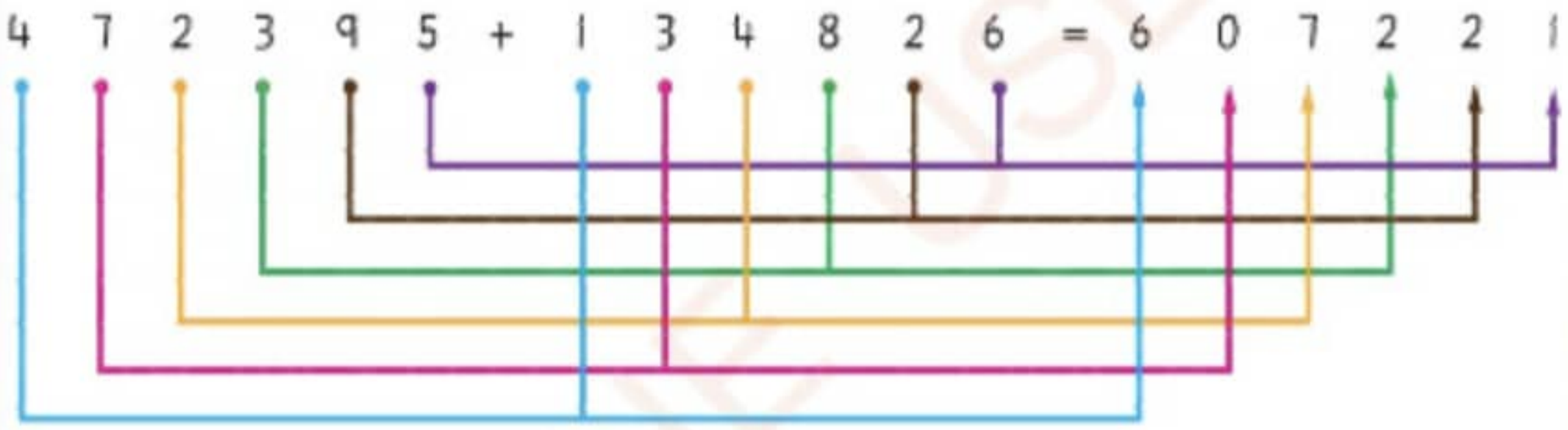


Therefore, 58271 + 32989 = 91260.

Example 3

472395 + 134826 =

Solution



Therefore, 472395 + 134826 = 607221.

Exercise 3

1. 7256 + 1796 =
2. 7531 + 1929 =
3. 8595 + 914 =
4. 3598 + 179 =
5. 4513 + 3299 =
6. 2643 + 993 =
7. 4951 +1891 =
8. 7536 + 19 =
9. 768 + 7861 =
10. 7816 +1832 =

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11. 7576 + 817 =

12. 2619 + 1786 =

13. 1745 + 899 =

14. 1652 + 476 =

15. 15433 + 798 =

16. 53415 + 21045 =

17. 64124 +16283 =

18. 75520 + 9221 =

19. 66612 + 24588 =

20. 162845 + 134027 =

21. 293586 + 26323 =

22. 567183 + 484921 =

23. 943573 + 56327 =

24. 720852 + 279168 =

25. 2964 + 298576 =

Adding whole numbers vertically

Example 1

$$\begin{array}{r} 2361 \\ +\ 3899 \\ \hline \\ \hline \end{array}$$

Solution

Align the given numbers vertically according to the place value of each digits.

Steps

1. Add the ones: 1+ 9 =10. Write 0 in the ones place. Regroup 10 ones into 1 tens and shift it to the tens place.

$$\begin{array}{r} {\scriptstyle 1} \\ 2361 \\ +\ 3899 \\ \hline 0 \\ \hline \end{array}$$

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2. Add the tens: 1+ 6 + 9 =16. Write 6 in the tens place. Regroup 10 tens into 1 hundreds and shift it to the hundreds place.

```
  11
 2361
+3899
------
  60
------
```

3. Add the hundreds: 1 + 3 + 8 =12. Write 2 in the hundreds place. Regroup 10 hundreds into 1 thousands and shift it to the thousands place.

```
  111
 2361
+3899
------
 260
------
```

4. Add the thousands: 1 + 2 + 3 = 6. Write 6 in the thousands place.

```
  111
 2361
+3899
------
6260
------
```

Therefore, the answer is 6260.

Example 2

```
  228186
+ 374899
--------

--------
```

Steps

1. Add the ones: 6 + 9 = 15. Write 5 in the ones place. Regroup 10 ones into 1 tens and shift it to the tens place.
2. Add the tens: 1 + 8 + 9 = 18. Write 8 in the tens place. Regroup 10 tens into 1 hundreds and shift it to the hundreds place.

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3. Add the hundreds: 1 + 1 + 8 = 10. Write 0 in the hundreds place. Regroup 10 hundreds into 1 thousands and shift it to the thousands place.
4. Add the thousands: 1 + 8 + 4 = 13. Write 3 in the thousands place. Regroup 10 thousands into 1 ten thousands and shift it to the ten thousands place.
5. Add the ten thousands: 1 + 2 + 7 = 10. Write 0 in the ten thousands place. Regroup 10 ten thousand into 1 hundred thousands and shift it to the hundred thousands place.
6. Add the hundred thousands: 1 + 2 + 3 = 6. Write 6 in the hundred thousands place.

Therefore, the answer is 603085.

This question can also be done by using the card number as follows:

Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
2 + 1	2 + 1	8 + 1	1 + 1	8 + 1	6
+ 3	7	4	8	9	9
6	10	13	10	18	15
6	0	3	0	8	5

Therefore, the answer is 603085.

Exercise 4

1. 6027 + 1784 = ____
2. 57336 + 12217 = ____
3. 48289 + 3457 = ____

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4. 85519 + 2719

5. 856956 + 134689

6. 88456 + 154689

7. 489765 + 94538

8. 365284 + 476329

9. 624872 + 9341

10. 872146 + 34987

11. 167426 + 594

12. 877925 + 98

13. 100009 + 9991

14. 89725 + 9999

15. 80298 + 19712

16. 998456 + 154689

17. 6024872 + 3975137

18. 8925778 + 7897987

19. 1000009 + 999991

20. 199001 + 950999

21. *Study the methods used by Ali, Rose and Dule to add 2,792 and 1,213. Use an altenative method that is easier to you to add the numbers.*

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Ali: 2792 + 1213

Add the ones: 3 + 2 = 5.

Add the hundreds: 1 + 7 + 2 = 10.

Add tens : 9 + 1 = 10.

Add thousands : 1 + 2 + 1 = 4.

Therefore, 2792 + 1213 = 4005.

Rose: 2792 + 1213 =

Breakdown the numbers as follows:

2792 + 1213 = 2790 + 2 + 3 + 10 + 1200

= 2700 + 90 + 2 + 3 + 10 + 1200

= 2700 + 1200 + 100 + 2 + 3

= 3900 + 100 + 5 = 4005

Therefore, 2792 + 1213 = 4005.

Dule: 2792 + 1213 =

Write the numbers in expanded form as follows:

2792 + 1203

= 2000 + 700 + 90 + 2 + 1000 + 200 + 10 + 3

= 2000 + 1000 + 700 + 200 + 90 + 10 + 2 + 3

= 3000 + 900 + 100 + 5 = 4005.

Therefore, 2792 + 1213 = 4005.

Word problems involving addition of whole numbers

Example 1

A meat factory bought 137,527 heads of cattle the year before last, and 222,062 last year. How many heads of cattle did the factory buy in the two years?

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Solution

Number of heads of cattle bought the year before last	137527
Number of heads of cattle bought last year	+ 222062
Total number of heads of cattle bought in the two years	359589

Therefore, the factory bought 359,589 heads of cattle in the two years.

Example 2

A certain national park in Tanzania has 78,696 zebras and 2,323 buffaloes. What is the total number of the animals?

Solution

Number of zebras	78696
Number of buffaloes	+ 2323
Total number of the animals	81019

Therefore, the total number of animals is 81,019.

Exercise 5

1. Two printing companies produced books for primary schools.If the first company produced 5,900 books and the second 3,600 books, how many books did the two companies produce in total?
2. A farmer sold 35,752 litres of milk in January. Also, he sold 27,888 litres in February. How many litres of milk were sold in the two months?
3. A farmer has 1,982 timber trees and 7,291 fruit trees. How many trees does the farmer have in total?

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4. In a certain hospital, 4,618 malaria patients and 689 eye patients were treated. How many patients were treated in total?
5. A farmer stored 25,677 kilograms of maize in the first season. In the second season, she stored 21,913 kilograms. How many kilograms of maize were stored by the farmer in the two seasons?
6. In a certain year, 9,436 cars were inspected in Mtwara region, while in Ruvuma region 10,170 cars were inspected. How many cars were inspected in the two regions?
7. Kakali primary school has 2,588 desks, Dadawa primary school has 8,813 desks, while Rukuku primary school has 10,554 desks. How many desks are there in all the three primary schools?
8. Juma harvested 5,630 mangoes from the first tree. He then harvested 480 mangoes from the second tree. How many mangoes did he harvest in total?
9. A bus carried two baskets of oranges. The first basket had 2,376 oranges and the second had 4,638 oranges. How many oranges did the bus carry?
10. In a certain region, 6,523 pupils joined standard one in 2015. In 2016, 7,212 pupils joined standard one, and in 2017, a total of 7,430 joined standard one. How many pupils joined standard one in the three years?

Subtraction of whole numbers

When subtracting whole numbers, consider the place value of each digit in a number. Start subtracting ones, tens and so on.

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Subtraction of whole numbers without regrouping

Subtracting whole numbers horizontally

Example 1

6874 – 2430 =

Steps

1. Subtract the ones: 4 – 0 = 4 Write 4 in the ones place.
2. Subtract the tens: 7 – 3 = 4 Write 4 in the tens place.
3. Subtract the hundreds: 8 – 4 = 4. Write 4 in the hundreds place.
4. Subtract the thousands: 6 – 2 = 4 Write 4 in the thousands place.



Therefore, 6874 – 2430 = 4444.

Example 2

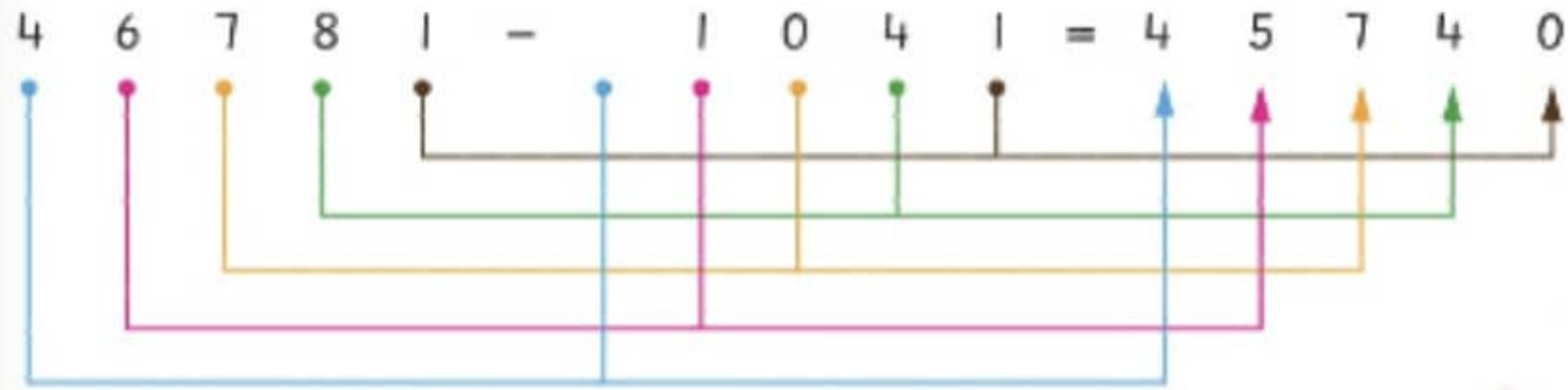
46781 – 1041 =

Steps

1. Subtract the ones: 1 – 1 = 0. Write 0 in the ones place.
2. Subtract the tens: 8 – 4 = 4. Write 4 in the tens place.
3. Subtract the hundreds: 7 – 0 = 7. Write 7 in the hundreds place.
4. Subtract the thousands: 6 – 1 = 5. Write 5 in the thousands place.

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5. Subtract the ten thousands: 4 − 0 = 4. Write 4 in the ten thousands place.



Therefore, 46811 − 10411 = 45740.

Exercise 6

1. 9942 − 42 =
2. 4376 − 3245 =
3. 3760 − 2550 =
4. 3049 − 2017 =
5. 8973 − 7642 =
6. 4256 − 3132 =
7. 65458 − 21344 =
8. 87999 − 33422 =
9. 623440 − 2010 =
10. 568237 − 18113 =
11. 784969 − 543867 =
12. 987645 − 234 =
13. 2000652 − 331 =
14. 17897988 − 687888 =
15. 6999456 − 2811222 =
16. 3334237 − 2223126 =
17. 9999999 − 7765555 =
18. 1079878 − 17561 =
19. 999789 − 347 =
20. 1000005 − 2 =

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Subtracting whole numbers vertically

Example 1

$$\begin{array}{r} 9854 \\ -\ 8743 \\ \hline \\ \hline \end{array}$$

Solution

Align the numbers according to the place value of each digit. This can be done as shown in the table.

	Thousands	Hundreds	Tens	Ones
	9	8	5	4
−	8	7	4	3
	1	1	1	1

Therefore, the answer is 1111.

Example 2

$$\begin{array}{r} 86597 \\ -\ 54326 \\ \hline \\ \hline \end{array}$$

Steps

1. *Subtract the ones: 7 − 6 = 1. Write 1 in the ones place.*
2. *Subtract the tens: 9 − 2 = 7. Write 7 in the tens place.*

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3. Subtract the hundreds: 5 – 3 = 2. Write 2 in the hundreds place.
4. Subtract the thousands: 6 – 4 = 2. Write 2 in the thousands place.
5. Subtract the ten thousands: 8 – 5 = 3. Write 3 in the ten thousands place.

Therefore, the answer is 32271.

Example 3

$$\begin{array}{r} 984265 \\ -\ 632143 \\ \hline \\ \hline \end{array}$$

Solution

The steps used in the horizontal and vertical subtraction are the same. Start subtracting ones, tens, hundreds, up to ten thousands.

$$\begin{array}{r} 984265 \\ -\ 632143 \\ \hline 352122 \end{array}$$

Therefore, the answer is 352122.

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Exercise 7

1. 7824 − 5612

2. 4851 − 3720

3. 4343 − 3231

4. 8899 − 8768

5. 6533 − 6322

6. 7291 − 3231

7. 69635 − 58313

8. 75757 − 25252

9. 99991 − 4340

10. 237972 − 112451

11. 823574 − 1230

12. 912345 − 810212

13. 695872 − 34751

14. 987672 − 34521

15. 652437 − 410316

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Subtraction of whole numbers by regrouping

Subtracting whole numbers horizontally

Example 1

5371 − 3282 =

Steps

1. Subtract the ones: 1− 2; 1 is insufficient in subtracting 2. Thus, take 1 group of tens from 7 tens. Regroup it to get 10 ones then add 1 ones: 10 + 1 =11. Now, 11 − 2 = 9. Write 9 in the ones place. Remember, 1 tens was regrouped into 10 ones.Thus, 6 tens remained in the tens place.

2. Subtract the tens: 6 − 8; 6 is insufficient in subtracting 8. Thus, take 1 group of hundreds from 3 hundreds. Regroup it to get 10 tens then add 6 tens: 10 + 6 = 16. Now, it becomes 16 − 8 = 8. Write 8 in the tens place. Remember, 1 hundreds was regrouped into 10 tens. Thus, 2 hundreds remained in the hundreds place.

3. Subtract the hundreds: 2 − 2 = 0. Write 0 in the hundreds place.

4. Subtract the thousands: 5 − 3 = 2. Write 2 in the thousands place.

Therefore, 5371 − 3282 = 2089.

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Example 2

982140 – 771671 =

Steps

1. Subtract the ones: 0 – 1; 0 is insufficient in subtracting 1. Thus, take 1 group of tens from 4 tens. Regroup it to get 10 ones, then add 0 ones: 10 + 0 =10. Now, 10 – 1 = 9. Write 9 in the ones place. Remember, 1 tens was regrouped into 10 ones. Thus, 3 tens remained in the tens place.
2. Subtract the tens: 3 – 7; 3 is insufficient in subtracting 7. Thus, take 1 group of hundreds from 1 hundreds and regroup it to get 10 tens then add 3 tens: 10 + 3 =13. Now, 13 – 7 = 6. Write 6 in the tens place. Remember, 1 hundreds was regrouped into 10 tens. Thus, 0 hundreds remained in the hundreds place.
3. Subtract the hundreds: 0 – 6; 0 is insufficient in subtracting 6. Take 1 group of thousands from 2 thousands. Regroup it to get 10 hundreds then add 0 hundreds: 10 + 0 =10. Now,

   10 – 6 = 4. Write 4 in the hundreds place. Remember, 1 thousands was regrouped into 10 hundreds. Thus, 1 thousands remained in the thousands place.
4. Subtract the thousands: 1 – 1 = 0. Write 0 in the thousands place.
5. Subtract the ten thousands: 8 – 7 = 1. Write 1 in the ten thousands place.
6. Subtract the hundred thousands: 9 – 7 = 2. Write 2 in the hundred thousands place.

Therefore, 982140 – 771671 = 210469.

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Exercise 8

1. 7100 − 4291 =
2. 68721 − 6667 =
3. 6816 − 5824 =
4. 4133 − 2145 =
5. 65458 − 26344 =
6. 9026 − 3146 =
7. 7761 − 3872 =
8. 3465 − 1456 =
9. 84223 − 5843 =
10. 85446 − 2254 =
11. 44346 − 37016 =
12. 672354 − 51132 =
13. 976540 − 888666 =
14. 164945 − 99 =
15. 876124 − 697982 =

Subtracting whole numbers vertically

Example 1

$$\begin{array}{r} 6\ 854 \\ -\ 3\ 887 \\ \hline \\ \hline \end{array}$$

Steps

1. Subtract the ones: 4 − 7; 4 is insufficient in subtracting 7. Thus, take 1 group of tens from 5 tens. Regroup it to get 10 tens then add 4 ones: 10 + 4 = 14. Now, 14 − 7 = 7. Write 7 in the ones place. Remember, 1 tens was regrouped into 10 ones. Thus, 4 tens remained in the tens place.

$$\begin{array}{r} {\scriptstyle 4\,14} \\ 6\,8\not{5}\not{4} \\ -\ 3887 \\ \hline 7 \\ \hline \end{array}$$

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2. Subtract the tens: 4 – 8; 4 is insufficient in subtracting 8. Thus, take 1 group of hundreds from 8 hundreds. Regroup it to get 10 tens then add 4 tens: 10 + 4 = 14. Now, 14 – 8 = 6. Write 6 in the tens place. Remember, 1 hundreds was regrouped into 10 tens. Thus, 7 hundreds remained in the hundreds place.

14
7 14
6 8 5 4
– 3887
67

3. Subtract the hundreds: 7 – 8; 7 is insufficient in subtracting 8. Thus, take 1 group of thousands from 6 thousands. Regroup it to get 10 hundreds then add 7 hundreds: 10 + 7 =17. Now, 17 – 8 = 9. Write 9 in the hundreds place. Remember, 1 thousands was regrouped into 10 hundreds. Thus, 5 thousands remained in the thousands place.

1714
5 7 14
6 8 5 4
– 3887
967

4. Subtract the thousands: 5 – 3 = 2. Write 2 in the thousands place.

1714
5 7 14
6 8 5 4
– 3887
2967

Therefore, the answer is 2967.

Example 2

87964
– 49897
_____

_____

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Solution

The steps used in the horizontal and vertical subtractions are the same. Start subtracting ones by ones up to ten thousand by ten thousands.

$$\begin{array}{r} 87964 \\ -\,49897 \\ \hline 38067 \end{array}$$

Therefore, the answer is 38067.

Exercise 9

1. $\begin{array}{r} 3822 \\ -\,1713 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 1245 \\ -\,158 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 45385 \\ -\,34392 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 2083 \\ -\,1976 \\ \hline \\ \hline \end{array}$

5. $\begin{array}{r} 30405 \\ -\,1992 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 8433 \\ -\,6377 \\ \hline \\ \hline \end{array}$

7. $\begin{array}{r} 3005 \\ -\,1992 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 64536 \\ -\,58323 \\ \hline \\ \hline \end{array}$

9. $\begin{array}{r} 55376 \\ -\,25445 \\ \hline \\ \hline \end{array}$

10. $\begin{array}{r} 873977 \\ -\,762582 \\ \hline \\ \hline \end{array}$

11. $\begin{array}{r} 682165 \\ -\,8987 \\ \hline \\ \hline \end{array}$

12. $\begin{array}{r} 875924 \\ -\,683678 \\ \hline \\ \hline \end{array}$

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13. $\begin{array}{r} 600000 \\ -\ 54164 \\ \hline \\ \hline \end{array}$

14. $\begin{array}{r} 87625 \\ -\ 9734 \\ \hline \\ \hline \end{array}$

15. $\begin{array}{r} 666666 \\ -\ 78947 \\ \hline \\ \hline \end{array}$

Word problems involving subtraction of whole numbers

Example 1

The total number of cattle and goats in a district is 2,139,500. The number of cattle is 1,104,563. How many goats are there in the district?

Solution

Total number of cattle and goats	2139500
Number of cattle	− 1104563
Number of goats	1034937

Therefore, there are 1,034,937 goats in the district.

Example 2

Bakers bake 72,324 loaves of bread everyday. On a certain day they baked only 45,458 loaves of bread. How many loaves were not baked on that day to meet the daily number?

Solution

Number of loaves baked everyday	72324
Number of loaves baked in a day	− 45458
Number of loaves which were not baked in that day	26866

Therefore, 26,866 loaves of bread were not baked in that day.

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Exercise 10

1. Maria's shop has 2,972 mobile phones and landlines. If there are 1,235 mobile phones, how many landlines are there?
2. A mango vender had four thousand three hundred and eighteen mangoes. He sold one thousand four hundred and thirty-five mangoes to a merchant. How many mangoes did he remain with?
3. A fisherman caught seven thousand three hundred and eighty-one fish. He sold five thousand seven hundred and ninety-two fish. How many fish did he remain with?
4. Remi bought two boxes containing six thousand four hundred and forty tomatoes. One thousand three hundred and two tomatoes were rotten. How many tomatoes were not rotten?
5. A school needs three thousand seven hundred forty-nine desks. Parents contributed two thousand nine hundred and ninety-seven desks. How many desks are still needed by the school?
6. The school bought seven thousand two hundred and fifty-two exercise books. Juma and Tatu were asked to carry the exercise books to the store. Juma carried three thousand and thirty-nine exercise books. How many exercise books were carried by Tatu?
7. Pupils planted three thousand four hundred and eighty trees in a certain year. After two years, five hundred and eighty-seven trees had dried up. How many trees remained healthy?

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8. A cooperative union produced 287,891 bricks out of the required 482,674 bricks. How many bricks were not produced?
9. A contractor was paid 845,750 shillings for the construction of a school toilet. If the contruction costs was 974,500 shillings, how much money was needed to complete her payments?
10. A certain ward has six thousand eight hundred and seventy-four voters. Three thousand and eighty-seven voters turned up to vote. How many voters did not turn up to vote?
11. A company was asigned to deliver two thousand eight hundred and seven iron sheets. The company delivered two thousand three hundred and ninety-eight iron sheets. How many iron sheets were not delivered?
12. Kulwa harvested five thousand and forty-seven bags of beans in the first year. He harvested three thousand four hundred and thirty-eight bags in his second year. Find the difference of the harvest in the two years.

Activity: Recognising addition and subtraction of whole numbers

Learning resources

Online programmes

Description

Use online programmes to explore various examples of adding and subtracting whole numbers.

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Summary

1. When adding or subtracting numbers, align the numbers either horizontally or vertically. Start by adding the ones, followed by the tens, then the hundreds digits and so on.
2. When adding or subtracting numbers, always start from the rightmost digit and end with the leftmost digit.
3. In solving word problems, consider the following:
   (a) Read the problem to identify the mathematical operations involved.
   (b) Perform the operations on the values mentioned in the problem.
   (c) Conclude by answering the question.

Revision exercise

1. $\begin{array}{r} 6142 \\ +\ 4226 \\ \hline \\ \hline \end{array}$

2. $\begin{array}{r} 8937 \\ -\ 27 \\ \hline \\ \hline \end{array}$

3. $\begin{array}{r} 55394 \\ +\ 1295 \\ \hline \\ \hline \end{array}$

4. $\begin{array}{r} 47326 \\ -\ 32198 \\ \hline \\ \hline \end{array}$

5. $\begin{array}{r} 2696 \\ +\ 2398 \\ \hline \\ \hline \end{array}$

6. $\begin{array}{r} 638345 \\ -\ 225221 \\ \hline \\ \hline \end{array}$

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7. $\begin{array}{r} 55394 \\ +\ 1295 \\ \hline \\ \hline \end{array}$

8. $\begin{array}{r} 7623 \\ -\ 278 \\ \hline \\ \hline \end{array}$

9. 6145 + 312 =

10. 8912 + 88 =

11. 54273 + 12632 =

12. 76832 + 2126 =

13. 9051 – 4030 =

14. 4177 – 408 =

15. 548123 – 253700 =

16 59635 – 926 =

17. *Salum needs four thousand five hundred and seventy bricks to build his house. He has one thousand two hundred and seventy bricks. How many more bricks does he need to build his house?*

18. *Tabora residents planned to plant nine thousand nine hundred and ninety trees in two years. They planted five thousand and seventy trees in the first year. How many trees did they plant in the second year?*

19. *Mwalimu Mponela had 7,568 exercise books. He gave his pupils 5,368 exercise books. How many exercise books did he remain with?*

20. *A school has a total of 3,270 pupils. Among them, 1,560 are girls. How many boys does the school have?*

21. *A ward has 6,328 inhabitants and another ward has 2,590 inhabitants. What is the total population in the two wards?*

22. *John raises 4,569 chickens and Ashura raises 5,200 chickens. How many chickens do they raise altogether?*

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Chapter Three

Measurements of length, mass and volume

Introduction

In this chapter, you will learn about the measurements of length, mass and volume. You will also be able to estimate and compare lengths, masses and volumes of different objects. The competencies developed will enable you to perform different activities related to measurements of length, mass, and volume.

Measurements of length

Length is the distance between two points. It can be measured by using different tools. The measuring tools of length are of two types, namely non-standard and standard tools.

Non-standard tools for measuring length

The use of non-standard measuring tools of length depends on the type of objects being measured. For example, the length of a chalkboard or table can be measured using non-standard tools such as handspan, stick, rope or footsteps.

Non-standard tools like a handspan do not give exact measurements because the obtained length will depend on the size of the person's handspan. If two persons with different handspans measure the same table, they will get different lengths.

Therefore, a handspan is a non-standard tool. The following picture shows a pupil measuring the length of a table by handspan.

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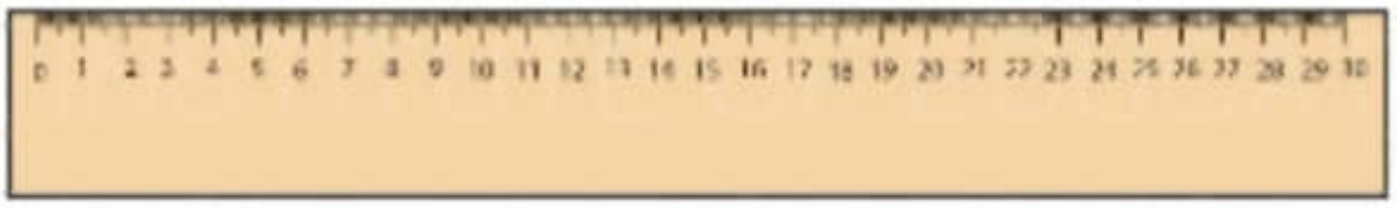
Activity 1: Measuring the length of a classroom

1. Measure the length of the classroom using footstep.
2. Record the length of the classroom.
3. Ask your friend to measure and record the length of the classroom using similar tool used in task 1.
4. Compare the measurements you obtained and that of your friend.
5. Explain about the similarities or differences of the obtained measurements.

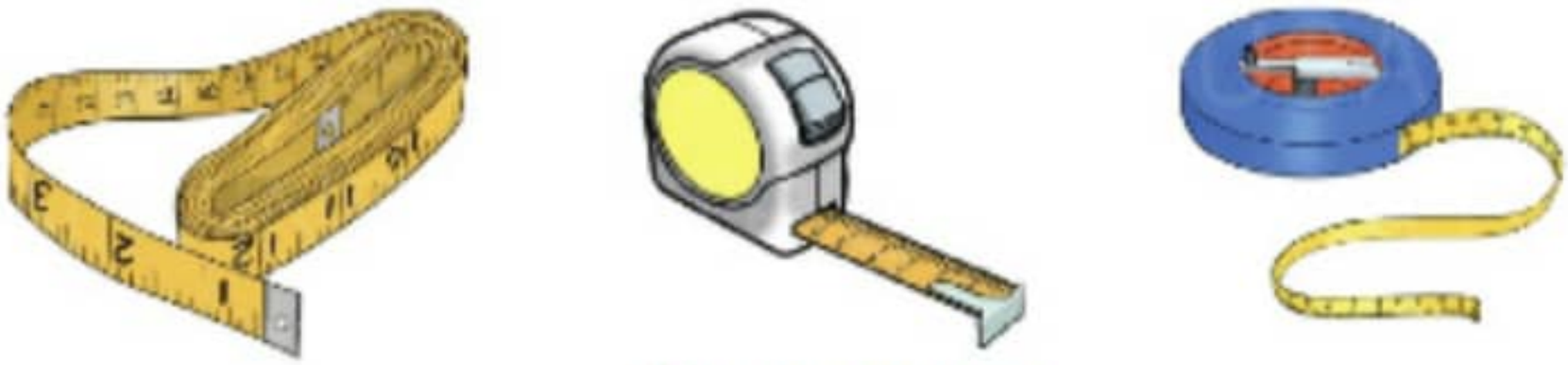
Standard tools used for measuring length

Standard tools for measuring length provide exact results which are acceptable worldwide. They include tape measure, ruler and metre stick. These tools give measurements in metric units. The common metric units of length include millimetres (mm), centimetres (cm), metres (m) and kilometres (km). The standard tools used for measuring length use these unit. The following picture shows some of the standard tools for measuring length.

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Ruler



Tape measures

Activity 2: Measuring length of different objects

1. Identify any four objects in your environment whose lengths can be measured by using non-standard and standard tools.
2. Choose one object then measure its length using both non-standard and standard tools.
3. Record its length using both non-standard and standard tools. How comparable are they?

Activity 3: Measuring the length of an exercise book

1. Measure the length of your exercise book in centimetres using a ruler.
2. Record the length measured in task 1.
3. Repeat task 1 by using a tape measure.
4. Record the length measured in task 3.
5. Compare the measurements obtained in task 3 and 4.
6. Explain your observation from task 5?

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Measuring length in centimetres and millimetres

Measurements in centimetres and millimetres are used more often when object are measured by using a ruler. One centimetre is equal to ten millimetres. The smallest measurement unit of a ruler is the millimetre.

Example 1

What is the length of the piece of cloth in the following picture?



Answer

The length of the piece of cloth is 5 cm.

Example 2

What is the length of the piece of wood in the following picture?



Answer

The length of the piece of wood is 3 cm and 5 mm.

Exercise 1

Measure the length of each object in cm and mm from the following list.

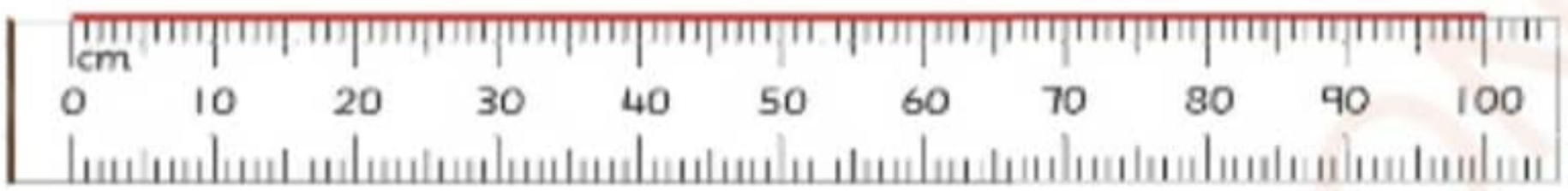
1. Table
2. Desk
3. Pencil

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4. Exercise book
5. Ruler
6. Mathematics book
7. Eraser
8. Mathematical set kit

Measuring length in metres

The length of different objects can also be measured in metres. One metre is equal to one hundred centimetres, that is 1 m = 100 cm. The given picture shows a ruler which is 100 centimetres or 1 metre long.



Activity 4: Measuring length of different objects

1. Measure the length of each object in metres from the following list.

	Objects	Length
(a)	Teacher's table	
(b)	Classroom door	
(c)	Desk	

2. Record the length of each object.
3. Which object has the largest length? Write its value.

Measuring length in kilometres

The length of a long distance can be measured in kilometres. This includes the distance between two places which are far apart. For instance, the distance between two villages or towns is usually measured in kilometres. One kilometre is equal to one thousand metres. This relationship can be written as 1 km = 1000 m.

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Exercise 2

1. Which measurement unit among cm, m and km is appropriate for measuring the following?
   (a) your height
   (b) distance from your class to the school toilet
   (c) distance from your school to home
   (d) length of a piece of chalk
2. Amina measured the length of a chalkboard using a ruler. John measured the length of the same chalkboard using his handspan. Who obtained the most correct measurement? Give reasons for your answer.
3. The distance from one school to a nearby dispensary is 800 m. The distance from the same school to a nearby market is one kilometre. Which one is far from the school between the dispensary and the market?
4. What is the length of the pencil in the following picture.



Measurements of mass

Mass is a measure of the amount of matter in an object. It can be measured by using different measuring tools. The tools for measuring mass are divided into two types, non-standard tools and standard tools.

Non-standard tools for measuring mass

Non-standard tools for measuring the mass gives a rough estimate of the amount measured. Some of the tools used

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include spoons, cups, boxes, sacks and buckets. The following picture shows two buckets that contain rice.



Bucket A



Bucket B

Which bucket has more rice? Explain your answer.

Standard tools for measuring mass

Standard tools used for measuring the masses of different things are called weighing balances. The following picture shows examples of weighing balances.



Units for measuring mass

The basic units used for measuring mass are gram and kilogram. A gram is a smaller measurement unit than a kilogram. They are both used for measuring masses of different items in our environment. The unit of gram is usually used for items whose mass is less than one kilogram. For heavier items, the unit of kilogram is used. The relationship between the two units is as follows:

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1 kilogram (kg) = 1000 grams (g).

$\frac{1}{2}$ kg = 500 grams (g).

$\frac{1}{4}$ kg = 250 gram (g)

Activity 5: Measuring mass of items in grams and kilograms

1. In your environment, identify items whose masses can be measured.
2. Select two items and measure their masses.
3. Measure about 750 g of the first item, then about 1 kg of the second.
4. From the measurements taken, which item is heavier? Give reasons.

Exercise 3

1. Measure the mass of the following items:
   (a) An empty bottle of soda
   (b) Exercise book
   (c) Mathematical set
2. From the masses of items you have measured in Question 1:
   (a) Which item is the lightest?
   (b) Which one is the heaviest?
3. List about 5 items that can be measured by using the standard measuring tools of mass.
4. Arrange the following masses in the descending order: 1600 g, 500 g, 25 kg, 50 kg, and 650 g.

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5. Zipora says the mass of 1000 grams is heavier than that of 1 kilogram. Is Zipora collect? Explain your answer.

6. Which measurement unit between gram and kilogram is appropriate for measuring:
   (a) The mass of your mathematics textbook.
   (b) A bottle filled with two litres of cooking oil.
   (c) A bags of rice.
   (d) The mass of a students.

Measurement of Volumes

Volume is the amount of space occupied by an object. It can be measured by using non-standard and standard tools.

Non-standard tools used for measuring volume

Common tools used for measuring volume include spoons, cups and buckets. The following picture shows pupils measuring the volume of water by using cups and buckets.



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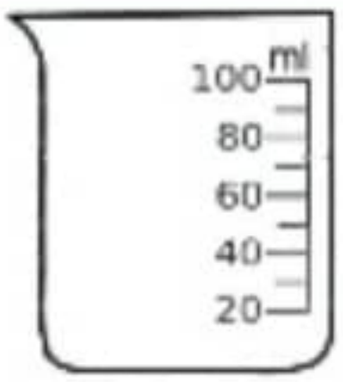
The pupils use cups to fill buckets of the same size with water. Each pupil uses the same number of cups to fill the bucket. Will the volume of water in each bucket be the same? Explain your answer.

Activity 6: Measuring the volume of water

1. Take two empty containers of the same size.
2. Counting the number of cups, fill one container with water.
3. Ask your friend to do task 2 by using the same measuring tools used in task 2.
4. Record the number of cups used to fill your container and that of your friend.
5. Do the number of cups used to fill the containers differ? Explain.
6. Identify other three items in your environment whose volume can be measured by using this activity?

Standard tools for measuring volume

Standard tools for measuring volume give exact values. These tools include beakers, cylinders and bottles. The following picture shows some of the standard tools used for measuring volume.

Tool	Name
	Beaker

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	Measuring cylinder
	Bottle

Units for measuring volume

The common units for measuring volumes of items are millilitres and litres. Millilitre is the smaller unit than the litre. They are both used to measure volumes of different items in our daily lives. Millilitres are usually used for items with less volume than one litre while litres are used for larger volumes. Their relationship is as follows:

1 litre (L) = 1000 millilitres (ml)

$\frac{1}{2}$ litre = 500 ml

$\frac{1}{4}$ of a litre = 250 (ml)

Activity 7: Measuring the volume of water

1. Take a measuring cylinder.
2. Pour some water into the cylinder until it reaches halfway.
3. Read the volume of water in the cylinder.

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Exercise 4

1. Arrange the following volumes from the smallest to the largest; 450 ml, 40 ml, 2 L, and 4 ml.

2. Examine the following table and then answer the questions that follow:

Jug	Litre	Bottle	Beam balance
Ruler	Box	Syringe	Bucket

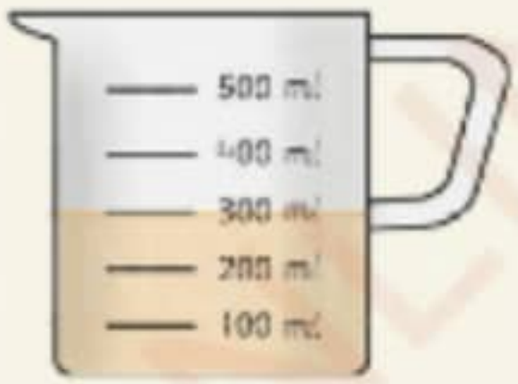
   (a) Which measuring tools can be used for measuring volume?

   (b) Which unit measures volume?

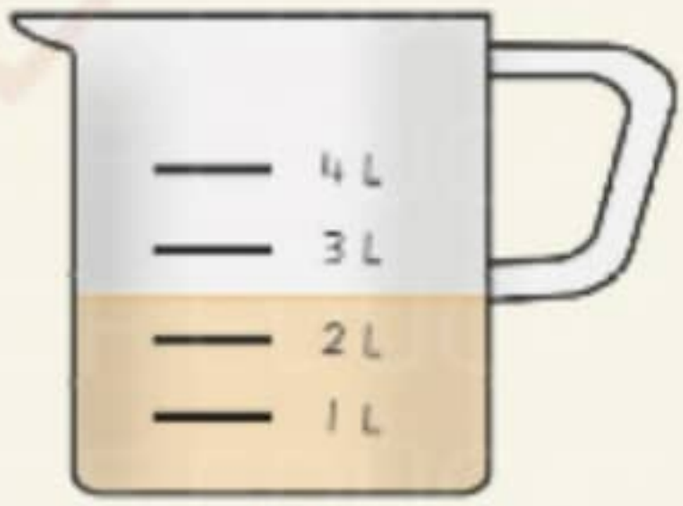
   (c) List the measuring tools that measure larger volumes

   (d) List the measuring tools that measure smaller volumes.

3. Observe the following jugs and then write the highest amount each jug can hold.



(a)



(b)

4. A cow produces 20 litres of milk every day. How many litres of milk will it produce in three days?

5. A pupil bought a bottle of 1 litre of juice. The pupil then gave 500 ml of juice from the bottle to another pupil. How many millilitres of juice did the pupil remain with?

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Activity 8: Investigate different measuring tools in our daily lives

1. Visit a nearby shop in your environment where people sell different items.
2. Observe the way people measure lengths, masses and volumes of different items.
3. Identify two items from the list of your observation.
4. Which tools do people use to measure the lengths, masses and volumes of the identified items in task 3?
5. Record the lengths, masses and volumes of the identified items.

Activity 9: Learn more about measurements from online sources

Summary

1. The basic unit of length is metre (m).
2. The basic unit of mass is kilogram (kg).
3. The basic unit of volume is litre (L).
4. The relationship between metre and centimetre is 1 ml = 100 cm.
5. The relationship between kilograms and grams is 1 kg = 1000 g.
6. The relationship between litre and millilitre is 1 L = 1000 ml.

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Vocabulary	
Length	Distance between two points
Mass	Measure of the amount of matter in an object
Volume	Amount of space occupied by an object
Standard tools	Tools which give accurate and exact measurements
Non-standard tools	Tools which give measurements that are not exact

Revision exercise

1. Choose correct units of measurement from list B that can be used for measuring the lengths in list A.

List A

(a) The length of the teachers' office _____

(b) Length of piece of chalk _____

(c) The length of the road from Dodoma to Dar es Salaam _____

List B

Metre (m)	Centimetre (cm)	Kilometre (km)
Kilogram (kg)	Mililitre(ml)	Litre(L)

2. Which of the following items can be estimated to weigh one kilogram?

(a) Bicycle (b) Pineapple (c) A piece of paper

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3. In the following pictures, write the units of mass (either gram (g) or kilogram (kg)) that you can use to estimate the mass of the following items. (a) is provided as an example.

	Picture	Appropriate unit (g or Kg)
(a)	 Egg	Gram (g)
(b)	 Bottle cap	_____
(c)	 Cow	_____
(d)	 Pencil	_____

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4. If two stones weighing 500 g and 250 g are used to find the mass of rice, how much does the rice weigh?

5. You are given two stones weighing 250 g and one stone weighing 500 g. How will you measure one kilogram using the given stones?

6. If you are given the following volumes: 150 ml, 400 ml, and 500 ml, what combination of volumes will you use to measure 550 millilitres?

7. Mzee Kinene has 750 ml of cattle medicine left in a 1000 ml bottle. How much medicine does he need to fill the medicine bottle?

8. Amina bought one litre of water. If she drank 500 ml during lunch, how many millilitres of water were left?

9. Aisha was asked to measure the distance of a netball pitch. Which of the following units is more appropriate for representing this distance: cm, m or km? Give reasons for your choice.

10. Kameme's ruler is broken so that it begins at 3 cm mark. If he wants to draw a line which is 5 cm long, at what mark of the ruler will the line end?

11. Which unit will you use to express the mass of an orange?

12. If a bag of potatoes weighs 2,000 g and a bag of rice weighs 1 kg, which bag is heavier?

13. Clare bought 500 ml of cooking oil from school. She used 250 ml, what volume of cooking oil remained?

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Chapter Four
Shapes and figures

Introduction

In this chapter, you will learn different types of plane figures. You will learn a point, line, line segment, and a ray. You will also construct plane figures using points, lines and line segments. The competencies developed will enable you to perform various real life activities such as in cooking, sewing, construction, carpentry, and painting.

A point, line, line segment and a ray

A point, line segment, line, and a ray are basic figures which are used in constructing various geometrical figures.

A point

A point is a geometrical figure which represents location of a place. A point has no size and it is shown by a dot. It has no length and width. Points are represented by capital letters. The following are examples of points.



The points A, B, and C in the examples represent different location of places.

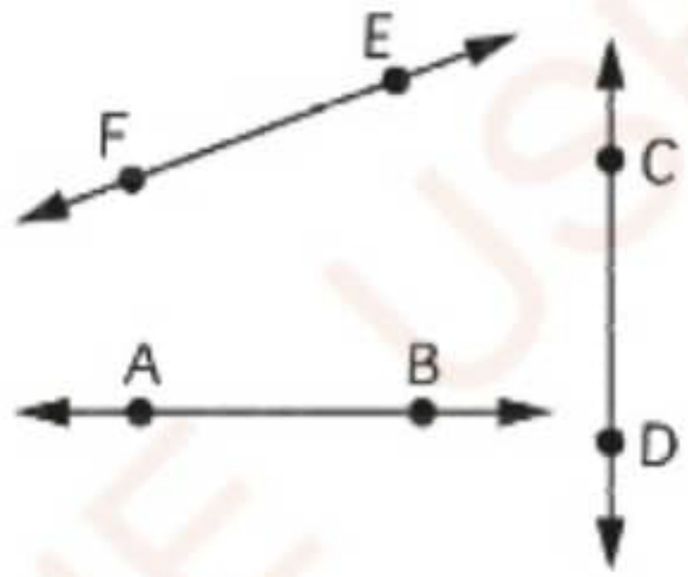
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Activity 1: Locating points on a plane

1. Use a pencil to locate five distinct points on a piece of paper.
2. Name the points by using capital letters.

A line

A line is made of a set of points joined together, and extends on opposite sides infinitely. A line is often indicated by a symbol $\leftrightarrow$ to show the infinite nature. A straight line can be denoted by any two distict points on the line. The following figures show examples of straight lines.



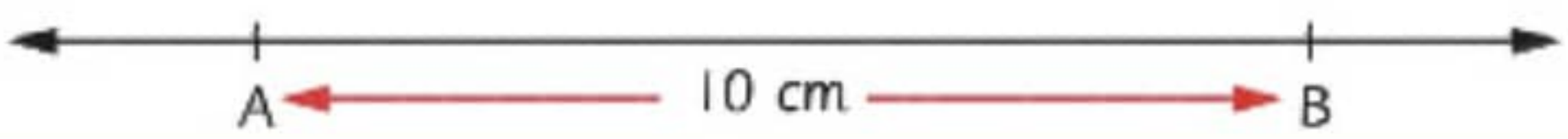
Therefore, $\overleftrightarrow{EF}$ or $\overleftrightarrow{FE}$ represents a straight line EF or FE, respectively. Similarly, $\overleftrightarrow{AB}$ or $\overleftrightarrow{BA}$ and $\overleftrightarrow{CD}$ or $\overleftrightarrow{DC}$ are straight lines.

Example 1

Draw a line which is longer than 10 cm from point A to point B.

Solution

A line drawn is longer than 10 cm from point A to B.



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Line segment

A line segment is a straight line that is constructed by joining two distinct points. The length of a line segment can be determined because it has end points. A line segment whose end points are A and B is denoted by $\overline{AB}$ or $\overline{BA}$ or line segment AB or line segment BA.

Activity 2: Drawing a line segment

1. Connect points A and B, C and D, and E, and F, using a pencil and free hand.



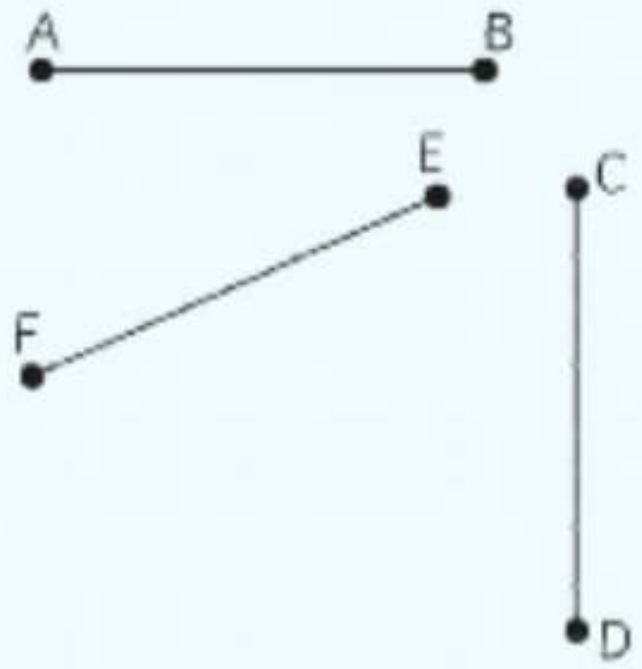
2. Connect points A and B, C and D, and E and F using a pencil and ruler.



3. What did you observe in task 1 and 2?

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4. In which task, the following lines were constructed?



Features of a line segment:

A line segment has the following features:

(a) It is straight.

(b) It has no thickness.

(c) It has two endpoints.

Example 1

In each of the following diagrams, find out if it is a line segment, a line or not.

(a)



(b)



(c)



Answer

(a) Is a line because it has no end points.

(b) Is a line segment because it has two end points.

(c) Is neither a line nor a line segment because it has one starting point and an arrow on the other end.

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Example 2

Write the name of each of the following line segments:

(a)



(b)



(c)



Answer

(a) $\overline{DQ}$ (b) $\overline{CD}$ (c) $\overline{AB}$

Example 3

Write the names of all possible line segments in the following figure:



Answer

$\overline{AB}$, $\overline{AC}$, $\overline{AD}$, $\overline{BC}$, $\overline{BD}$ and $\overline{CD}$.

Example 4

How many line segments are there in the following figure?



Answer

There are eight line segments.

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Exercise 1

1. Draw line segments of the following lengths:

   $\overline{PQ} = 4$ cm

   $\overline{CD} = 5$ cm

   $\overline{LM} = 6$ cm

   $\overline{JK} = 7$ cm

   $\overline{XY} = 8$ cm

   $\overline{EF} = 9$ cm

   $\overline{MN} = 10$ cm

2. Which of the following represents a line segment?

   (d) $\overrightarrow{AB}$ (b) $\overline{AB}$ (c) $\overline{AB}$ (d) $\overline{AB}$

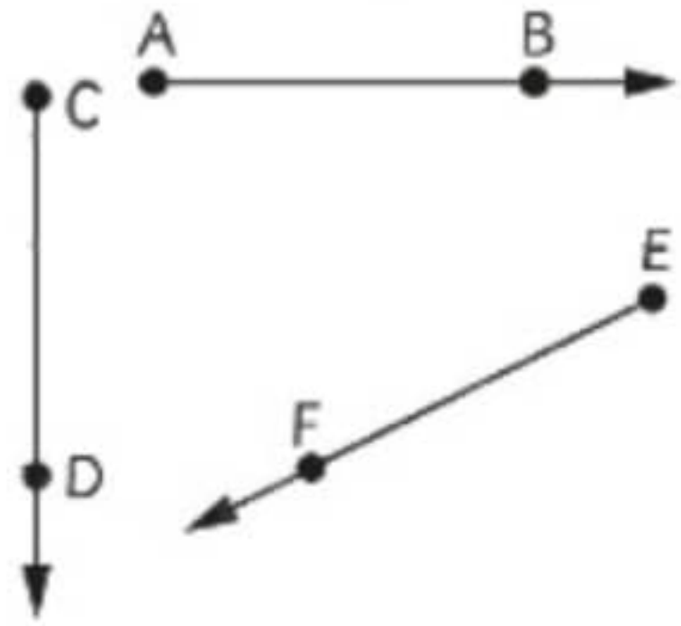
3. How many line segments does a square have?

4. What is the difference between a line segment and a line?

5. What are the examples of line segments in real life?

A ray

A ray is a line with a fixed starting point but has no end point. A ray is formed when a line segment is drawn and extended at one of its end points and marked with an arrow. Examples of rays are as shown in the following diagram:



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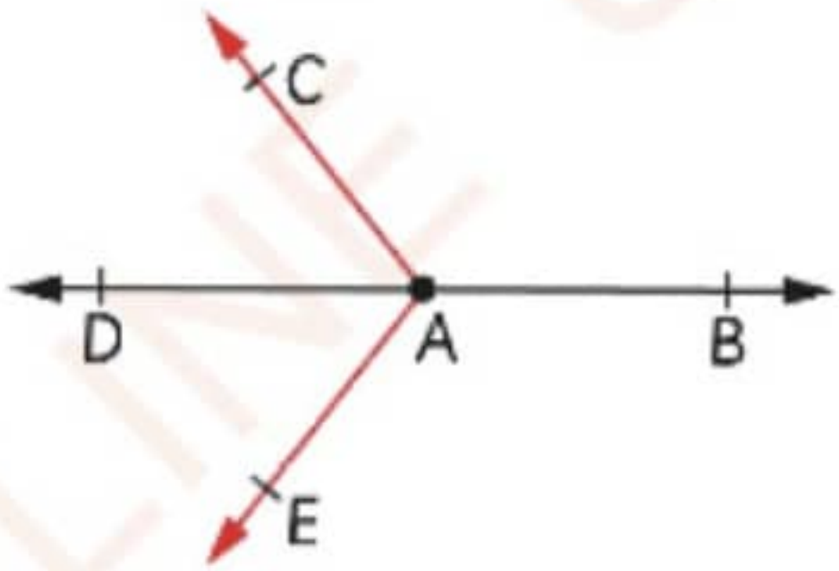
A ray is represented by the symbol $\rightarrow$ on the top of its notation. For example, the ray AB is written as $\overrightarrow{AB}$, and the ray BA is written as $\overrightarrow{BA}$.

Activity 3: Drawing a ray

1. Use a ruler to draw a line segment AB.
2. Extend the line at any point and put an arrow-head to show the direction.
3. Repeat tasks 1 and 2 to draw three more rays and name them.

Example 1

What is the endpoint of a ray AC?



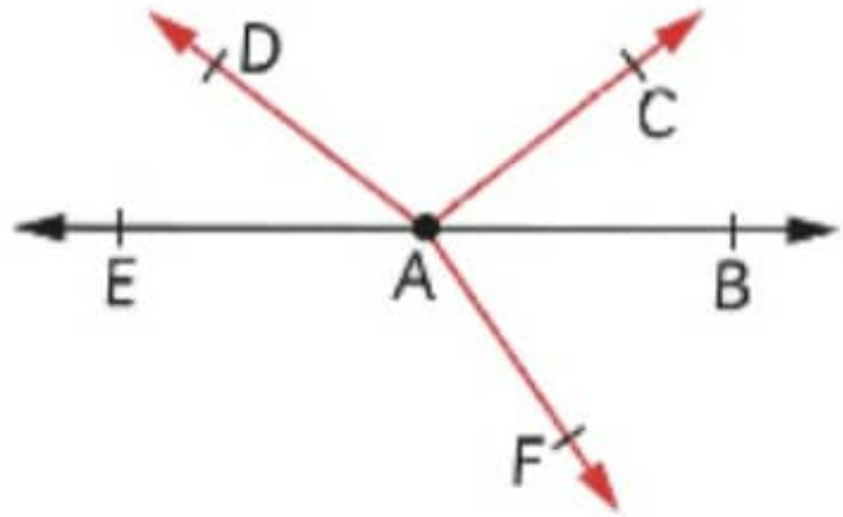
Answer

Since the point of origin of a ray is also called its endpoint. Therefore, the end point of the ray AC is point A.

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Example 2

Which rays are opposite to each other in the following figure?

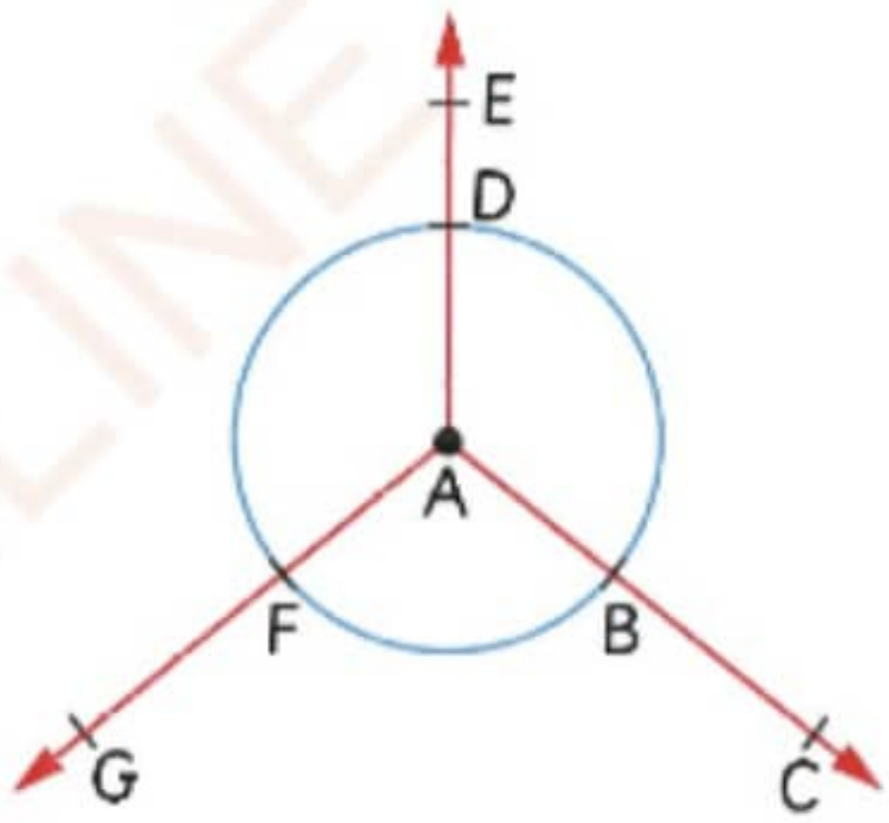


Answer

Ray AB and ray AE are opposite rays in the given figure. The rays start from point A and proceed in the opposite direction to form a straightline EB.

Example 3

Write the names of all rays formed from the following figure:



Answer

From the given figure, the rays are:

$\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{BC}$, $\overrightarrow{AD}$, $\overrightarrow{AE}$, $\overrightarrow{AF}$, $\overrightarrow{AG}$ and $\overrightarrow{FG}$

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Activity 4: Drawing a straight line

1. Use a ruler to draw two rays, one pointing towards the left and the other towards the right.
2. Connect the initial points of the two rays in task 1 to get a straight line.

Exercise 2

1. Draw and label two line segments.
2. Draw and label three straight lines.
3. State the diference between a straight line and a line segment.
4. Can you draw a 4 cm line segment without a ruler? Give reasons.
5. Draw rays XY and YX.
6. Give the difference between a line segment and a ray.
7. Which instrument can be used to measure a line segment of length 5 cm?
8. Determine the number of line segments in the following line.



9. Draw a line segment PQ and state the name of the end points.
10. Join three line segments AB, BC, and CA. Then, state the name of the figure formed.

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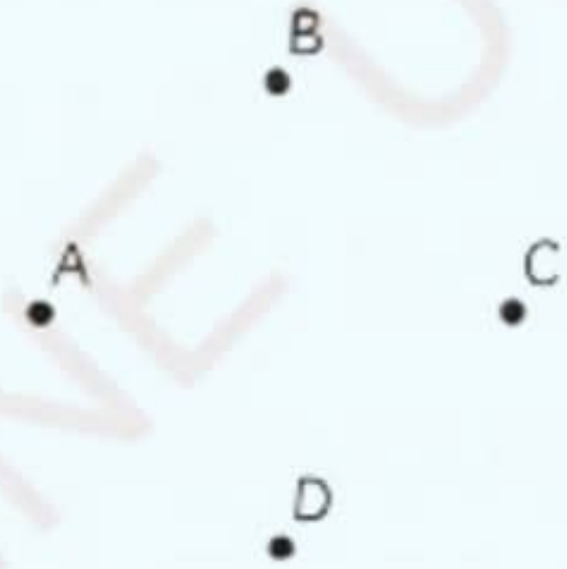
11. How many rays can you draw from a single point?
12. Give at least three examples that can be considered to represent a ray in our daily life.

Constructing plane figures

A plane figure is a two-dimensional figure which has only length and width. It has no thickness or depth. All plane figures have flat shapes and exist on flat surfaces. There are many different types of plane figures, including squares, rectangles, triangles, circles, and ovals.

Activity 5: Constructing plane figures using points

1. Study the following points.



2. Use a pencil and a ruler to connect points A and B, C and D, A and D, and D and C.
3. Measure and record the lengths of line segments AB, AD, CD and BC.
4. Name the plane figure formed in task 2.

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Exercise 3

1. How many line segments are there in a triangle?
2. Connect the points E, F, G, and H to form the following plane figures:
   (a) Two triangles.
   (b) Four triangles.



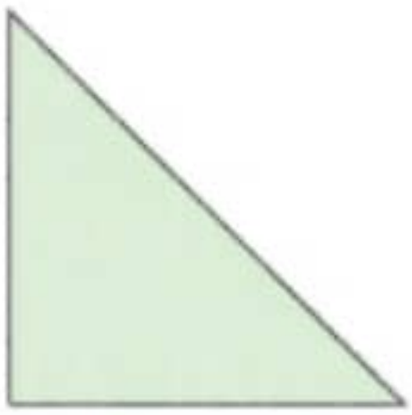
3. How many line segments does a kite have?
4. How many line segments are in the following figure?



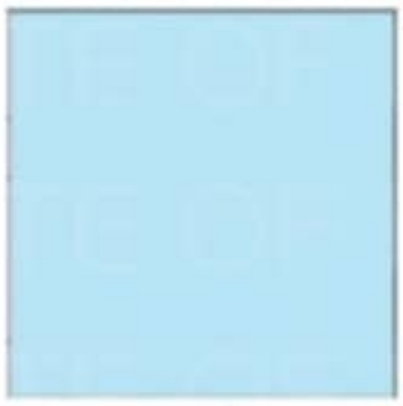
Names of plane figures

Plane figures are named according to their shapes, angles or sides. The following are examples of plane figures and their names:

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Triangle



Square



Rectangle



Circle



Pentagon



Hexagon

A square and a rectangle are plane figures which have four sides each. Their difference is that all the sides of the square have equal length, while in a rectangle, only the pairs of opposite sides have equal lengths.

Activity 6: Drawing plane figures by using ICT

1. Identify a computer programme for drawing plane figures.
2. Use the programme in task 1 to draw squares, triangles, circles, and rectangles of different sizes.
3. Draw five plane figures with at least five sides, and name them.
4. Draw more plane figures to enhance your competence.

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Example

Classify the shape of each of the following traffic sign as a plane figure:

(a)



(b)



(c)



(d)



Answer

(a) There are 8 sides; it is an octagon.

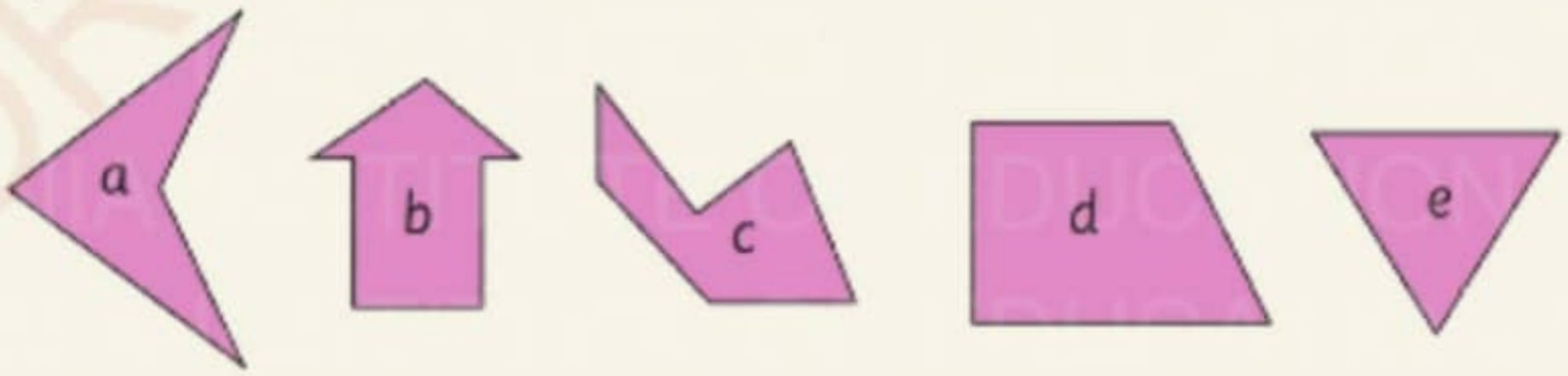
(b) There are 4 sides; it is a rectangle (or quadrateral).

(c) There are 3 sides; it is a triangle.

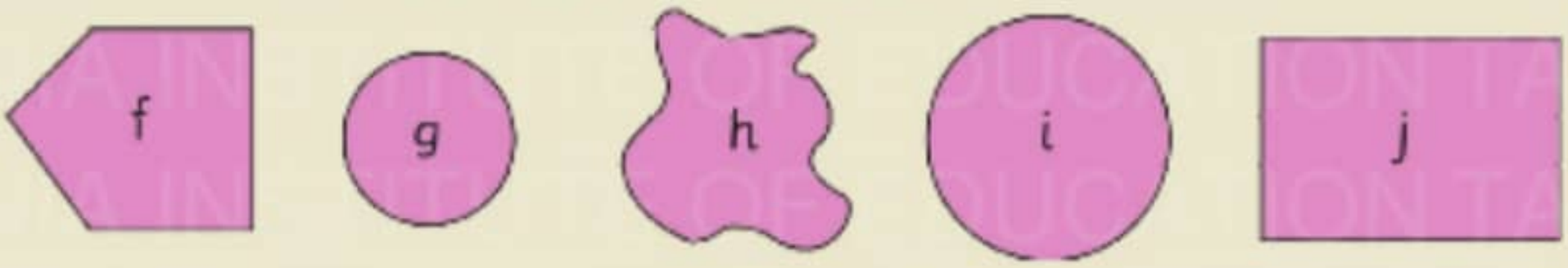
(d) It is a circle.

Exercise 4

1. Carefully study the plane figures a to j and answer the questions that follow:



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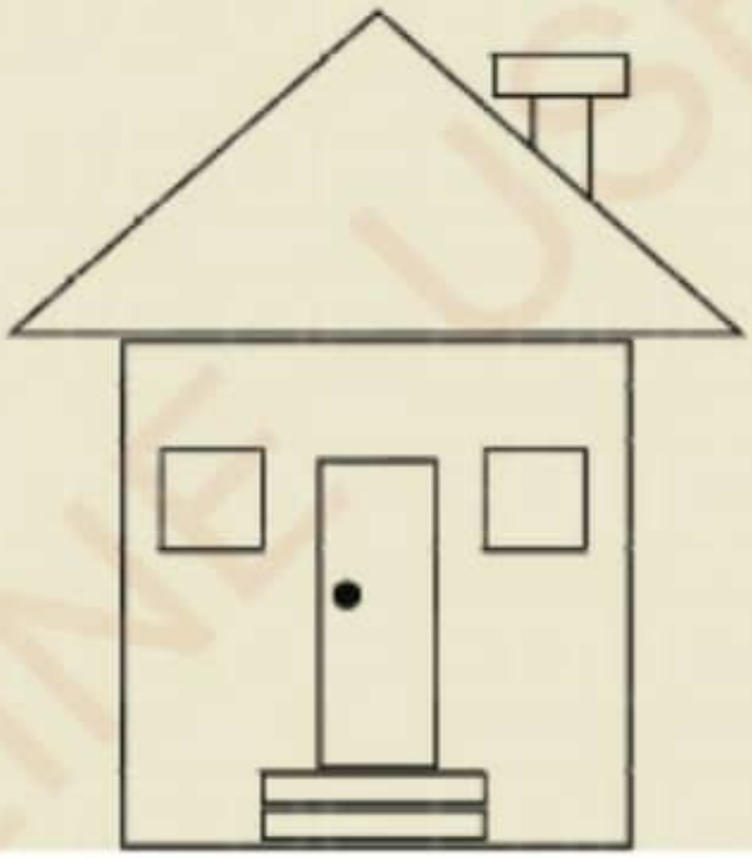
(a) Draw all the figures with four sides.

(b) How many figures are triangles?

(c) Write down the letters labelled in the circles.

(d) Make the figures identified with letters labelled a, b, d, e, and f by cutting a piece of paper or manila card.

2. The following diagram shows a house made of plane figures. Draw all the plane figures found in the diagram:



Activity 7: Recognising plane figures by using ICT

Use online programmes to identify different types of plane figures and match them with real objects.

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Summary

1. Two dimensional figures are such as squares, rectangles, triangles, and hexagons.
2. A square has sides which are equal in length, while a rectangle has pairs of opposite sides with equal length.
3. A line segment is drawn by joining two points.
4. A ray has an initial point and extends one side infinitely.

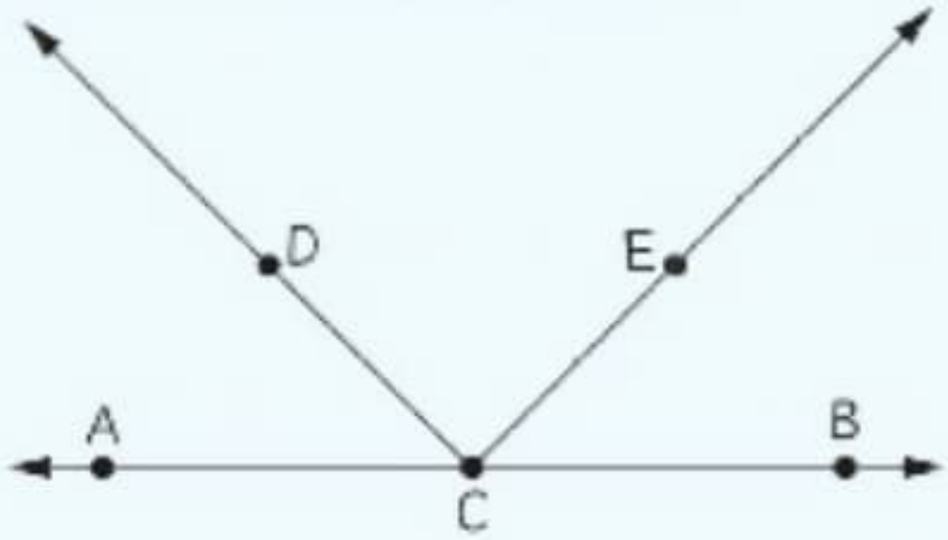
Vocabulary

Hexagon	a plane figure enclosed by six sides
Rectangle	a four-sided figure whose opposite sides are equal in length
Square	a four-sided figure whose sides are equal in length

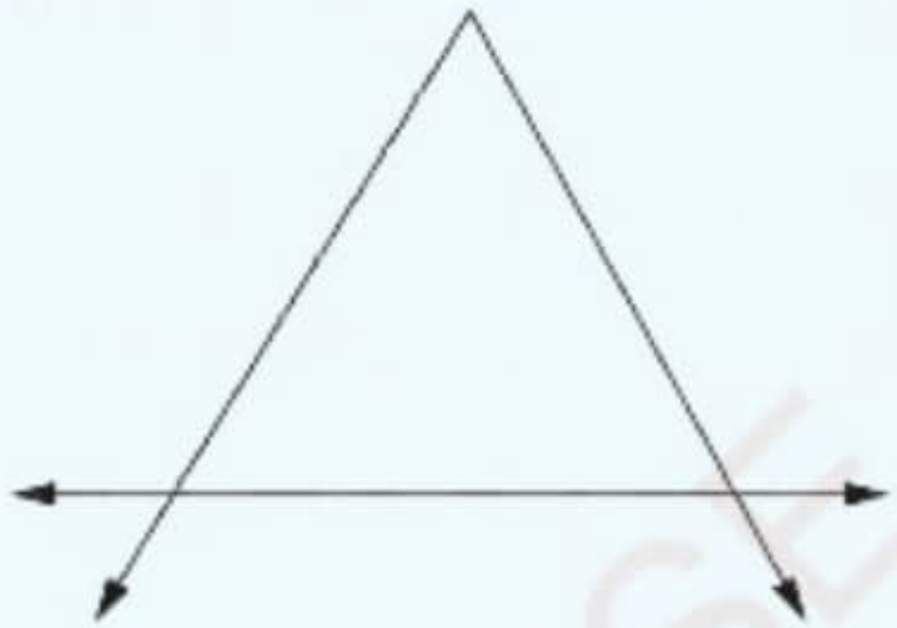
Revision exercise

1. Draw and label two line segments of length 4 cm and 5 cm, respectively.
2. Draw a ray PQ.
3. How many ends does a ray have?
4. Write the names of the rays found in the following figure.

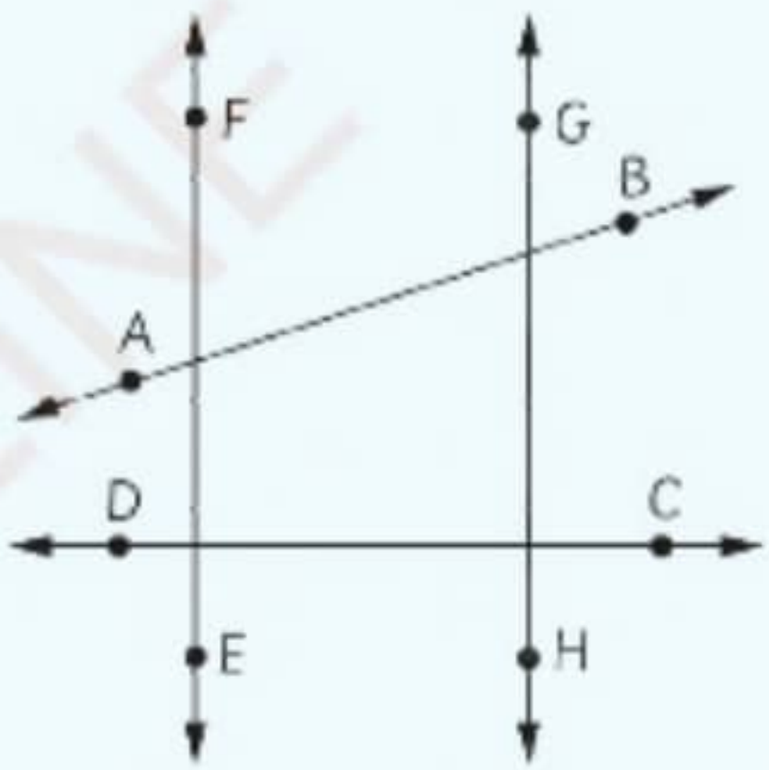
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5. How many straight lines does the following figure have?



6. Write down the names of the straight lines found in the following figure.



7. How many angles do the following figures have?
   (a) Triangle (b) Pentagon
   (c) Hexagons

8. Draw any two plane figures.

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9. Which of the following figures are not rectangular? Give reasons for your answer:

(a)



(b)



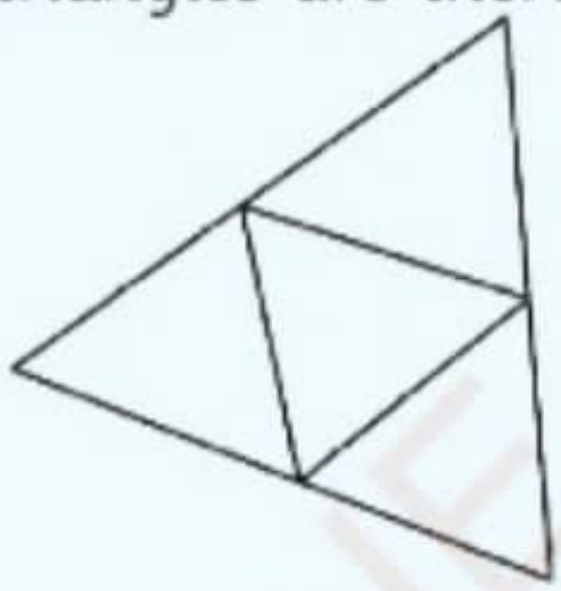
(c)



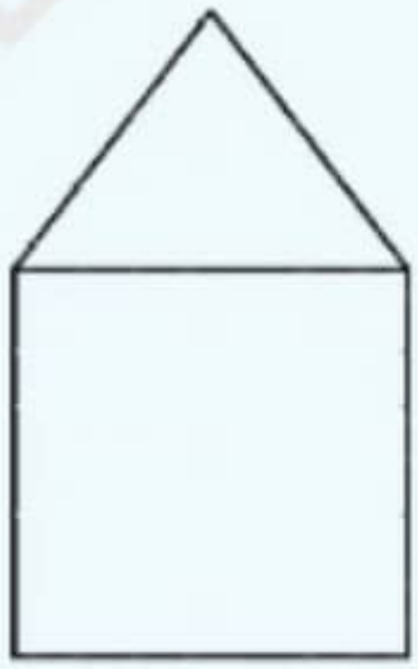
(d)



10. How many triangles are there in the following figure?



11. Study the following diagram, and then answer the questions that follow:

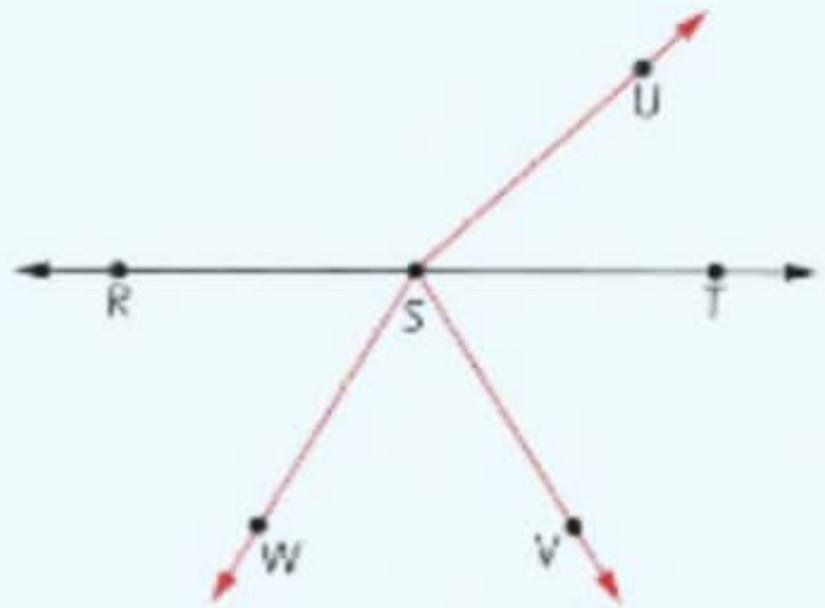


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(a) How many line segments are in the figure?

(b) Name all plane figures found in the diagram.

12. Use the following figure to answer the questions that follows:



(a) Name all points.

(b) Name all the line segments.

(c) Name all the rays.

(d) Name a straight line in three ways.

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Chapter Five
Fractions

Introduction

In this chapter you will divide objects or group of objects to learn fractions. In addition, you will add and subtract fractions with the same denominator. The competencies developed will enable you to divide different things such as medicine, food and money as well as adding or subtracting fractions.

Recognising fractions

A fraction is a part of a thing that makes up a whole thing. When you cut an orange into slices and share it with your friends, each piece is called a fraction of the orange. Examples of fractions are $\frac{1}{2}$, $\frac{1}{3}$, or $\frac{2}{5}$. The upper number is called the numerator while the lower one is called the denominator.

Activity 1: Recognising fractions

Steps

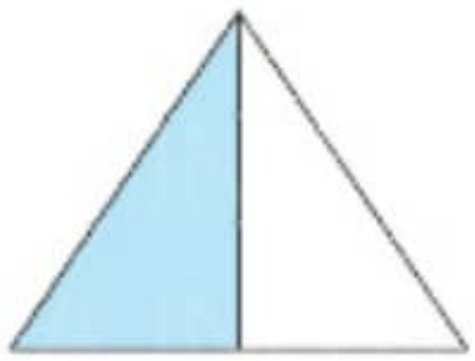
1. Take a knife, marker pen, and an orange or watermelon.
2. Use a knife to cut the orange or watermelon into several equal pieces. Start with halves or quarters.
3. Count the number of slices obtained in step 2.
4. Use a marker pen to label each slice with its corresponding fraction.
5. Repeat steps 2, 3, and 4 to cut an orange into different fractions. For example cut it into quarters, thirds or fifths.

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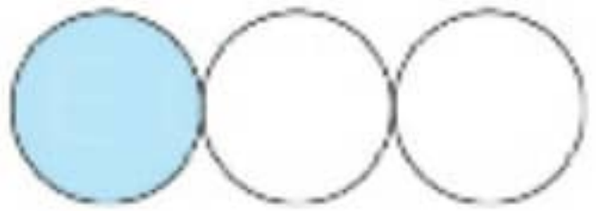
Example 1

Write in fraction the shaded part of each figure.

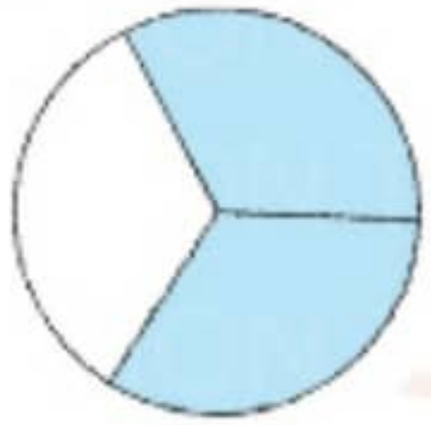
(a)



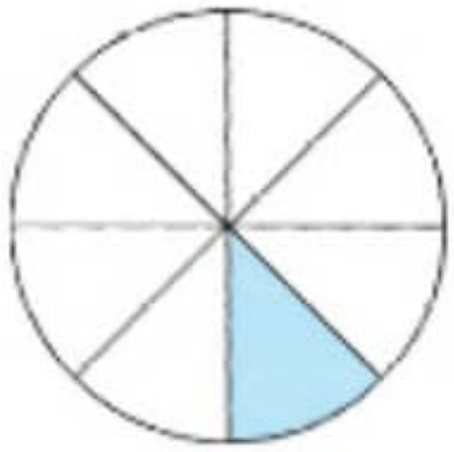
(b)



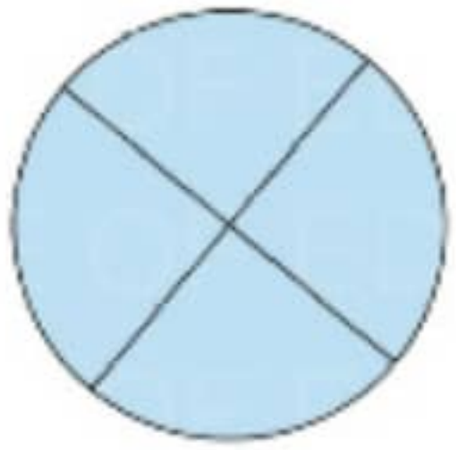
(c)



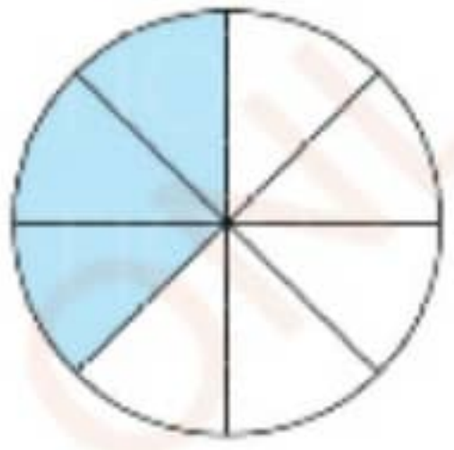
(d)



(e)



(f)



Answer

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{8}$ (e) $\frac{4}{4}$ (f) $\frac{3}{8}$

Example 2

A mother gave mangoes to her three children as follows: Rashid got 2 mangoes, Irene got 2 mangoes, and Likweme got 2 mangoes. What fraction of mangoes did Rashid get?

Solution

Rashid got 2 mangoes

Irene got 2 mangoes

Likweme got 2 mangoes

Total mangoes: $2 + 2 + 2 = 6$

Therefore, Rashid got $\frac{2}{6}$ of the mangoes.

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Exercise 1

1. Draw a circle and then divide it and shade it to show the following fractions:

   (a) $\frac{1}{8}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$

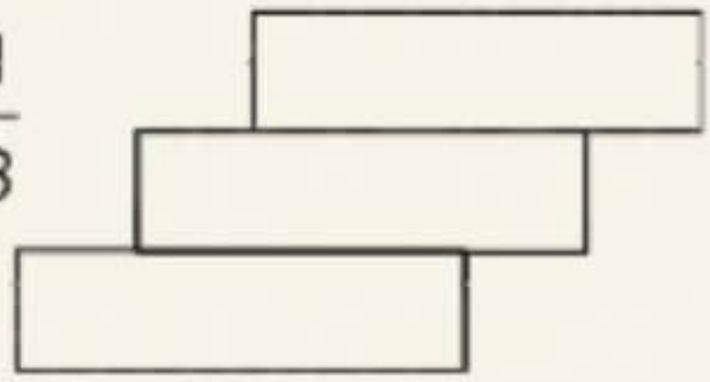
2. How many $\frac{1}{3}$ are there in a whole thing?

3. How many $\frac{1}{5}$ are there in a whole thing?

4. Mussa cut an orange into two equal pieces as shown in the following figure. He gave one piece to his sister Neema. What fraction of the orange did each get?



5. Shade each figure from (a) to (e) to show the fraction indicated in each figure.

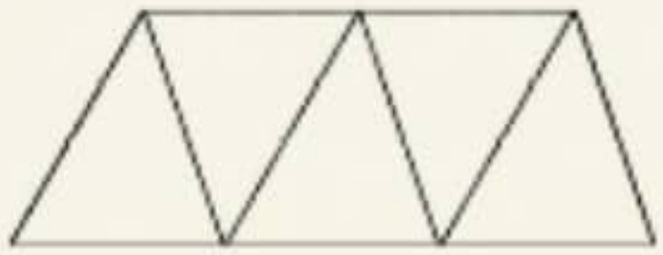
   (a) $\frac{2}{4}$  (b) $\frac{1}{3}$ 

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(c) $\frac{6}{8}$



(d) $\frac{3}{5}$

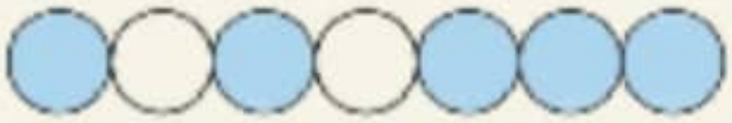


(e) $\frac{5}{12}$



6. Write the fractions which represent the shaded parts of the drawings (a) to (d).

(a)



(b)



(c)



(d)



7. Musa divided a soap bar into four equal pieces. He gave one piece to his sister. What fraction of the bar did he give to his sister?

8. What is the fraction of two goats out of six goats?

9. There are eight mangoes, two of them are rotten. What is the fraction of the rotten mangoes?

10. A farmer picked six papayas and wants to distribute them equally to his three children. What fraction of all the papayas will each child get?

11. Selemani had one papaya. He decided to divide the papaya into three equal pieces with his two friends. What part of the papaya did each of them get?

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12. Sadiki has twenty bananas. He wants to distribute the bananas to his four children equally. What fraction of all the bananas will each child get?

Addition of fractions with the same denominator

To add fractions with the same denominator simply add the numerators and keep the denominator the same.

Example 1

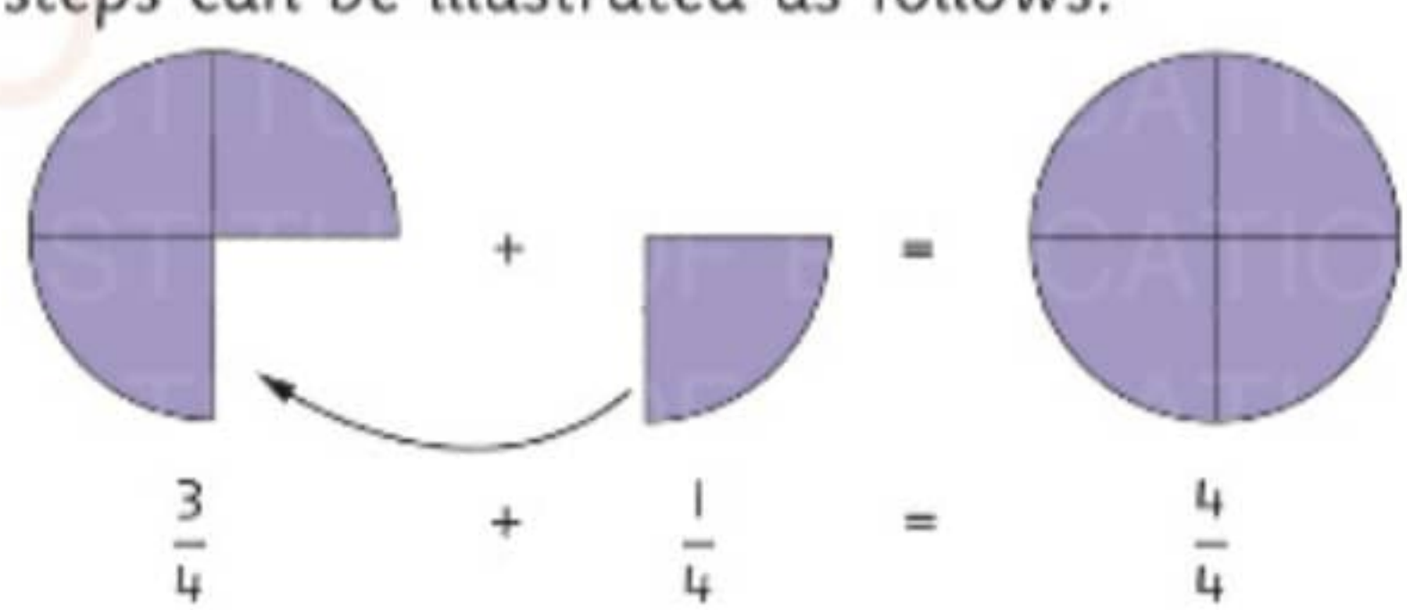
Use drawings to answer the following question.

$$\frac{3}{4} + \frac{1}{4} =$$

Steps

1. Draw a shaded three-quarters of a circle.
2. Write in numerals the three-quarters of the circle.
3. Draw and shade a quarter of the circle.
4. Write the quarter of a circle in numerals.
5. Add the shaded parts of the circle.
6. Draw a shaded whole circle to represent the sum of three-quarters and one-quarter.
7. Write the fraction of the whole circle.

The previous steps can be illustrated as follows:



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The answer is one shaded circle divided into four equal parts. To add fractions with the same denominator, add the numerators as you add ordinary numbers. The denominator does not change because the parts make up one thing. Thus, $\frac{3}{4}+\frac{1}{4}=\frac{3+1}{4}=\frac{4}{4}$

Therefore, $\frac{3}{4}+\frac{1}{4}=\frac{4}{4}$.

Example 2

$\frac{13}{27}+\frac{7}{27}=$

Solution

Add the numerators as you usually add numbers; the denominator does not change.

Thus, $\frac{13}{27}+\frac{7}{27}=\frac{13+7}{27}=\frac{20}{27}$.

Therefore, $\frac{13}{27}+\frac{7}{27}=\frac{20}{27}$.

Exercise 2

1. Answer the following questions without using drawings:

(a) $\frac{2}{5}+\frac{2}{5}=$ (b) $\frac{3}{9}+\frac{5}{9}=$ (c) $\frac{1}{4}+\frac{1}{4}=$

(d) $\frac{3}{6}+\frac{2}{6}=$ (e) $\frac{1}{5}+\frac{2}{5}=$ (f) $\frac{5}{8}+\frac{1}{8}=$

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(g) $\frac{3}{10}+\frac{5}{10}=$ (h) $\frac{1}{7}+\frac{5}{7}=$ (i) $\frac{1}{2}+\frac{1}{2}=$

(j) $\frac{2}{6}+\frac{4}{6}=$ (k) $\frac{1}{4}+\frac{2}{4}=$ (l) $\frac{1}{15}+\frac{6}{15}=$

(m) $\frac{3}{9}+\frac{4}{9}=$ (n) $\frac{3}{7}+\frac{3}{7}=$ (o) $\frac{0}{5}+\frac{2}{5}=$

(p) $\frac{1}{3}+\frac{1}{3}=$ (q) $\frac{2}{7}+\frac{4}{7}=$ (r) $\frac{11}{20}+\frac{7}{20}=$

(s) $\frac{1}{10}+\frac{3}{10}=$ (t) $\frac{16}{35}+\frac{9}{35}=$

2. *Answer the following questions by using drawings:*

(a) $\frac{2}{4}+\frac{0}{4}=$ (b) $\frac{1}{3}+\frac{1}{3}=$

(c) $\frac{3}{8}+\frac{1}{8}=$ (d) $\frac{2}{5}+\frac{1}{5}=$

(e) $\frac{4}{6}+\frac{2}{6}=$ (f) $\frac{5}{7}+\frac{1}{7}=$

(g) $\frac{2}{10}+\frac{5}{10}=$ (h) $\frac{4}{9}+\frac{3}{9}=$

(i) $\frac{5}{15}+\frac{7}{15}=$ (j) $\frac{6}{20}+\frac{3}{20}=$

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Word problems involving addition of fractions

Example 1

Yona filled $\frac{3}{9}$ of a tank with water, and Halima filled $\frac{2}{9}$ of the tank. What fraction of the tank did they fill?

Solution

Yona filled $\frac{3}{9}$ of the tank.

Halima filled $\frac{2}{9}$ of the tank.

Both of them filled: $\frac{3}{9}+\frac{2}{9}=\frac{3+2}{9}$

$=\frac{5}{9}$

Therefore, both Yona and Halima filled $\frac{5}{9}$ of the tank with water.

Example 2

A household had one bag of sugar, Anna used $\frac{3}{8}$ of the sugar and John used $\frac{4}{8}$ of the sugar in the bag. What fraction of the sugar was used by both Anna and John?

Solution

Anna used $\frac{3}{8}$ of the sugar and John used $\frac{4}{8}$ of the sugar.

In total they used: $\frac{3}{8}+\frac{4}{8}=\frac{3+4}{8}$

$=\frac{7}{8}$

Therefore, Anna and John used $\frac{7}{8}$ of the sugar.

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Exercise 3

1. A mother gave her son a quarter of chapati. Later, she gave him three quarters of the chapati. What fraction of chapati was given to the son?

2. Masanja ate $\frac{1}{6}$ of a loaf bread and Amani ate $\frac{2}{6}$ of the same loaf. What fraction of the bread did they eat?

3. A farmer divided his field into 8 equal parts. He used $\frac{4}{8}$ of the field to plant maize and $\frac{3}{8}$ to plant beans. What part of the field did he plant maize and beans?

4. Villagers sold $\frac{2}{5}$ of the cotton to their cooperative union, and $\frac{3}{5}$ of the cotton to another cooperative union. What fraction of the cotton did the villagers sell?

5. Tausi spent $\frac{1}{7}$ of her salary to pay taxes. She also spent $\frac{3}{7}$ of her salary on food. What fraction of her salary did she spend?

6. Maunda bought two fish. The first fish weighed $\frac{2}{4}$ kilogram and the other weighed $\frac{2}{4}$ kilogram. Find the total mass of the fish.

7. Residents of Maendeleo Village planned to build a health centre. The villagers produced $\frac{5}{10}$ of the required bricks and

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$\frac{3}{10}$ of the bricks were provided by the village chairperson. What fraction of the required bricks does the village have?

8. Nuru sold $\frac{3}{10}$ of the oranges on the first day and $\frac{6}{10}$ on the second day. What fraction of the oranges did he sell in two days?

9. The school bought books. Among the books, $\frac{4}{12}$ were Mathematics, $\frac{2}{12}$ Kiswahili, and $\frac{3}{12}$ Science and Technology. What fraction of the books was bought for the three subjects?

10. Jane's mother bought a cake for her birthday. She divided it into six equal pieces and gave two pieces to Jane. What fraction of the cake was given to Jane?

Subtraction of fractions with the same denominator

To subtract fractions with the same denominator, subtract the numerators and keep the denominator the same.

Example 1

Use drawings to find the correct answer to the following question:

$$\frac{3}{3} - \frac{1}{3} =$$

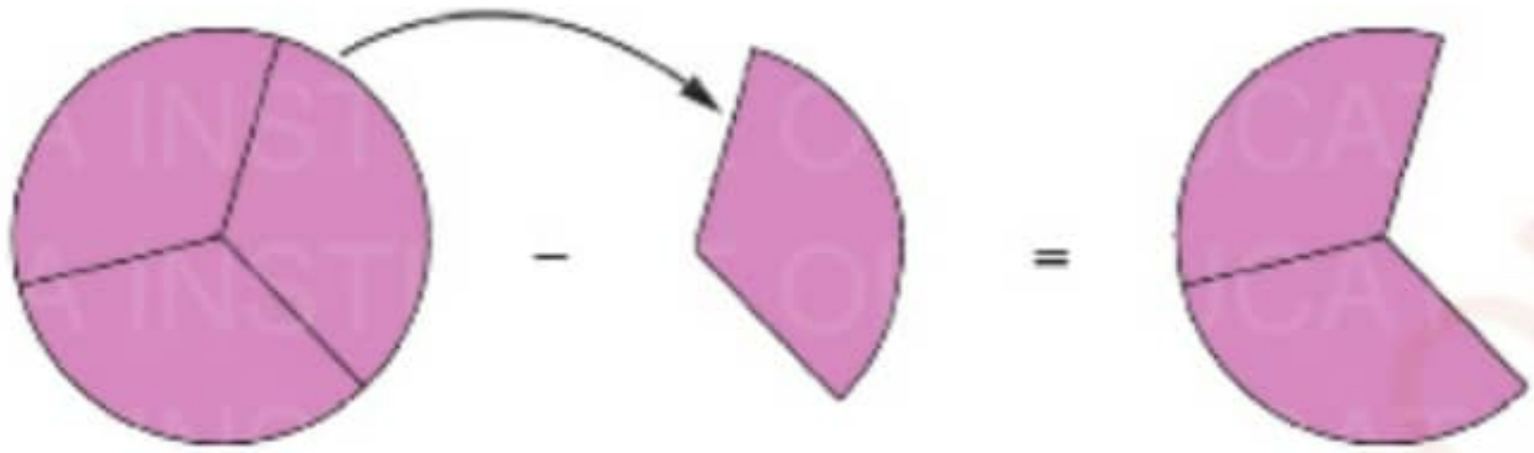
Steps

1. Draw a circle and divide it into three equal parts.
2. Shade the three parts of the circle in step 1. Write the fraction of the shaded part in numerals.

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3. Remove one shaded part of the circle in step 2. Write its fraction in numerals.
4. Draw the circle again to show the remaining shaded part after removing one part in step 3.
5. Write the fraction of the remained shaded part in step 4.

The previous steps can be illustrated as follows:



$\frac{3}{3}$ – $\frac{1}{3}$ = $\frac{2}{3}$

To subtract fractions with the same denominator subtract the numerators as you subtract ordinary numbers. The denominator does not change because the parts make up one thing.

Thus, $\frac{3}{3}-\frac{1}{3}=\frac{3-1}{3}=\frac{2}{3}$.

Therefore, $\frac{3}{3}-\frac{1}{3}=\frac{2}{3}$.

Example 2

Use drawings to find the answer to the following question:

$\frac{5}{6}-\frac{1}{6}=$

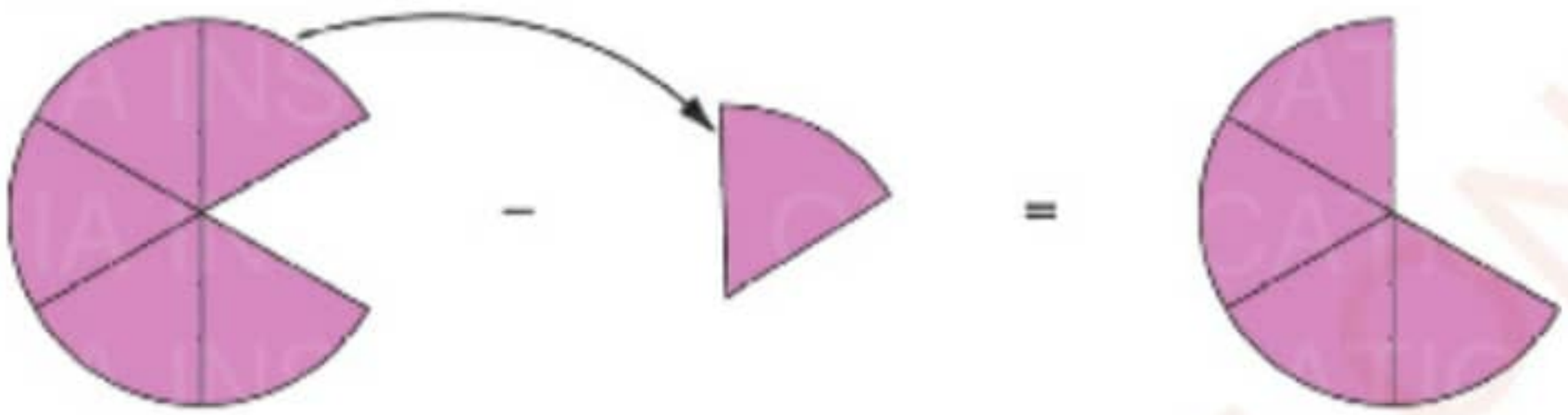
Steps

1. Draw and shade five out of six parts of a circle.
2. Write the fraction of the shaded parts in numerals.

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3. Remove one out of six parts of the circle.
4. Write in numerals the part removed.
5. Draw and shade four out of six of the remaining parts of the circle.
6. Write in numerals the fractions of the remained circle.

The previous steps can be illustrated as follows:



$\frac{5}{6}$ − $\frac{1}{6}$ = $\frac{4}{6}$

$$\frac{5}{6}-\frac{1}{6}=\frac{5-1}{6}=\frac{4}{6}$$

Therefore, $\frac{5}{6}-\frac{1}{6}=\frac{5-1}{6}=\frac{4}{6}$.

Example 3

Find the value of $\frac{11}{12}-\frac{7}{12}=$

Solution

Subtract numerators as you perform usual subtraction of numbers. The denominator does not change.

$$\frac{11}{12}-\frac{7}{12}=\frac{11-7}{12}=\frac{4}{12}$$

Therefore, $\frac{11}{12}-\frac{7}{12}=\frac{11-7}{12}=\frac{4}{12}$.

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Exercise 4

1. $\frac{5}{8} - \frac{3}{8} =$
2. $\frac{2}{5} - \frac{1}{5} =$
3. $\frac{4}{9} - \frac{1}{9} =$
4. $\frac{6}{7} - \frac{4}{7} =$
5. $\frac{2}{3} - \frac{1}{3} =$
6. $\frac{14}{14} - \frac{1}{14} =$
7. $\frac{3}{6} - \frac{1}{6} =$
8. $\frac{7}{8} - \frac{5}{8} =$
9. $\frac{6}{13} - \frac{3}{13} =$
10. $\frac{3}{5} - \frac{2}{5} =$
11. $\frac{4}{5} - \frac{1}{5} =$
12. $\frac{5}{7} - \frac{3}{7} =$
13. $\frac{2}{6} - \frac{2}{6} =$
14. $\frac{2}{2} - \frac{1}{2} =$
15. $\frac{5}{10} - \frac{2}{10} =$
16. $\frac{4}{4} - \frac{2}{4} =$
17. $\frac{5}{11} - \frac{1}{11} =$
18. $\frac{16}{50} - \frac{5}{50} =$
19. $\frac{24}{25} - \frac{20}{25} =$
20. $\frac{75}{100} - \frac{50}{100} =$
21. $\frac{35}{40} - \frac{25}{40} =$
22. $\frac{41}{75} - \frac{23}{75} =$
23. $\frac{17}{20} - \frac{9}{20} =$
24. $\frac{75}{100} - \frac{25}{100} =$
25. $\frac{24}{40} - \frac{20}{40} =$

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Word problems involving subtraction of fractions

Example 1

Five-eighths of a farm is planted eucalyptus trees. The rest of the farm is planted orange trees. What part of the farm is planted orange trees?

Solution

The whole farm is divided into 8 parts; same as $\frac{8}{8}$.

The area planted eucalyptus trees is $\frac{5}{8}$.

The area planted orange trees = whole farm − the area planted eucalyptus trees.

$$= \frac{8}{8} - \frac{5}{8}$$

$$= \frac{8-5}{8} = \frac{3}{8}$$

Therefore, the part of the farm planted oranges trees is $\frac{3}{8}$.

Example 2

Katu cleaned four-twelfths of his classroom. What part of the classroom was not cleaned?

Solution

The whole class has 12 parts; same as $\frac{12}{12}$.

The cleaned part is $\frac{4}{12}$.

The part that was not cleaned = the whole class − the part that was cleaned.

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$$= \frac{12}{12} - \frac{4}{12}$$

$$= \frac{12-4}{12}$$

$$= \frac{8}{12}$$

Therefore, the part of the class that was not cleaned is $\frac{8}{12}$.

Exercise 5

1. *Half of Ngosha's farm is cultivated. What fraction of his farm is not cultivated?*
2. *Aisha harvested oranges from her farm. If a quarter of the oranges were spoiled, what part of the oranges were not spoiled?*
3. *Bahati divided a watermelon into 10 equal pieces. If 7 out of the 10 pieces were sold, what fraction of the watermelon was not sold?*
4. *Mariam ate one-sixths of a whole sugarcane. What fraction of the sugarcane remained?*
5. *Kandi had* $\frac{5}{9}$ *of a loaf of bread. He gave Ali* $\frac{4}{9}$ *of the bread. What part of the bread did he remain with?*
6. *Kabula answered* $\frac{7}{12}$ *of Mathematics questions in the morning and* $\frac{2}{12}$ *in the afternoon. What fraction of the questions remained?*

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7. Joyce harvested $\frac{3}{10}$ of the onions in a certain farm. Debora harvested $\frac{4}{10}$ of the onions from the same farm. What is the difference between the fraction of the onions harvested by Joyce and Debora?
8. Three quarters of Kwame's farm is adjacent to the main road. Which part of Kwame's farm is not adjacent to the main road?
9. Bakari used $\frac{1}{10}$ of his salary to pay house rent, and $\frac{2}{10}$ to pay water and electricity bills. What fraction of the salary did he remain with?
10. The villagers planted fruit tree seedlings. If $\frac{2}{7}$ of all the trees withered, what fraction of the trees flourished?

Activity 2: Visual representation of fractions

Description: Use basic drawing software to:

1. Create visual representations of simple fractions. Example $\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}$
2. Draw different shapes and divide them into equal parts. Shade and label the fractions.

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Summary

1. Some fractions of shapes have special names; for example, $\frac{1}{4}$ is called a quarter, $\frac{1}{2}$ is a half, and $\frac{1}{3}$ is a third.
2. The number above the division line is called numerator and the one below the line is called denominator.
3. Adding and subtracting fractions with the same denominator, the denominator does not change.

Vocabulary

Half	$= \frac{1}{2}$	part out of two equal parts
Quarter	$= \frac{1}{4}$	part out of four equal parts
Three quarters	$= \frac{3}{4}$	3 parts out of four equal parts
Third	$= \frac{1}{3}$	part out of three equal parts
Two thirds	$= \frac{2}{3}$	2 parts out of three equal parts

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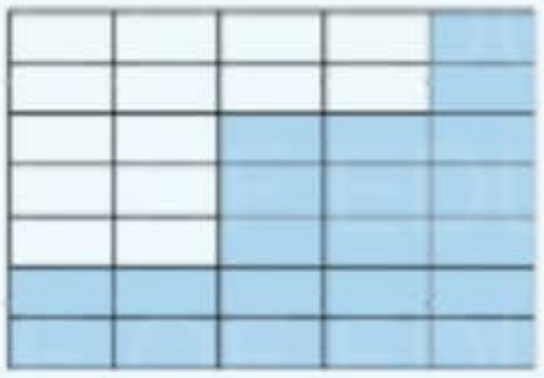
Revision exercise

1. $\frac{3}{9}+\frac{5}{9}=$
2. $\frac{2}{5}+\frac{2}{5}=$
3. $\frac{2}{6}+\frac{3}{6}=$
4. $\frac{3}{7}+\frac{3}{7}=$
5. $\frac{3}{6}-\frac{1}{6}=$
6. $\frac{3}{10}+\frac{5}{10}=$
7. $\frac{9}{16}+\frac{2}{16}=$
8. $\frac{3}{5}+\frac{2}{5}=$
9. $\frac{2}{6}-\frac{2}{6}=$
10. $\frac{4}{10}+\frac{5}{10}=$
11. $\frac{25}{25}-\frac{20}{25}=$
12. $\frac{35}{50}-\frac{10}{50}=$
13. $\frac{5}{8}-\frac{1}{8}=$
14. $\frac{6}{7}-\frac{4}{7}=$
15. $\frac{3}{25}+\frac{4}{25}=$
16. $\frac{6}{20}+\frac{11}{20}=$
17. $\frac{9}{15}+\frac{2}{15}=$
18. $\frac{8}{9}+\frac{1}{9}=$
19. $\frac{6}{7}-\frac{2}{7}=$
20. $\frac{30}{30}-\frac{15}{30}=$
21. $\frac{13}{50}+\frac{30}{50}=$
22. $\frac{40}{100}-\frac{16}{100}=$
23. $\frac{15}{35}-\frac{6}{35}=$
24. Write in numerals the fraction which represent the shaded parts of the drawings (a), (b) and (c)

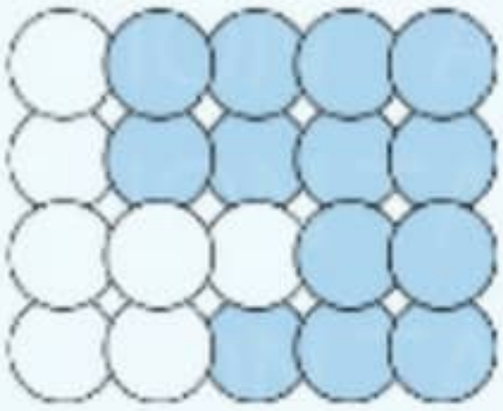
(a)



(b)



(c)



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25. Jacob bought two chickens. One weighed $\frac{2}{4}$ a kilogram and the other $\frac{2}{4}$ a kilogram. Find the total mass of the chickens.

26. One out of six of the tank was filled with water. What fraction of the tank was not filled with water?

27. John divided a pineapple into 20 equal pieces. If $\frac{17}{20}$ of the pineapple was sold, what part of the pineapple was not sold?

28. In certain school pupils in class three have blue and black bags. If $\frac{1}{3}$ of the pupils have blue bags, what fraction of the pupils have black bags?

29. A teacher has a piece of sugarcane with four equal parts. He wants to devide it among his four pupils. What fraction of the sugarcane will each pupil get?



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30. A mathematics quiz was given to standard three A pupils. If $\frac{35}{40}$ of the pupils did the quiz, what fraction of the pupils did not do the quiz?

31. Pupils cultivated $\frac{4}{10}$ of the garden on the first day and $\frac{5}{10}$ of the garden on the second day. What fraction of the garden was cultivated for two days?

32. Burhan has 8 apples. If $\frac{5}{8}$ of the apples were rotten, what fraction of the apple was not rotten?

33. Ali used $\frac{4}{7}$ of his money to buy clothes and $\frac{1}{7}$ for buying box. What fraction of his money remained?

34. Fatuma gave Rose $\frac{2}{5}$ of her candies, and Kabela $\frac{1}{5}$. What fraction of the candies was given to her friends?

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Chapter Six
Measurements of time

Introduction

In this chapter, you will learn about time measurements. You will read the time in 12-hour and 24-hour formats. You will also learn how to recognise the number of hours in a day, days in a week, weeks in a month, and months in a year. The competencies developed will enable you to manage time and be punctual in various scheduled activities. Examples of such activities include keeping a record of school, media and transport schedules.

Units of time

Time is measured in seconds, minutes, hours, days, weeks, months or years. The relationship between the units of time is as follows:

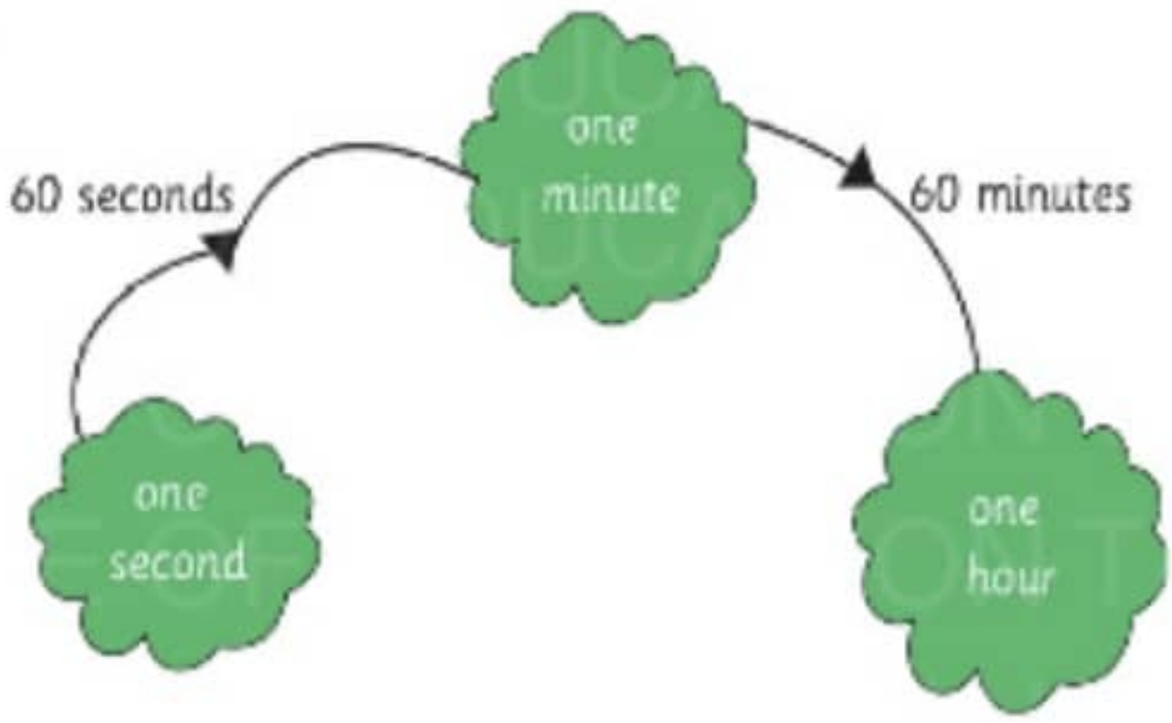
(a) There are 60 seconds in one minute.

(b) There are 60 minutes in one hour.

(c) There are 24 hours in one day.

(d) There are 7 days in one week.

(e) There are 12 months in one year.



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Tools for measuring time

Tools for measuring time are clocks, watches and calendars.

There are two types of clocks, namely analogue and digital.

A clock is used for measuring time based on the twenty-four hours of a day. A calendar is used for measuring days, weeks and months of a year.



Analogue watch



Digital clock



Calendar

Exercise 1

The following table shows the time spent by pupils in a game. Read the information and answer the questions that follow.

Name	Time
Musa	1 minute and 24 seconds
Neema	1 minute and 40 seconds
Sofia	103 seconds
Hassan	1 minute and 36 seconds

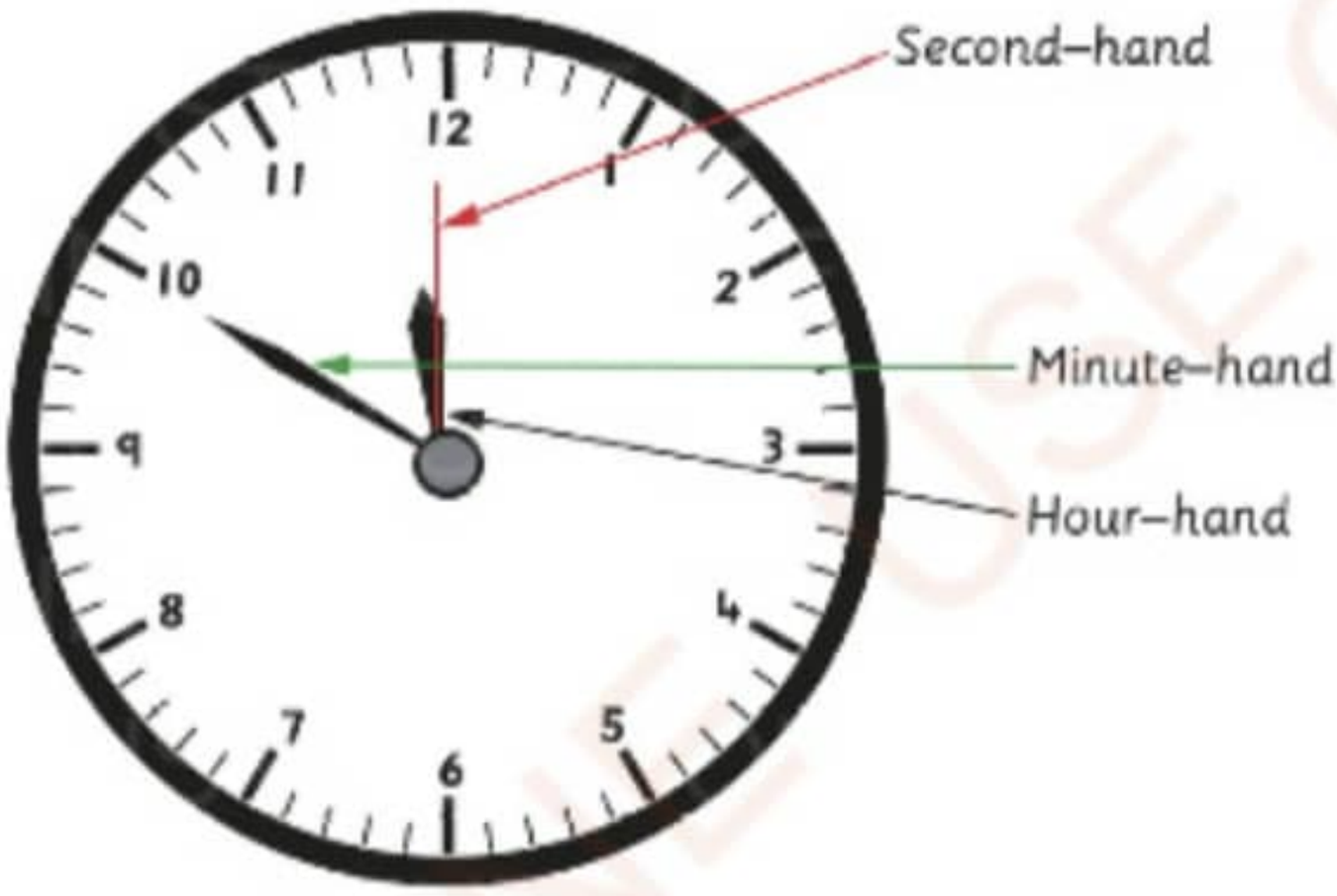
1. Who spent the longest time?
2. Who spent the least amount of time?
3. How many seconds did Neema spend?

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4. How many seconds did Musa spend?
5. How many seconds did Hassan spend?
6. How many minutes and seconds did Sofia spend?

Reading time in the 12-hour format

When reading an analogue clock, the shortest arrow is the hour-hand. The longer arrow is the minute-hand; while the longest and thinnest is the second-hand. All the clock arrows rotate in the clockwise direction.

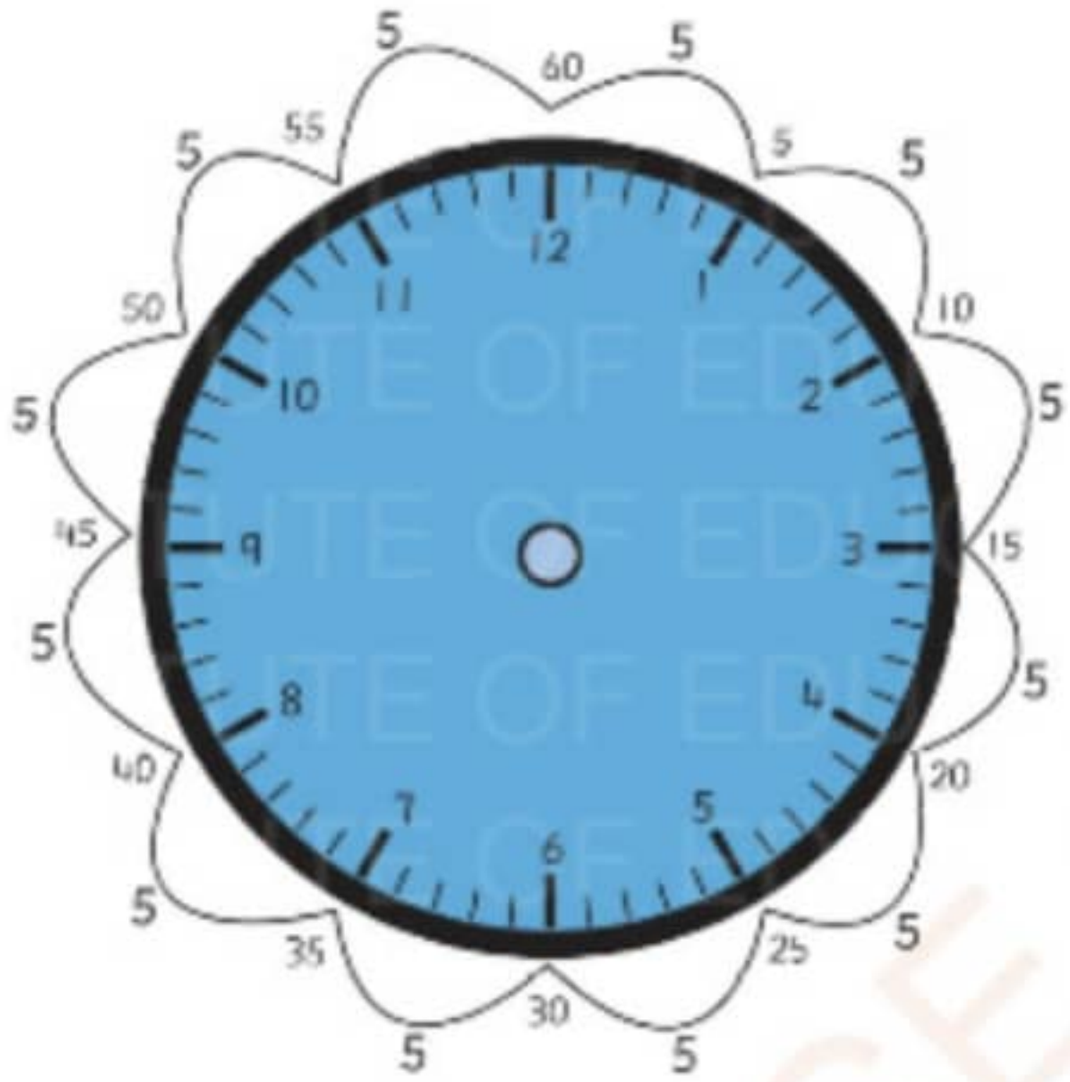


The face of a digital clock displays two pairs of digits. The pairs are separated by a colon (:). The first pair is for hours and the other for minutes. This is illustrated in the following figure.



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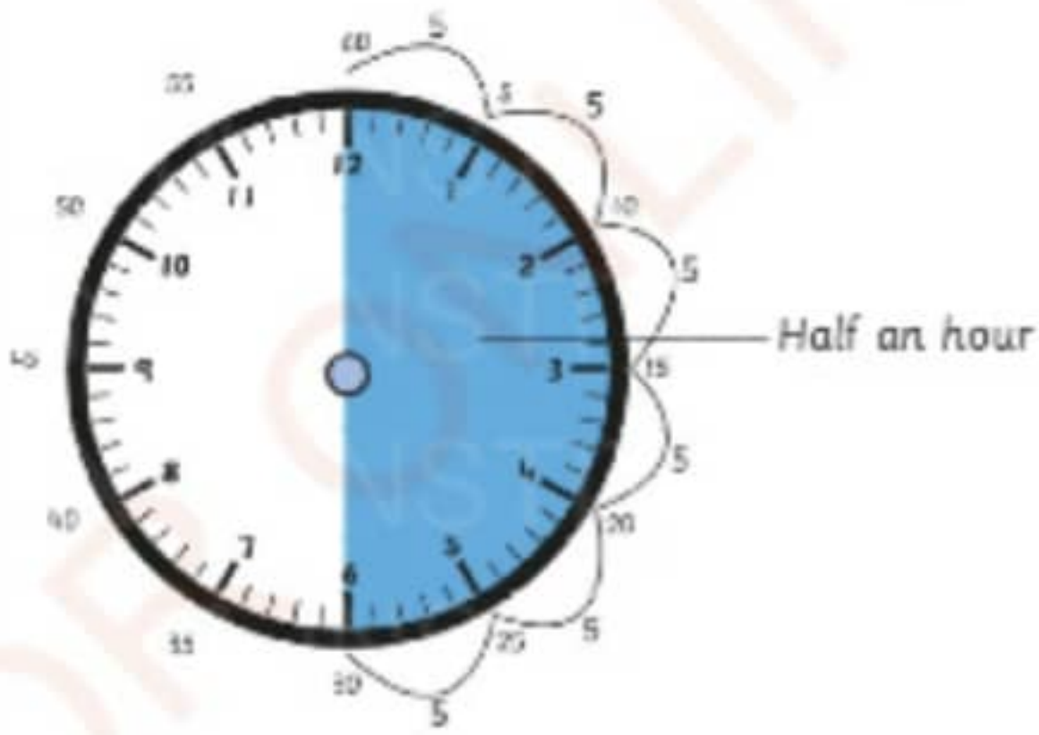
Study the following diagram, and learn how the clock face is divided.



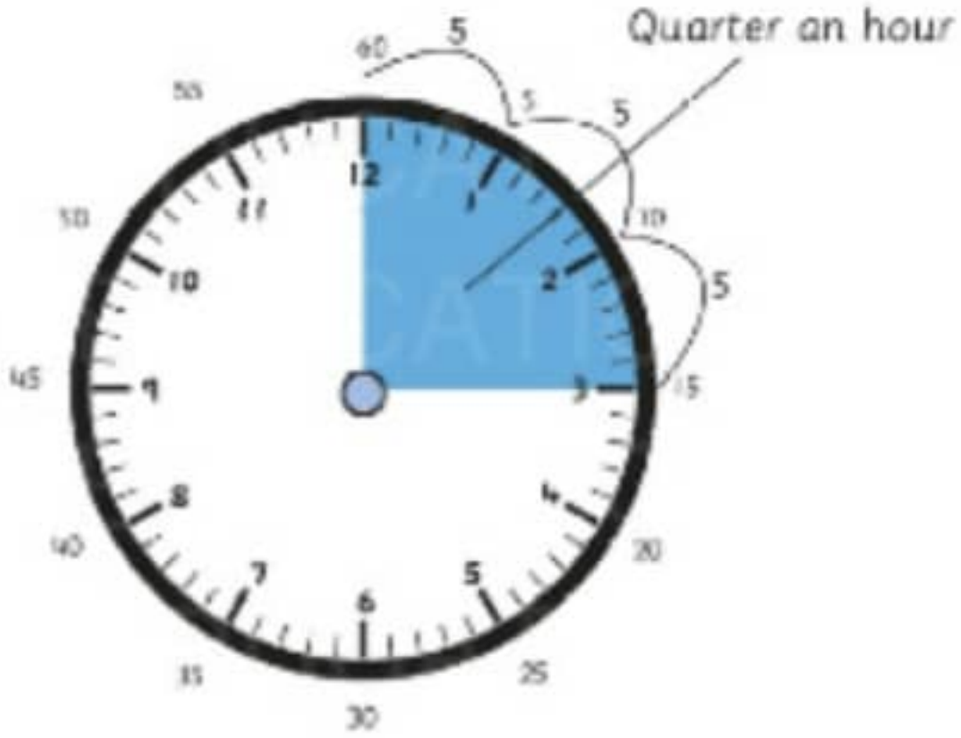
60 minutes represent one hour

The minute-hand takes five minutes to rotate from one number to the next on the clock face.

You can read 30 minutes as half an hour, and 15 minutes as quarter an hour. This is illustrated in the following figures.



30 minutes is half an hour



15 minutes is quarter an hour

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Example 1

Draw an analogue clock face that shows 04:00.

Steps:

1. Take a circular object like a 200 shilling coin.
2. Use the circular object and a pencil to draw a circle on the paper.
3. Divide the circle into 12 equal parts. Put twelve thick marks on the circle to form a clock face.
4. Write the numbers 1, 2, 3 up to 12 facing the thick marks you have placed as shown in the following figure.



5. Draw an hour hand arrow pointing at number 4 from the centre of the circle.
6. Draw a minute-hand arrow pointing at number 12 from the center of the circle.
7. Draw a second-hand arrow pointing number 12.



8. The clock face shows 4 O'clock.

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Reading the time in hours and minutes

Start reading hours followed by minutes. To read minutes, start at number 12 in the clockwise direction, until the minute-hand is reached.

Example 2

Draw an analogue clock face that shows 07:15.

Answer



Example 3

Study the following analogue and digital clock faces, then read the time.

(a)



01:00

(b)



02:00

(c)



06:00

Answer

(a) 1 O'clock (b) 2 O'clock (c) 6 O'clock

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Example 4

Read the following analogue clock faces with no numbers and digital clock faces with numbers. Write down the times shown.

(a)



01:15

(b)



10:30

(c)



08:45

Answer

(a) Fifteen minutes past one

(b) Thirty minutes past ten or half past ten

(c) Fourty-five minutes past eight

Example 5

Read the following clock faces and then write the time in numerals and in words.

(a)



(b)



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(c)



(d)



(e)



(f)



(g)



(h)



(i)



(j)



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Answer

(a) Ten minutes past 10 (10:10)

(b) Twenty minutes past 6 (06:20)

(c) Thirty-five minutes past 7 (07:35) or 25 minutes to 8

(d) Five minutes past 2 (02:05)

(e) Fourty-five minutes past 12 or quarter to 1 (15 minutes to 1)(12:45)

(f) Fourty-five minutes past 5 or quarter to 6 (15 minutes to 6) (05:45)

(g) Fourty-five minutes past 9 or quarter to 10 (15 minutes to 10)(09:45)

(h) Fifty minutes past 2 or 10 minutes to 3 (02:50)

(i) Fourty minutes past 6 or 20 minutes to 7 (06:40)

(j) Fifty-five minutes past 4 or 5 minutes to 5 (04:55)

Exercise 2

1. Study the following clock faces and then answer the questions that follow:

A.



B.



(a) Identify the type of clock face in A?

(b) What time does the clock face in B show?

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(c) What time does clock face in A show?

(d) What is the difference between clock faces in A and B?

2. Read the following clock faces and then write the time in numerals and in words.

(a)



(b)



(c)



(d)



(e)



(f)



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3. Draw arrows on the following clock faces to show the time indicated:

(a)



5 minutes past 7
(7:05)

(b)



35 minutes past 2
(2:35)

(c)



55 minutes past 6 or 5 minutes to 7
(6:55)

(d)



45 minutes past 2 or Quarter to 3
(2:45)

4. Write the following times in words:

(a) 08:45 _____

(b) 12:10 _____

(c) 11:40 _____

(d) 02:25 _____

(e) 04:35 _____

(f) 05:05 _____

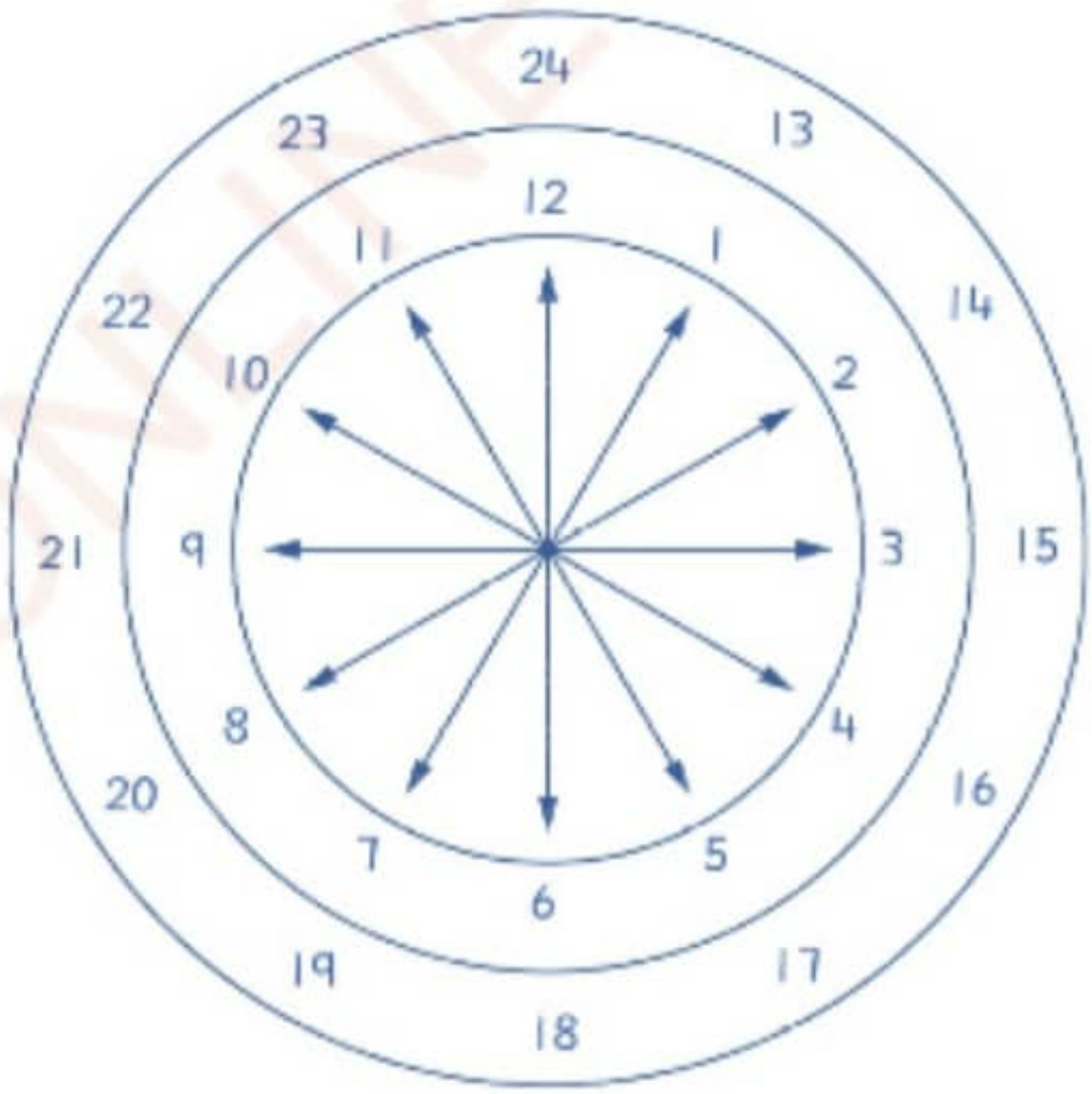
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5. Draw digital and analogue clock faces to show each of the following times:
   (a) Seven o'clock
   (b) Nine o'clock
   (c) Half past two
   (d) Forty five minutes past three

6. Write the following times in numerals:
   (a) Half past one _____
   (b) Twenty-five minites past four _____
   (c) Eleven o'clock _____
   (d) Forty-five minutes past six _____

Reading time in the 24-hour format

The following figure shows the 24-hour format that is not read on analogue clock faces.



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A day has 12 hours and a night has 12 hours. In the 24 hours format, a day starts after 12:00 midnight. The 24 hour format is written in four digits. The first two digits on the left indicate hours and the remaining digits indicate minutes.

The following table shows the relationship between time in 12-hour and 24-hour format.

In 12-hour format	In 24-hour format
12:00 midnight	0000
12:01 am	0001
1:00 am	0100
2:00 am	0200
3:00 am	0300
4:00 am	0400
5:00 am	0500
6:00 am	0600
7:00 am	0700
8:00 am	0800
9:00 am	0900
10:00 am	1000
11:00 am	1100
12:00 noon	1200
12:01 pm	1201
1:00 pm	1300
2:00 pm	1400
3:00 pm	1500

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4:00 pm	1600
5:00 pm	1700
6:00 pm	1800
7:00 pm	1900
8:00 pm	2000
9:00 pm	2100
10:00 pm	2200
11:00 pm	2300
12:00 midnight	0000

Note: A.m. is a short form of the Latin word "ante-meridian" (before mid-day) and p.m. for "post-meridian" (after mid-day).

Example 1

Write 12:30 pm in the 24-hour format and read it.

Answer

12:30 pm is written as 1230 hours in the 24-hour format. It is read as "twelve thirty hours".

Example 2

Write 0235 hours in the 12-hour format.

Answer

0235 hour is written as 2:35 am in the 12-hour format.

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Example 3

Write 10:45 pm in 24-hour format and read it.

Answer

10:45 pm is written as 2245 hours in 24 hour format. It is read as "twenty-two forty-five hours".

Exercise 3

1. Change 3:47 am to 24-hour format.
2. Write 8:00 pm in 24-hour format.
3. Write 7:00 pm in 24-hour format.
4. Change the 0400 hours to 12-hour format.
5. Write 1700 hours in 12-hour format.
6. Write 2200 hours in 12-hour format.

Exercise 4

1. Write the following times in the 24-hour format:

(a)	9:00 am	(b)	1:00 pm
(c)	6:01 pm	(d)	3:45 pm
(e)	11:00 am	(f)	4:00 pm
(g)	12:50 am	(h)	8:00 am
(i)	8:34 am	(j)	5:55 am

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2. Write the following times in the 12-hour format.

(a) 0012 hours (b) 1710 hours

(c) 1650 hours (d) 0712 hours

(e) 2010 hours (f) 2324 hours

(g) 1015 hours (h) 0815 hours

(i) 1918 hours (j) 0000 hours

3. How many hours are there from 1800 hours to 2400 hours?

4. What time is 2300 hours in the 12-hour format?

5. Sheba told Bura that his watch showed 2:15 am. What time is that in the 24-hour format?

6. A doctor worked for 11 hours. If she started working at 1:30 am, at what time did she finish work in the 24-hour format?

7. The tourists arrived at Kilimanjaro airport at 1625 hours. Write the time in the 12-hour format.

8. A traffic police officer arrived at the police station at 11:45 am. What time did he arrive in a 24-hour format?

9. Asha had a headache on Friday at 1808 hours. Write the time in the 12-hour format.

10. Sofia and Ramadhani wake up to prepare themselves for school at 6:30 am everyday. Write this time in the 24-hour format.

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Relationship between units of time

The following diagram shows the relationship of the number of days in a week, weeks in a month, and months in a year.



Note: Time can be measured by counting the number of days, weeks, months or years.

(a) A day has 24 hours.

(b) A week has 7 days.

(c) A month has 4 weeks.

(d) A month has either 28, 29, 30 or 31 days.

(e) A year has 12 months.

Names of week days

One week has seven days. The names of the days are as shown in the following table:

Names of week days						
Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday

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Names of months in a year

A year has twelve months. Each month has its own name as shown in the following table.

January	February	March	April	May	June
1	2	3	4	5	6
31 days	28 or 29 days	31 days	30 days	31 days	30 days
July	August	September	October	November	December
7	8	9	10	11	12
31 days	31 days	30 days	31 days	30 days	31 days

Exercise 5

1. How many days are there in 24 hours?
2. How many week(s) are there in 7 days?
3. How many weeks are there in a month?
4. How many days are there in a month?
5. Twelve months are equal to _____ year(s).
6. Bakari bought an airtime for his mobile phone which lasted for 48 hours. How many days do the hours represent?

Calendar

A calendar displays dates, weeks, months and years. It also highlights national and international public holidays.

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A calendar helps to identify two types of years: a short year and a leap year. A leap year is the year in which February has 29 days. A short year is the year in which February has 28 days.

The following are the calendars for the year, 2019 and 2020.

2019

Tanzania

JANUARY

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

FEBRUARY

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28		

MARCH

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

APRIL

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

MAY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

JUNE

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

JULY

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

AUGUST

S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

SEPTEMBER

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

OCTOBER

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

NOVEMBER

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

DECEMBER

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

2019 Holidays for Tanzania

Jan 1 New Year's Day
Jan 12 Zanzibar Revolution Day
Apr 7 Karume Day
Apr 19 Good Friday
Apr 21 Easter Sunday
Apr 22 Easter Monday
Apr 26 Union Day
May 1 Labour Day
Jun 4 End of Ramadan (Eid al-Fitr)
Jul 7 Saba Saba Day
Aug 8 Nane Nane Day
Aug 11 Feast of the Sacrifice (Eid al-Adha)
Oct 14 Nyerere Day
Nov 9 Mauld Day
Dec 9 Independence Day
Dec 25 Christmas Day
Dec 26 Boxing Day

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2020

Tanzania

JANUARY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

FEBRUARY

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29

MARCH

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

APRIL

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

MAY

S	M	T	W	T	F	S
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

JUNE

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

JULY

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

AUGUST

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

SEPTEMBER

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30			

OCTOBER

S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

NOVEMBER

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

DECEMBER

S	M	T	W	T	F	S
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

2020 Holidays for Tanzania

Jan 1 New Year's Day
Jan 12 Zanzibar Revolution
Apr 7 Karume Day
Apr 10 Good Friday
Apr 12 Easter Sunday
Apr 13 Easter Monday
Apr 26 Union Day
May 1 Labour Day
May 24 End of Ramadan (Eid al-Fitr)
Jul 7 Saba Saba Day
Jul 31 Feast of the Sacrifice (Eid al-Adha)
Aug 8 Nane Nane
Oct 14 Nyerere Day
Oct 29 Maulid Day
Dec 9 Independence Day
Dec 25 Christmas Day
Dec 26 Boxing Day

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Exercise 6

1. Write the names of the months with 30 days.
2. Write the names of the months with 31 days.
3. How many Sundays are there in the month of July this year?
4. Mention the month in which normally schools close for the end of the year holiday.
5. How many days are there in February this year?

Activity: My day in clocks

Instructions: Use a basic drawing software(example, Paint) to create a digital timetable illustrating your daily activities. Draw both analogue and digital clock faces to depict the times you engage in each activity.

Summary

1. On the clock face, the short arrow shows hours, the long arrow shows minutes.
2. One hour is equal to 60 minutes.
3. Time can be measured by counting number of days, weeks, month or year

Vocabulary

Half an hour	30 minutes
A quarter of an hour	15 minutes

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Revision exercise

1. Draw the face of an analogue clock for each of the following times:
   (a) Quarter past six (6:15)
   (b) Ten o'clock (10:00)
   (c) Half past two (2:30)
   (d) Fourty-five minutes past three or quarter to four (3:45).

2. Write the following times in words:
   (a) 12:15 _____
   (b) 1:00 _____
   (c) 11:30 _____
   (d) 2:00 _____
   (e) 8:15 _____
   (f) 10:15 _____
   (g) 1:20 _____
   (h) 2:45 _____

3. Choose the item from list B that matches an item from list A.

List A	List B
(a) 90 minutes	Shows hours
(b) Hour-hand	Quarter an hour
(c) Minute-hand	Shows 12 hours
(d) 15 minutes	Shows minutes
(e) 30 minutes	60 minutes
	Half an hour
	Quarter to an hour
	One and a half hours

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4. Write the following times in the 12-hour format:

(a) 1930 hours
(b) 2324 hours
(c) 2127 hours
(d) 1513 hours
(e) 0712 hours
(f) 0835 hours

5. Write the following times in the 24-hour format:

(a) 8:34 pm
(b) 8:10 am
(c) 1:05 pm
(d) 11:30 am
(e) 5:55 am
(f) 3:51 pm
(g) 3:15 pm
(h) 4:00 pm

6. Write 4:45 am in the 24-hour format.

7. What time is 1945 hours in the 12-hour format?

8. How many hours are there from 1500 to 2300?

9. How many days are there in 72 hours?

10. Is this year a leap or a short year?

11. Write the total number of days in June, July and August.

12. Write the months that follow each other and each has 31 days.

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Chapter Seven

Tanzanian currency

Introduction

In this chapter, you will learn about the Tanzanian currency in shillings and cents. Also, you will add and subtract Tanzanian currency in shillings and cents. The competencies developed will enable you to use the Tanzanian currency in purchasing, budgeting, managing, and saving money.

Tanzanian currency

Tanzanian currency is in the form of shillings and cents. The value of one cent is smaller than the value of one shilling. One shilling is equal to one hundred cents. A cent is mostly observed in writing than its availability in everday use. Also, Tanzanian currency with different values are in the form of coins and banknotes.

Tanzanian coin

Tanzanian coins have different deniminations. In the circulation there are fifty, one hundred, two hundred and five hundred shillings coins. Tanzanian coins show various features including, faces of presidents, wildlife, historical monuments, and national symbols.

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Example 1

How do you differentiate a one hundred Tanzanian coin from other Tanzanian coins?

Answer

Each coin can be differentiated based on the number and features it contains. For example, the one hundred coin is written 100 on one of its sides. It also shows four running gazelles.

Example 2

What is the colour of the Tanzanian fifty shillings coin?

Answer

The Tanzanian fifty shillings coin is coated with the gold colour.

Example 3

Which coin has a picture of the first President of Tanzania?

Answer

It is the one hundred shillings coin.

Exercise 1

Carefully study the Tanzanian coins and answer the questions that follow.

(a)





_____ Shillings

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(b)





_____ Shillings

(c)





_____ Shillings

(d)





_____ Shillings

1. What is the value of each Tanzanian coin? Write the answer in the blank space.
2. Which coin has the highest value?
3. Which coin has the lowest value?
4. Which coins have you ever seen before?
5. Which coins have you ever used before?

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Tanzanian banknotes

Tanzania has banknotes with different denominations. The value of each banknote is given by a number. You can tell the difference between one banknote and another by looking at the different marks inscribed on them.

Exercise 2

Study the price in each of the following items and answer the questions that follow.

S/No	Price	Item
(a)	1000 shillings	 Plate
(b)	4000 shillings	 Fish
(c)	500 shillings	 Bunch of green vegetables
(d)	1000 shillings	 Loaf of bread

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(e)	5000 shillings		Cooking pot

1. Which item has the highest price?
2. Which one between a plate and cooking pot has a higher price?
3. Which item has the lowest price?
4. Which items have the same price?

Exercise 3

Study the following banknotes and answer the questions that follow:









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1. Mention the banknote which has the highest denomination.
2. Which banknote has the lowest denomination?
3. Mention three identification marks found on the front face of the five thousand shillings banknote.
4. Mention three identification marks found on the back face of the two thousand shilling banknote.
5. Mention five identification marks on the one thousand shilling banknote.

Writing Tanzanian currency in shillings and cents

When writing Tanzanian currency in words, begin with its value then shillings. When writing Tanzanian currency in short form begin with sh or shs, followed by its value. Also, when writing currency in cents, begin with its value, then cents.

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Example 1

(a) How many 50 cents are there in 1 shilling?

(b) Write 500 shillings and 50 cents in short form.

Solution

(a) 100 cents = 1shilling. Since 100 cts = 50 cts + 50 cts and sh 1 = 100 cts = 50 cts + 50 cts, it follows that, sh 1 is the same as twice 50 cts.

Therefore, there are two 50 cents in 1 shilling.

(b) In short form it is written as shs 500 and 50 cts.

Example 2

(a) Write shs 700 in words.

(b) Write in short form shs 1200 and 70 cts.

Answer

(a) shs 700 in words is written as seven hundred shillings.

(b) shs 1200 and 70 cts is written in short form as shs 1200 70 cts.

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Exercise 4

1. Four 50 cents is equal to how much shillings?
2. Read and write in words the value of money written in short form in the following table. Part (a) is provided as an example.

	Currency in shorts	Currency in words
(a)	shs 5800 35 cts	Five thousand eight hundred shillings and thirty-five cents
(b)	shs 75400 10 cts	
(c)	shs 23000 70 cts	
(d)	shs 632 37 cts	
(e)	shs 48350 05 cts	
(f)	shs 3600	
(g)	shs 80495 95 cts	
(h)	shs 1750 50 cts	
(i)	shs 10000	
(j)	shs 1100 20 cts	

3. Read and write in short form the value of money written in words. Part (a) is an example.

	Currency in words	Currency in short form
(a)	Two hundred shillings and twenty-five cents	shs 200 25 cts
(b)	Three hundred shillings	
(c)	Two thousand two hundred and fifty shillings and fifty cents	
(d)	Forty-five thousand shillings and thirty cents	
(e)	Seventy shillings and ten cents	

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4. Read and write the currency in short form in the following table. Part (a) is given as an example.

	Currency in words	Currency in short form
(a)	Fifty-five thousand shillings and eighty-five cents	shs 55000 85 cts
(b)	Six hundred and forty shillings and five cents	
(c)	One shilling and fifty cents	
(d)	Ninety-nine thousand eight hundred shillings and ninety cents	
(e)	Fifty shillings and sixty cents	

5. Rozi's monthly salary is 315,000 shillings and 80 cents. Write this salary in short form.

Addition of Tanzanian currency

Adding currency is the same as adding whole numbers. When adding currency, consider that one shilling is equal to hundred cents.

Example 1

shs 500 20 cts + shs 380 70 cts =

Steps

1. Add cents: 20 cts + 70 cts = 90 cts, Write 90 in the cents place.
2. Add shillings: shs 500 + shs 380 = shs 880. Write 880 in the shillings place.

Therefore, the answer is shs 880 90 cts.

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Example 2

	shs	cts
	4596	65
+	3987	75

Steps:

1. Add cents: 65 cts + 75 cts = 140 cts.
   Regroup 140 cts into sh 1 and 40 cts.
   Write 40 in the cents place and add 1 shilling to the shillings place.

	shs	cts
	4596	65
+	3987	75
		40

2. Add shillings:
   1 + 4596 + 3987 = sh 8584. Write 8584 in the shillings place.

	shs	cts
	4596	65
+	3987	75
	8584	40

Therefore, the answer is shs 8584 and 40 cts.

Example 3

Solution

	shs	11650
+	shs	78450

	shs	11650
+	shs	78450
	shs	90100

Therefore, the answer is shs 90100.

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Exercise 5

1. shs 5000 + shs 4000 =

2. shs 2438 + shs 4326 =

3. shs 1326 + shs 1318 =

4. shs 625445 50 cts + shs 357223 85 cts =

5. shs 14955 50 cts + shs 4955 55 cts =

6. shs 423293 45 cts + shs 549 314 45 cts =

7.
	shs	
	4521	
+ shs	2183	

8.
	shs	
	1326	
+ shs	2788	

9.
	shs	cts
	625	45
+	364	20

10.
	shs	cst
	5001	80
+	4999	45

11.
	shs	cts
	1060	05
+	2175	15

12.
	shs	cts
	714370	80
+	3056	35

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13.

	shs	cts
	433270	55
+	433865	45

14.

	shs	cts
	385534	05
+	453057	45

15.

	shs	60250
+	shs	28970

Word problems involving addition of Tanzanian currency

Example 1

Sabina sold two chicks. The first was sold at 33,620 shillings and the second at 22,350 Tanzanian shillings. How much money did Sabina get?

Solution

	shs	33620
+	shs	22350
	shs	55970

Therefore, Sabina got 55,970 Tanzanian shillings.

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Example 2

Mapunda spent 245,950 Tanzanian shillings for buying milk, and 152,850 Tanzanian shillings for transport. How much money did he spend on milk and transport altogether?

Solution

$$\begin{array}{rr} & \text{shs } 245950 \\ + & \text{shs } 152850 \\ \hline & \text{shs } 398800 \end{array}$$

Therefore, Mapunda spent 398,800 Tanzanian shillings.

Exercise 6

1. Saidi decided to sell two chickens. If the first chicken was sold for 13,600 Tanzanian shillings and the second was for 16,250 Tanzanian shillings. How much money did Saidi get altogether?
2. A vegetable vendor got 5000, 2000 and 1000 Tanzanian shillings notes and 200 Tanzanian shillings coin after selling vegetables. Find the total money the vendor got.
3. Yohana sold eggs for 8,600 shillings and a chicken for 17,500 Tanzanian shillings. How much money did he get?
4. Maria deposited 230,000 Tanzanian shillings in the bank. The bank added 4,550 shillings and 75 cents to her as a profit. How many shillings does Maria have in total?

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5. A certain youth group started a bee-keeping project. It spent 42,400 Tanzanian shillings to buy beehives and 48,200 Tanzanian shillings to buy honey harvesting equipment. How much money did it spend on starting the project?

6. A cook bought a bag of charcoal for 43,350 Tanzanian shillings and a bag of maize for 55,600 Tanzanian shillings. How much did the cook pay for the items?

7. A pupil bought a Mathematics book for 9,500 Tanzanian shillings and an exercise book for 2,600 Tanzanian shillings. How much money did the pupil spend on purchasing the books?

8. Bupe bought a bag of rice at 438,550 Tanzanian shillings. She got a profit of 96,350 Tanzanian shillings after selling it. How much money did she get after selling the bag of rice?

9. Jane got a loss of commodity for 2,500 Tanzanian shillings after selling the commodity for 87,500 Tanzania shillings. What was the buying price of the commodity?

10. Juhudi got 52,500 Tanzania shillings as a gift for passing mathematics examination. He also, received 21,700 Tanzanian shillings from his mother. If Juhudi had 18,550 Tanzanian shillings before, how much money did he have in total?

Subtraction of Tanzanian currency

Subtrating Tanzanian currency is done like subtracting whole numbers from right to left.

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Example 1

shs 4896595 cts – shs 25843 85 cts =

Steps

1. *Subtract cents:* 95 – 85 = 10 *cts. Write* 10 *cts in the cents place.*
2. *Subtract shillings:* 48965 – 25843 = *shs* 23122.
   Write shs 23122 *in the shillings place.*

Therefore, the answer is shs 23122 10 *cts.*

Example 2

shs	cts
83721	85
–61510	40

Steps

1. *Subtract cents:* 85 cts – 40 cts = 45 cts.
   Write 45 *cts in the cents place.*

shs	cts
83721	85
–61510	40
	45

2. *Subtract shillings:*
   83721 – 61510 = sh 22211
   Write sh 22211 *in the shillings place.*

shs	cts
83721	85
–61510	40
22210	45

Therefore, the answer is shs 22211 45 *cts.*

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Example 3

	shs	cts
	869335	30
−	427123	70

Steps

1. *Subtract cents: 30 − 70, It is not suffient. Take 1 shilling from 5 shillings and regroup it into 100 cents.*

   Add cents: 100 + 30 = 130 cts.

   Subtract cents: 130 − 70 = 60 cts.

   Write 60 cts in the cents place.

   Remember, 1 shilling from 5 was regrouped into 100 cents.

   Thus, 4 shillings remained.

	shs	cts
	869335	30
−	427123	70
		60

2. *Subtract: shillings:*

   869334 − 427123 = sh 442211.

   Write sh 442211 in the shillings place.

   Therefore, the answer is shs 442211 60 cts.

	shs	cts
	869335	30
−	427123	70
	442211	60

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Example 4

shs 4180
− shs 2494

Solution

shs 4180
− shs 2494
shs 1686

Exercise 7

1. shs 4561 − shs 2364 =
2. shs 5751 − shs 4030 =
3. shs 7620 − shs 5245 =
4. shs 90900 − shs 75300 =
5. shs 58675 − shs 39248 =
6. shs 89990 − shs 68990 =
7. shs 914955 60 cts − shs 612950 05 cts =
8. shs 870 300 40 cts − shs 413200 25 cts =

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9. shs 4330
− shs 1215

10. shs 6779
− shs 4125

11. shs 80560
− sh 60590

12. shs 99100
− shs 65700

13. shs 75750
− shs 25920

14. shs 35000
− shs 20090

15. shs 21800
− shs 15350

16. shs 42300
− shs 18800

17.

shs	cts
720561	65
− 310340	40

18.

shs	cts
564868	25
− 19853	75

19.

shs	cts
867662	20
− 453221	10

20.

shs	cts
867662	20
− 453221	10

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21.

	shs	cts
	21201	10
−	11102	60

22.

	shs	cts
	68654	14
−	18129	14

23.

	shs	cts
	4500	80
−	180	45

24.

	shs	cts
	100	10
−	90	20

25

	shs	cts
	945601	70
−	431102	60

Word problems involving subtraction of Tanzanian currency

Example 1

Anna had 80,800 Tanzanian shillings. She bought school uniform and exercise books at 20,000 Tanzanian shillings and 17,500 Tanzanian shillings, respectively. What was Anna's balance?

Solution

School uniform	shs 20000
Exercise books	+ shs 17500
	shs 37500

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Anna spent 37500 shillings on buying school uniform and exercise books.Subtract the amount she spent from the amount she had before buying school uniform and exercise books.

The amount she had	shs 80800
The money spent	– shs 37500
	shs 43300

Therefore, Anna's balance was 43,300 Tanzanian shillings.

Example 2

Peter sold one cow for 450,000 shillings on the day of the auction. After the auction, he gave his son shs 215000 for school needs. How much money did Peter remain with?

Solution

The price of cow	shs 450000
The amount he gave his son	– shs 215000
	shs 235000

Peter was left with 235000 Tanzanian shillings.

Exercise 8

1. Halima went to the shop with 5,000 Tanzanian shillings to buy sugar. She bought one kilogram of sugar for 3,325 Tanzanian shillings. How many shillings did Halima return home with?

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2. Musa got a profit of 2,070 Tanzanian shillings after selling maize at 97,650 Tanzanian shillings. How much money did Musa spend on the maize?
3. Bahati had 10,000 Tanzanian shillings which she gave to her two children for transport. If the first child was given 4,600 Tanzanian shillings, how much money was given to the second child?
4. Rehema had 75,000 Tanzanian shillings and spent 31,650 Tanzanian shillings. How many shillings did Rehema remain with?
5. Seif took 85,000 Tanzanian shillings from his savings in the bank. He paid shillings 50,449 and 25 cents for electricity and shillings 15,000 and 75 cents for water. How much money did Seif keep?
6. Mwanaheri was given 60,000 Tanzanian shillings by her father to buy schools needs. If she used 31,500 Tanzanian shillings to buy exercise books and 8,500 Tanzanian shillings to buy a textbook. How many shillings did Mwanaheri remain with?
7. Safina was given 1,000 Tanzanian shillings by her mother for expenses at school. She bought a bottle of drinking water at 450 Tanzanian shillings and chapati at 300 Tanzanian shillings. How much money did she remain with?
8. Pendo had 90,000 Tanzanian shillings. She bought bedsheets at 32,500 Tanzanian shillings and a gown at 40,000 Tanzanian shillings. How much money did she remain with?
9. Joseph had 350,000 Tanzanian shillings. He used 127,500 Tanzanian shillings. How much money was he left with?

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10. Fabian bought a television set for 435,250 Tanzanian shillings. What profit did he make after selling the television set at 582,950 Tanzanian shillings?

Activity: Learning Tanzanian currency through online programmes

Instruction: Use online programmes to learn addition and subtraction of shillings and cents.

Summary

1. When you separate shillings and cents in digits, cents are written as two digits.
2. One hundred cents is equal to one shilling.
3. When adding and subtracting currency, start with cents and then shillings.
4. Tanzanian currency is in the form of coins and banknotes with different values.
5. Money is used for making payments. In addition, it shows the value of things in the form of prices.

Vocabulary

Banknote	a piece of special paper representing a particular value of money
Coin	a piece of metal representing a particular value of money

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Revision exercise

Study the Tanzanian banknotes and then answer the questions 1 to 3.

(a)





(b)





1. Mention the banknotes (a) and (b).
2. Mention three identification marks which are found on banknote (a).
3. Mention three identification marks which are found on banknote (b).
4. Write the following amounts of money in short form.

Currency in words	Currency in short form
(a) One thousand one hundred shillings and twenty cents	
(b) Six hundred and forty shillings and five cents	

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5. shs 600 + shs 400 =

6. shs 4000 + shs 3700 =

7. shs 8700 + shs 900 =

8. shs 450 + shs 7300 + shs 100 =

9. shs 25300 + shs 60200 =

10. shs 52670 + shs 17380 =

11. shs 11500 + shs 80150 =

12. shs 36800 + shs 44000 =

13. shs 63435 10 cts – shs 8375 65 cts =

14. shs 145489 15 cts – shs 51432 45 cts =

15. shs 32456 10 cts + shs 63367 65 cts =

16. shs 25445 50 cts + shs 57223 85 cts =

17. shs 7535 – shs 6025 =

18. shs 4050 – shs 2500 =

19. shs 392450 – shs 138510 =

20. shs 714000 – shs 35600 =

21.
shs 6430
– shs 5240

22.
shs 9883
– shs 3265

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23.

	shs	cts
	90527	85
−	30210	80

24.

	shs	cts
	71171	00
−	15101	50

25.

	shs	cts
	67501	70
−	35322	35

26.

	shs	cts
	868660	45
−	534330	15

27.

	shs 27800
+	shs 39400

28.

	shs	cts
	156 280	15
+	138 270	30

29.

	shs 70880
+	shs 16970

30.

	shs	cts
	178850	45
+	235112	55

31.

	shs	cts
	5075	05
+	3350	50

32.

	shs	cts
	263009	65
−	124087	55

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33. Suzana earned 5,380 Tanzanian shillings on Saturday and 2,010 on Monday. How much money did she get in total?

34. Paza spent 45,000 Tanzanian shillings on buying milk and 52,500 Tanzanian shillings on transport. How much money did he spend in total?

35. Obedi needs 5,500 Tanzanian shillings to buy shoes. If he has 2,600 Tanzanian shillings, how many shillings does he require to be able to buy shoes?

36. Dorice bought a goat for 55,000 Tanzanian shillings and sold it for Tanzanian 70,500 Tanzanian shillings. How much profit did she get?

37. Safina was given 1,000 Tanzanian shillings for school expenses. She bought a bottle of drinking water at 500 Tanzanian shillings and sweets at 250 Tanzanian shillings. How much money did she remain with?

38. Nasibu gave out a 10,000 Tanzanian shillings note for bus ticket worth 6,500 Tanzanian shillings. How much did he receive back?

39. Last month Skola saved 5,600 Tanzanian shillings and this month she has saved 2,500 Tanzanian shillings. How much money does she have so far?

40. Ali used 25,000 Tanzanian shillings for buying shoes, 30,000 Tanzanian shillings for bus fare and 45,000 Tanzanian shillings for school uniform. How many shillings did he spend in total?

Date ↓	Impressions	Profit	CP
23.01.2025	75,971	$ 41.030	0.
22.01.2025	49,122	$ 23.078	0.
21.01.2025	56,597	$ 27.336	0.
20.01.2025	50,146	$ 18.909	0.
19.01.2025	37,928	$ 10.405	0.
18.01.2025	36,461	$ 8.744	0.
17.01.2025	45,379	$ 15.136	0.
16.01.2025	54,288	$ 22.469	0
15.01.2025	59,027	$ 19.485	0.
14.01.2025	64,152	$ 30.660	0.
13.01.2025	63,396	$ 28.994	0.
12.01.2025	45,738	$ 14.895	0.
11.01.2025	47,332	$ 14.892	0
10.01.2025	49,546	$ 14.808	0.
09.01.2025	51,119	$ 15.268	0.
08.01.2025	46,677	$ 13.182	0.
07.01.2025	48,710	$ 14.385	0.

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Application Status	Job Title	Date Applied	Let
Shortlisted - Session Location is DAR ES SALAAM UNIVERSITY COLLEGE OF EDUCATION (DUCE) LECTURE THEARE C at DAR ES SALAAM, Time: 07:00:00 25-01-2025 Interview No: PSRS/24/8613/26123 *>> View Timetable*	**Employer:** Mkoa wa Pwani **Job Post:** MWALIMU DARAJA LA III C - KISWAHILI	2024-08-01	V
Shortlisted - Session Location is DAR ES SALAAM UNIVERSITY COLLEGE OF EDUCATION (DUCE) LECTURE THEARE B at DAR ES SALAAM, Time: 07:00:00 01-02-2025 Interview No: PSRS/24/8576/21949 *>> View Timetable*	**Employer:** Mkoa wa Pwani **Job Post:** MWALIMU DARAJA LA III C - JIOGRAFIA (GEOGRAPHY)	2024-07-31	V

Showing 1 to 2 of 2 entries

Sio rahisi kufaulisha sayansi kwa shule za kata(wazazi)! Mazingira magumu ya kufundishia, tabia mbaya kwa wanafunzi, kukosa motivation kwa masomo ya sayansi, na vikwazo vingi

Nafundisha Biology na chemistry

Wanafunzi wangu wamejua kunigurahisha

Chemistry no F
Biology F moja,
Mwenzangu wa Physics F tatu tu!
Kati ya wanafunzi 72!

Siri ni resources kutoka wazaelimu!

Ahsanteni sana!! Edited 8:44 in the evening

Hongera sana mwalimu 🔥🔥 9:51 at night ✓

Message

ikipatikana tutaiweka 🙏🏾

8:59 in the morning ✓✓

You

ikipatikana tutaiweka 🙏🏾

Mungu awabarik mno kaka, nyie ndo tegemeo letu 🙏🏾

2:31 in the afternoon

Haya ndo matokeo ya shule yetu ya somo la commerce baada ya kutumia wazaelimu, na zaid ya waza elim sijawah tumia resources zingine, shule ni ya serikali ya kata, kabla ya wazaelimu sikuwapata mataokeo makubwa hivi, nawashukuru sana

A=7
B=18
C=4
D=0
F=0

2:36 in the afternoon

Hongera sana, Mwalimu. 🙏🏾 🙏🏾

2:55 in the afternoon ✓✓

Hello 8:21 in the morning

hello karibu 8:21 in the morning ✓✓

Nashindwa kijiunga na wazaelimu iliionekane kwenye wasap. 8:23 in the morning

channel? 8:58 in the morning ✓✓

Ndio ionekane kama vile magroup au majina kwenye wasap 1:48 in the afternoon

https://whatsapp.com/channel/0029VaCPHC4GpLHWlLdTrQ3p

2:57 in the afternoon ✓✓

ingia hapo na uifollow 2:57 in the afternoon ✓✓